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(54) **METHODS FOR THE PREPARATION OF TRYPSIN-RESISTANT POLYPEPTIDES FOR MASS SPECTROMETRIC ANALYSIS**

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(58) **Field of Classification Search**

CPC **G01N 33/6848**; **G01N 2333/69441**
See application file for complete search history.

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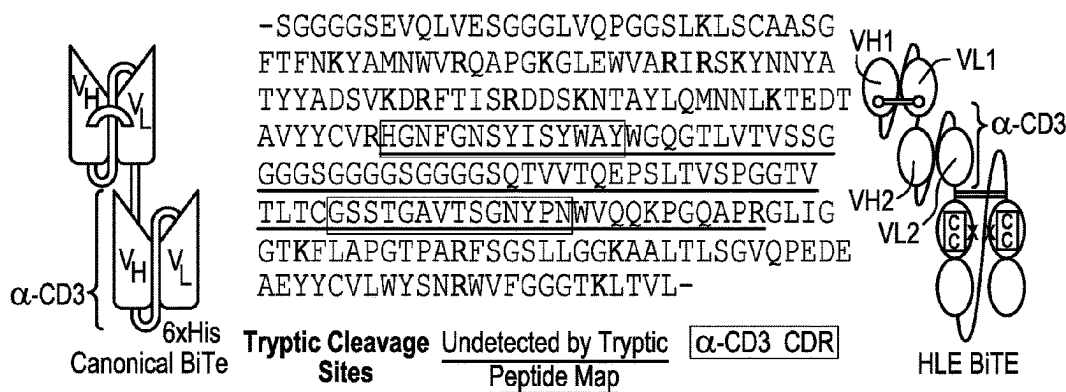
(57) **ABSTRACT**

The disclosed methods are directed to preparing and detecting polypeptides using neutrophil elastase, such as human neutrophil elastase. The polypeptides are optionally denatured, reduced, and/or alkylated before being subjected to a first digestion. A second digestion comprises the use of neutrophil elastase, such as human neutrophil elastase. The prepared fragments are then analyzed chromatographically, electrophoretically, or spectrometrically, or a combination of these methods. The methods are especially useful for the preparation of therapeutic polypeptides for analysis, especially those that are not easily cleaved, such as some bi-specific T-cell engager (BiTE®) molecules.

20 Claims, 10 Drawing Sheets

Specification includes a Sequence Listing.

Conserved α -CD3 Domain



(56)

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FIG. 1

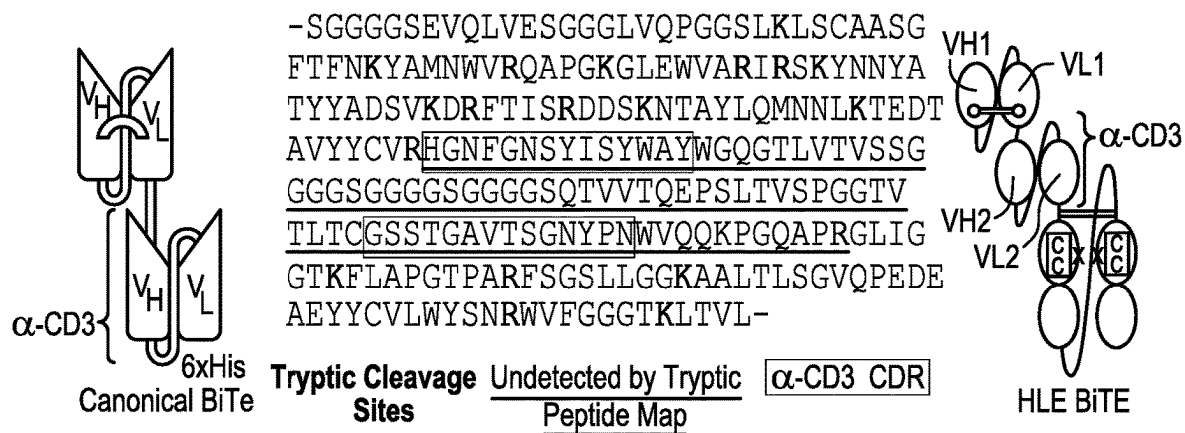
Conserved α -CD3 Domain

FIG. 2

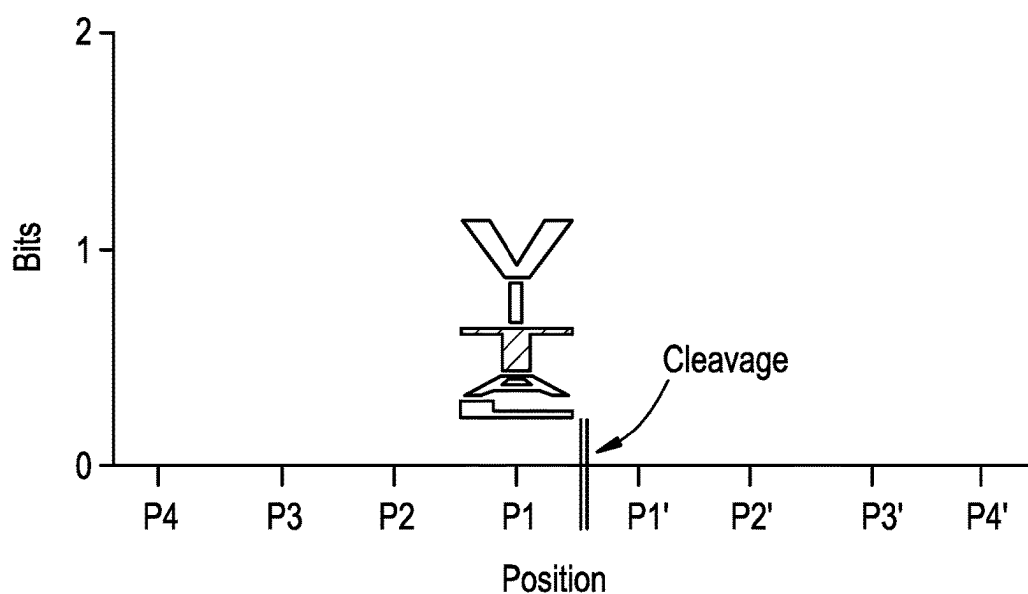


FIG. 3

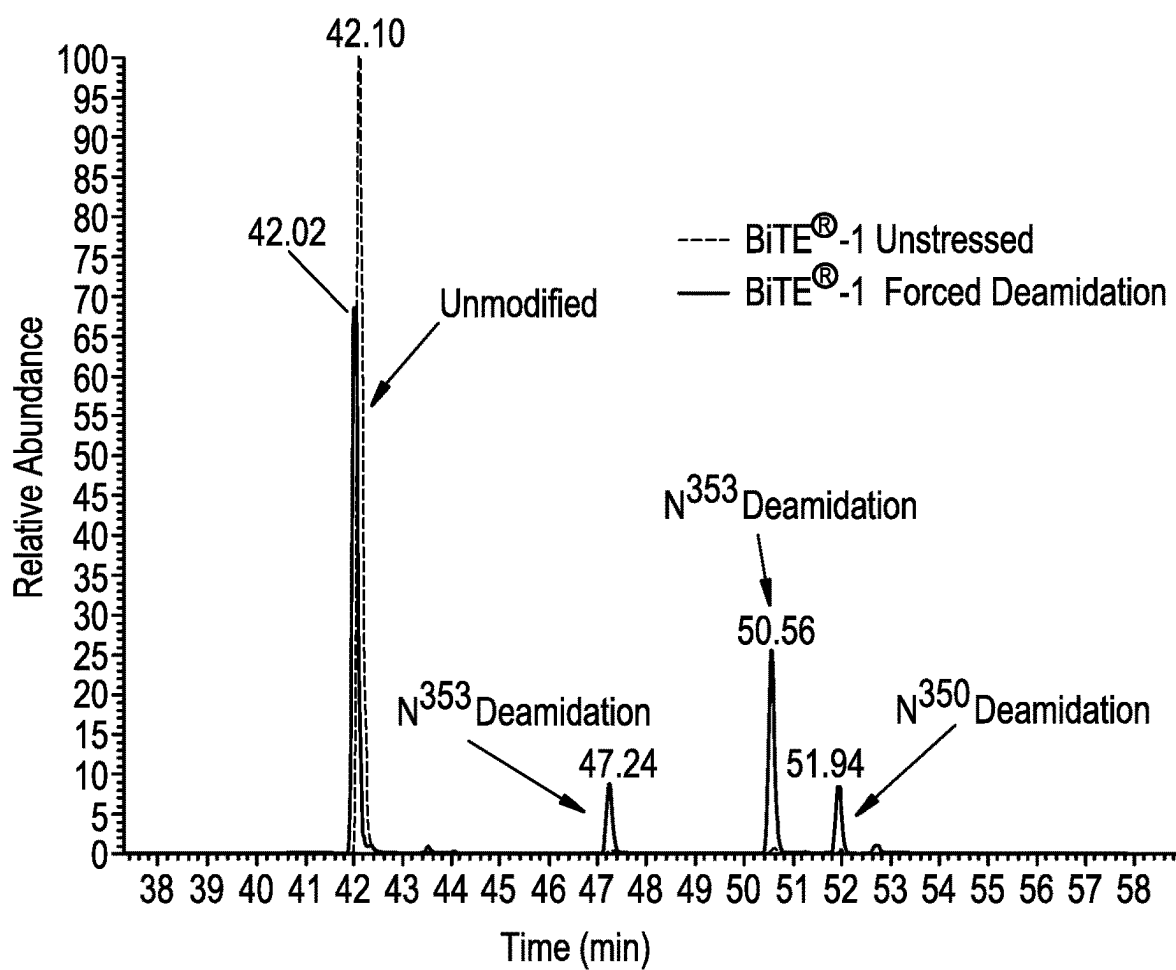


FIG. 4

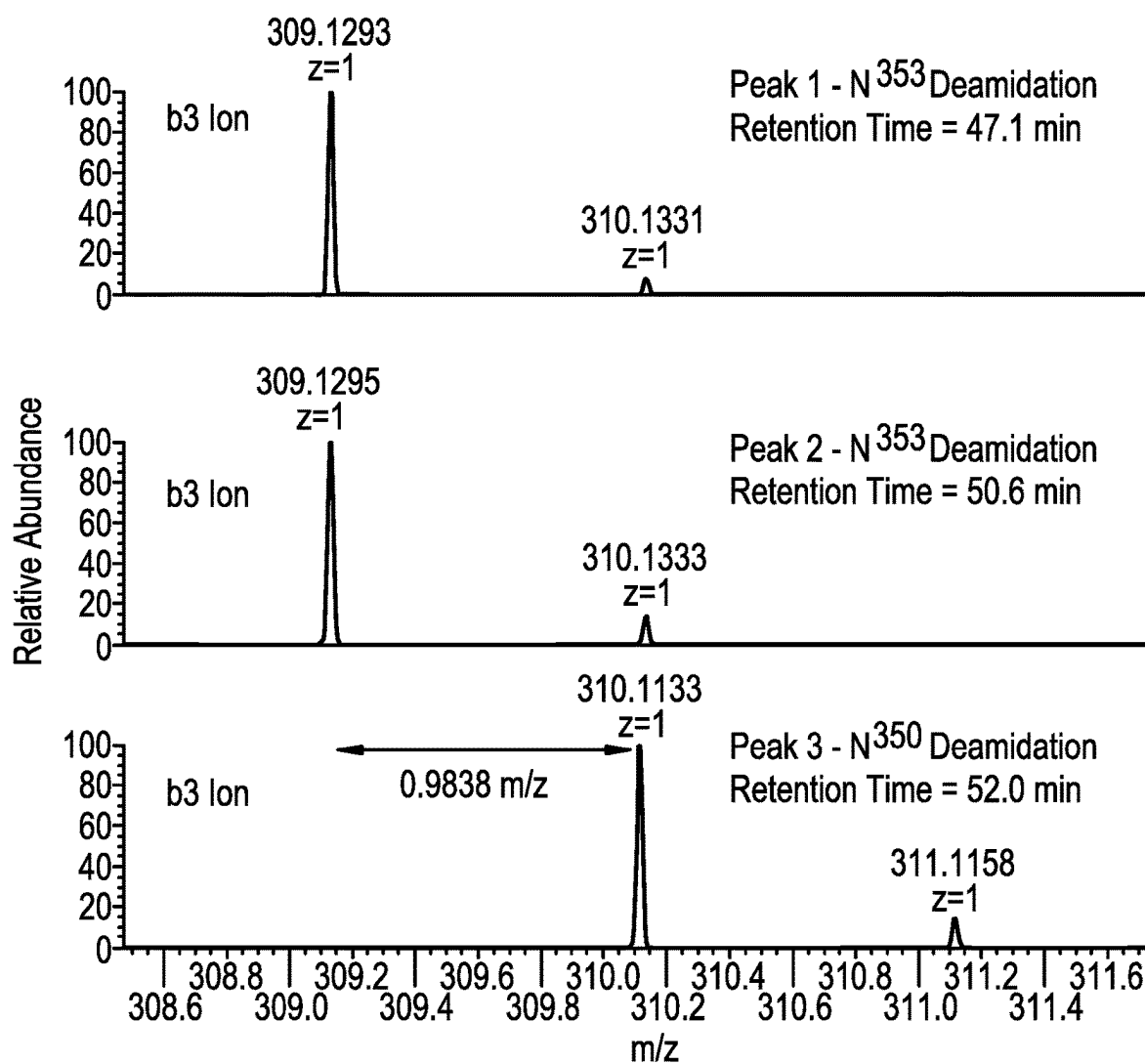


FIG. 5

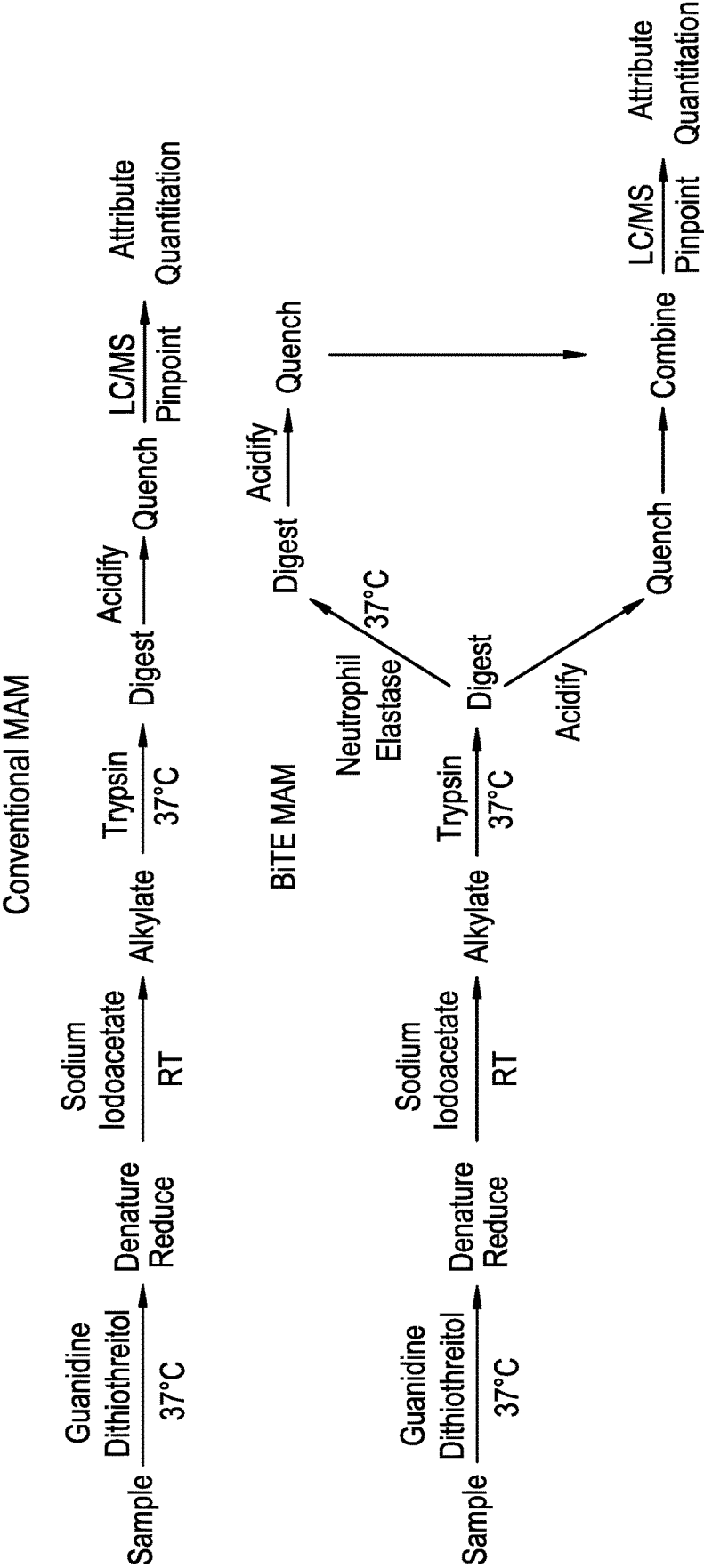
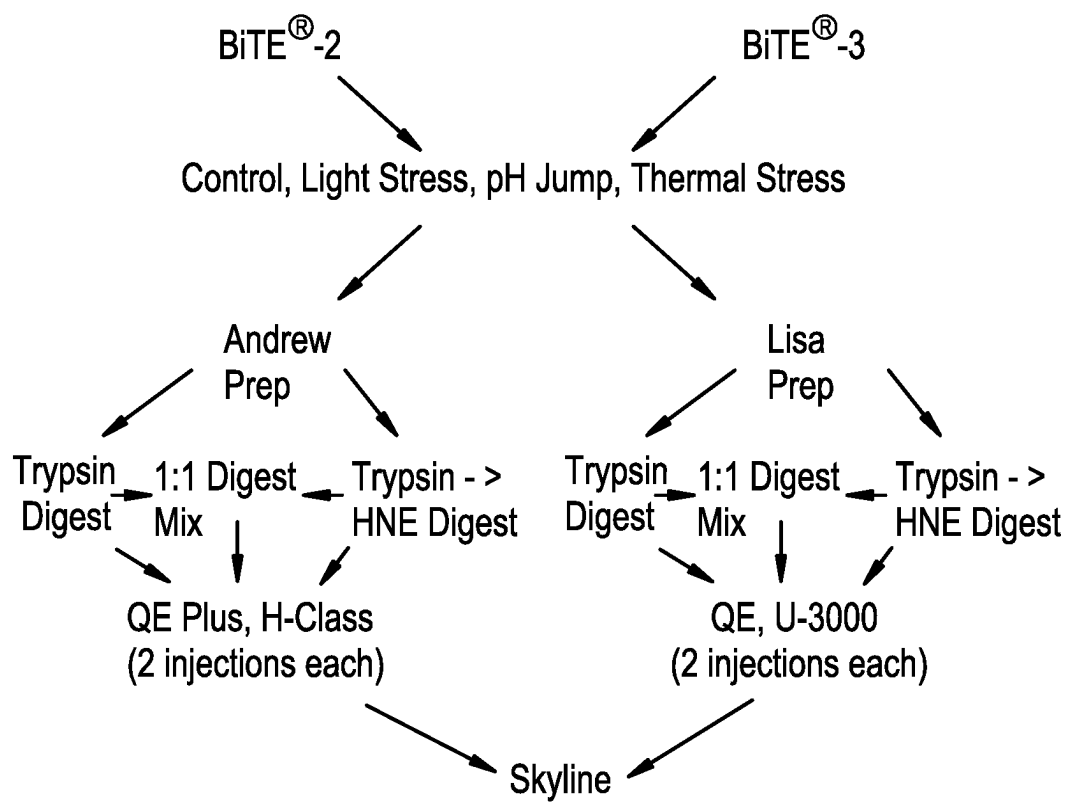


FIG. 6



Multi-Enzyme Filter-Based Sequential Digestion MAM

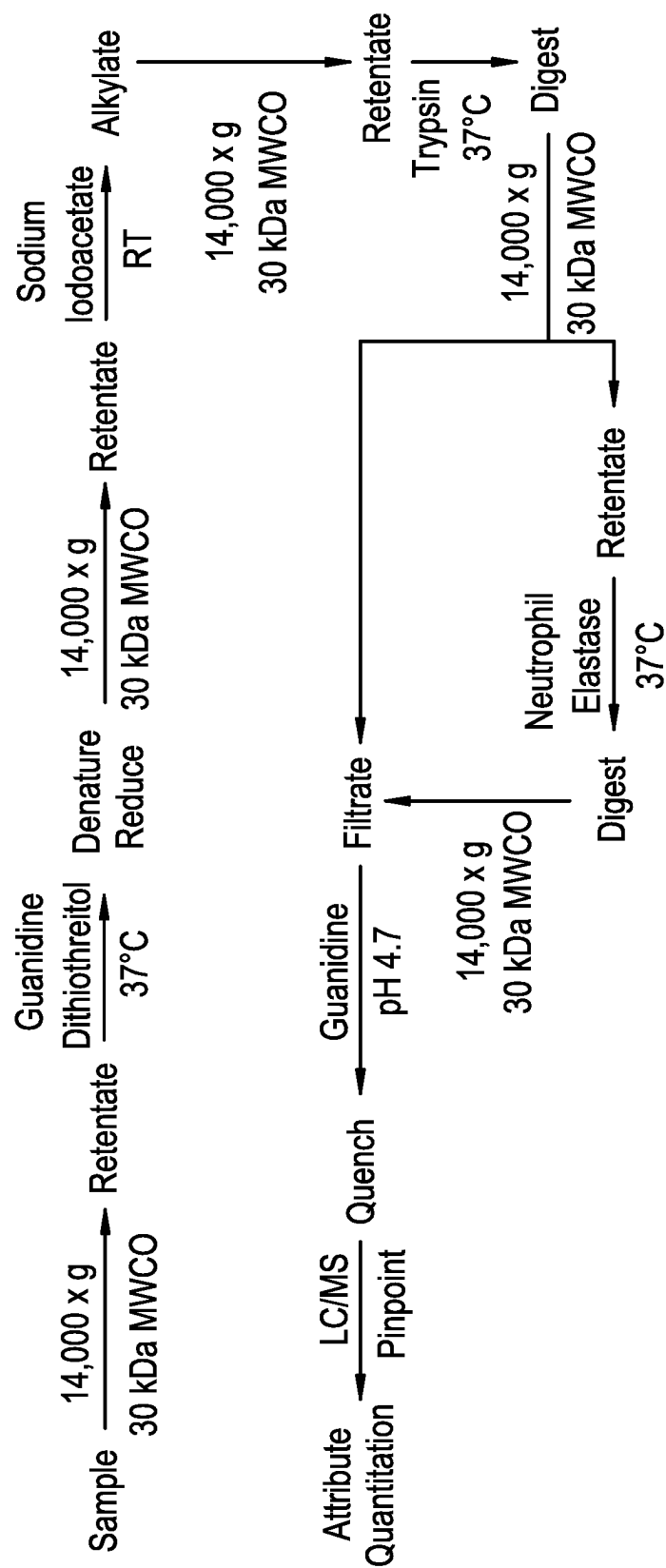
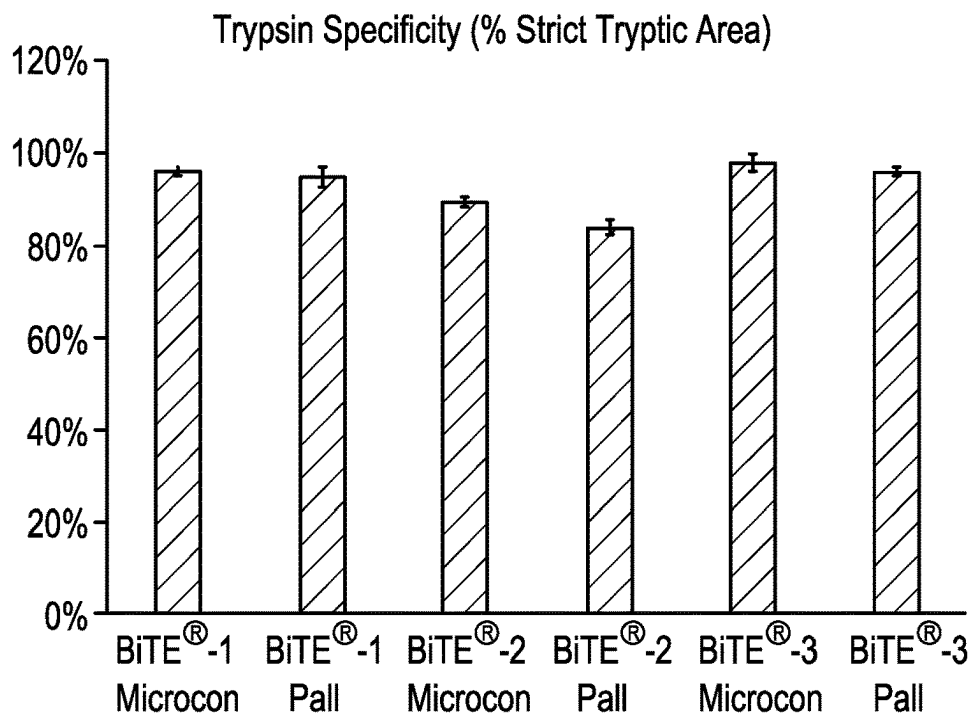


FIG. 8

% Strict Tryptic Area = Total Area of Identified Peptides (Strict Tryptic Search)/
Total Area of Identified Peptides (KRVITAL Cleavage Search)

FIG. 9

% Total Area Corresponding to 8 kDa Linker

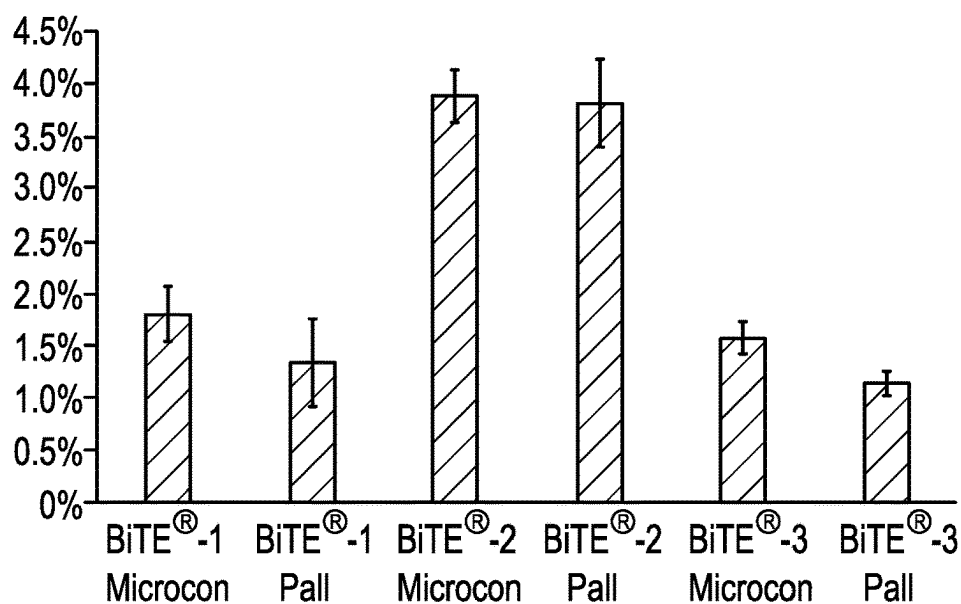


FIG. 10

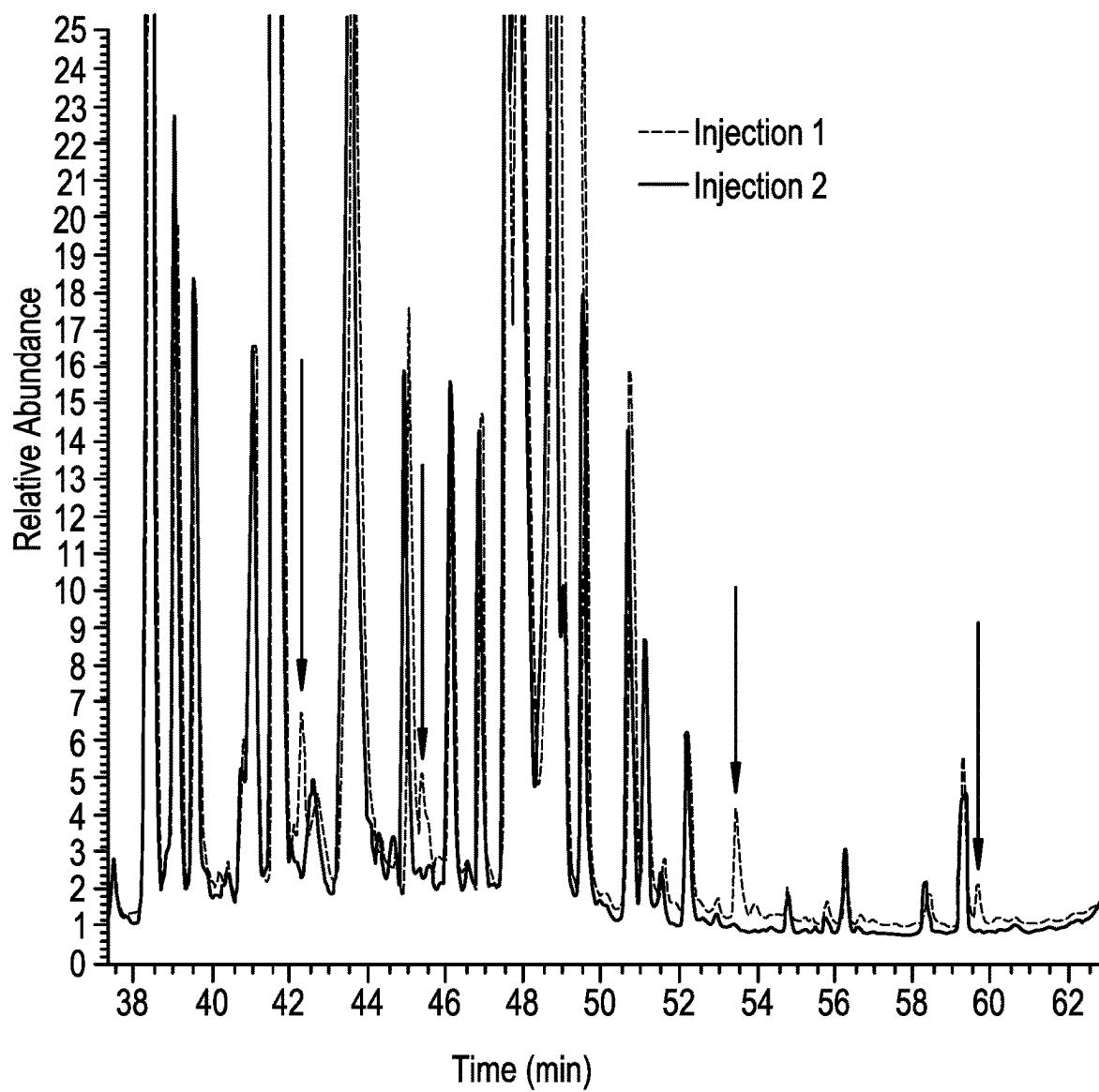


FIG. 11

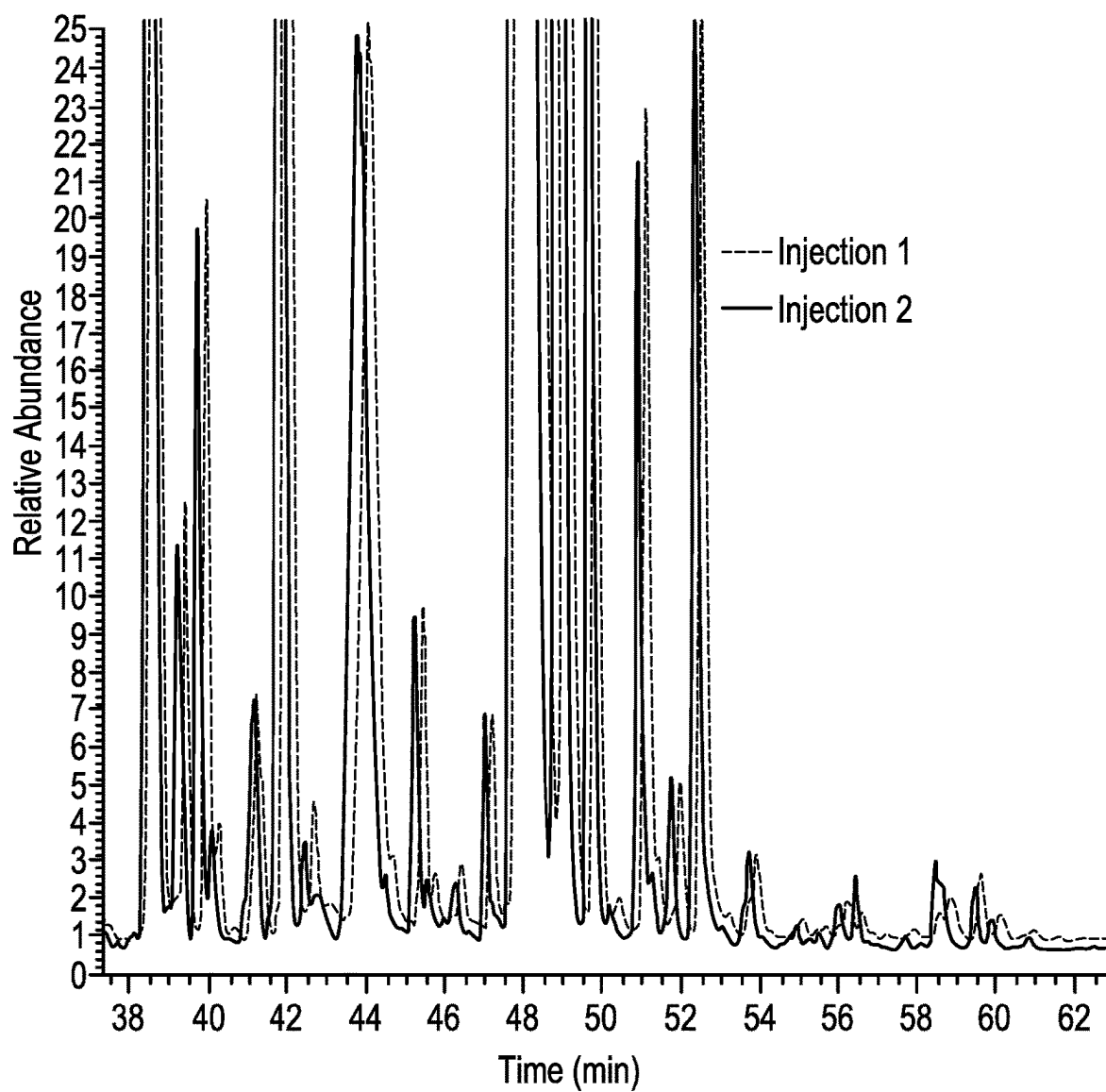
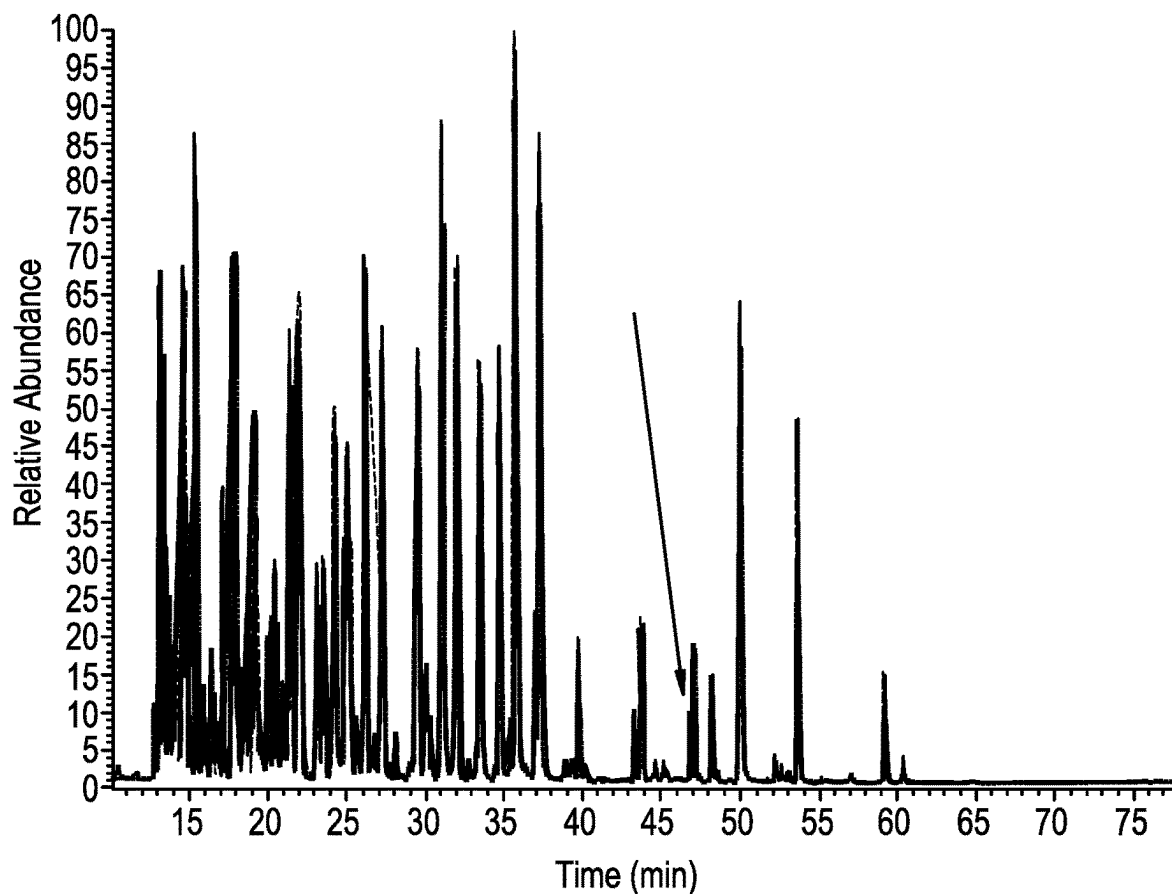


FIG. 12



- BiTE[®]-2 Protocol 4 Prep 1
- BiTE[®]-2 Protocol 4 Prep 2
- BiTE[®]-2 Protocol 4 Prep 3
- BiTE[®]-2 Protocol 5 Prep 1
- BiTE[®]-2 Protocol 5 Prep 2
- BiTE[®]-2 Protocol 5 Prep 3

1

METHODS FOR THE PREPARATION OF TRYPSIN-RESISTANT POLYPEPTIDES FOR MASS SPECTROMETRIC ANALYSIS

RELATED APPLICATIONS

This application is a National Stage application under 35 U.S.C. § 371 of International Application No. PCT/US2019/021927, having an international filing date of Mar. 12, 2019, which claims the benefit of U.S. Provisional Patent Application No. 62/642,444, filed Mar. 13, 2018, the entire contents of which are incorporated herein by reference in their entirety for all purposes.

SEQUENCE LISTING

The present application is being filed with a sequence listing in electronic format. The sequence listing provided as a file titled, "A-2151-WO-PCT sequence listing_ST25.txt," created 2019 Mar. 12, and is 261 KB in size. The information in the electronic format of the sequence listing is incorporated herein by reference in its entirety.

FIELD

The presented subject matter relates to the field of polypeptide analysis. Specifically, the presented subject matter relates to the preparation of samples for the detection of polypeptides comprised therein, such as therapeutic polypeptides. The disclosed methods use sequential digestion in the preparation of the polypeptides, wherein a last digestion comprises the use of neutrophil elastase.

BACKGROUND

Multi-attribute method (MAM) relies on enzymatic digest of molecules followed by mass spectrometry-based characterization and quantitation of attributes of interest. Modifications in the complementary-determining regions (CDRs) are of particular concern because they may impact potency and/or safety of the molecule. Trypsin, which cleaves as the C-terminus of lysine and arginine residues, is typically utilized as the enzyme of choice for peptide mapping and MAM for many reasons, including high digestion specificity and the frequency of occurrence of lysine and arginine residues; trypsin digestion results in peptides with basic residues on the C-terminus which also typically have an optimal length for mass spectrometry-based analysis. However, many of the bi-specific T-cell engager (BiTE®) molecules currently being developed at Amgen contain a conserved α -CD3 domain with two CDRs in close proximity to a long linker region. One of these CDRs (HGNFGNSYISYWAY) contains two asparagine residues which may be susceptible to deamidation and a tryptophan residue that may be susceptible to oxidation. This region of the molecule contains no residues amenable to trypsin digestion (FIG. 1). As such, attributes of interest in the CDR domains cannot be monitored by MAM due to the large size of this peptide (~8 kDa), the difficulty in chromatographically separating modified versions of the peptide, poor recovery and/or ionization of the peptide, and general challenges associated with interpreting the mass spectrometry data corresponding to peptides of this size. In addition, this linker region has no residues susceptible to commonly used secondary proteases such as Asp-N, Lys-C, and Glu-C. While chymotrypsin does cleave at tryptophan residues, utilization of this enzyme was not pursued both because of the enzyme's low specificity

2

and because oxidation at tryptophan residues may affect quantitation of this attribute. Likewise, many BiTE® molecules also have a shorter linker peptide (~5-7 kDa) in close proximity to target-specific CDRs that is similarly difficult to monitor with trypsin digestion. For example, a potential CDR aspartic acid isomerization site is located within the 5.7 kDa linker peptide. Collectively, these attributes may impact target and/or CD3 binding, which necessitates MAM-based monitoring.

SUMMARY

In a first aspect, provided herein are methods of preparing a polypeptide for analysis, comprising cleaving the polypeptide in a sample in a first digestion, wherein the cleaving produces at least two fragments of the polypeptide, which the at least two fragments of the polypeptide are subsequently digested by a neutrophil elastase; and analyzing the sample after digesting the at least two fragments of the polypeptide with the neutrophil elastase.

In a second aspect, provided herein are methods of preparing a polypeptide for analysis, comprising cleaving the polypeptide in a sample in a first digestion, wherein the cleaving produces at least two fragments of the polypeptide; dividing the sample comprising the at least two fragments into a first aliquot and a second aliquot; digesting the at least two fragments of the first aliquot with a neutrophil elastase; and analyzing the first aliquot and the second aliquot. In some sub-aspects, after digesting the at least two polypeptide fragments with the neutrophil elastase, the first and second aliquots are combined; in such cases, the first and second aliquot can comprise approximately equal amounts of the polypeptide.

In both first and second aspects, cleaving the polypeptide in the first digestion comprises proteolytic or chemical cleavage. In those sub-aspect comprising proteolytic cleavage, such cleavage is accomplished by a protease, wherein the protease has an activity different from the neutrophil elastase. In some sub-aspects, the protease is selected from the group consisting of trypsin, endoproteinase Glu-C, endoproteinase Arg-C, pepsin, chymotrypsin, chymotrypsin B, Lys-N protease, Lys-C protease, Glu-C protease, Asp-N protease, pancreatopeptidase, carboxypeptidase A, carboxypeptidase B, proteinase K, and thermolysin, and combinations thereof. In some sub-aspects of these first and second aspects, the protease is selected from the group consisting of trypsin, Asp-N, and Glu-C, and combinations thereof; in yet other sub-aspects, the protease is trypsin. However, when chemical cleavage is used to fragment the polypeptide in the sample in these first and second aspects, such cleavage is accomplished by a chemical, such as a chemical selected from the group consisting of cyanogen bromide, 2-Nitro-5-thiocyanobenzoate, hydroxylamine, and BNPS-skatole, and combinations thereof.

Furthermore, the polypeptide can be denatured, and/or reduced, and/or alkylated before cleaving the polypeptide in the sample in the first digestion. The neutrophil elastase can be, for example, human neutrophil elastase (EC 3.4.21.37). When analyzing the sample, the analysis can include at least one technique selected from the group consisting of chromatography, electrophoresis, spectroscopy, or spectrometry, and combinations thereof. For example, when the technique for analyzing the sample comprises chromatography, the chromatographic technique can be selected from the group consisting of gas chromatography, liquid chromatography, high performance liquid chromatography, ultra-performance liquid chromatography, size-exclusion chromatography, ion-

3

exchange chromatography, affinity chromatography, expanded bed adsorption chromatography, reverse-phase chromatography, hydrophobic interaction chromatography, and combinations thereof; when the technique is electrophoresis, the electrophoretic technique can be selected from the group consisting of gel electrophoresis, free-flow electrophoresis, electrofocusing, isotachopheresis, affinity electrophoresis, immunoelectrophoresis, counterelectrophoresis, and capillary electrophoresis, capillary zone electrophoresis, and combinations thereof; and when the technique is spectroscopic (or spectrometric), the spectroscopic (or spectrometric) technique can be selected from the group consisting of mass spectrometry, ultraviolet spectroscopy, visible light spectroscopy, fluorescent spectroscopy, and ultraviolet-visible light spectroscopy, and combinations thereof. In some particular sub-aspects, the technique comprises liquid chromatography-mass spectrometry; in other cases, the technique comprises capillary zone electrophoresis coupled to mass spectrometry.

In a third aspect, disclosed herein are methods of preparing a polypeptide for analysis, comprising denaturing, reducing, and alkylating the polypeptide; digesting the polypeptide with trypsin to produce trypsin-cleaved polypeptide fragments; dividing the trypsin-cleaved polypeptide fragments into a first aliquot and a second aliquot; digesting the trypsin-cleaved polypeptide fragments of the first aliquot with neutrophil elastase; combining the first aliquot and second aliquot at about a 1:1 ratio; and analyzing the combined aliquots. The neutrophil elastase can be, for example, human neutrophil elastase (EC 3.4.21.37). When analyzing the sample, the analysis can include at least one technique selected from the group consisting of chromatography, electrophoresis, spectroscopy, spectrometry, and combinations thereof. For example, when the technique for analyzing the sample comprises chromatography, the chromatographic technique can be selected from the group consisting of gas chromatography, liquid chromatography, high performance liquid chromatography, ultra-performance liquid chromatography, size-exclusion chromatography, ion-exchange chromatography, affinity chromatography, expanded bed adsorption chromatography, reverse-phase chromatography, hydrophobic interaction chromatography, and combinations thereof; when the technique is electrophoresis, the electrophoretic technique can be selected from the group consisting of gel electrophoresis, free-flow electrophoresis, electrofocusing, isotachopheresis, affinity electrophoresis, immunoelectrophoresis, counterelectrophoresis, and capillary electrophoresis, capillary zone electrophoresis, and combinations thereof; and when the technique is spectrometry, the spectroscopic (or spectrometric) technique can be selected from the group consisting of mass spectrometry, ultraviolet spectroscopy, visible light spectroscopy, fluorescent spectroscopy, and ultraviolet-visible light spectroscopy, and combinations thereof. In some particular sub-aspects, the technique comprises liquid chromatography-mass spectrometry; in other cases, the technique comprises capillary zone electrophoresis coupled to mass spectrometry.

In a fourth aspect, disclosed herein are methods of preparing a polypeptide for analysis, comprising providing a sample comprising the polypeptide; applying the sample to a filter having a molecular weight cut-off; digesting the sample on the filter with a first protease; digesting the sample on the filter with a second protease; and analyzing the sample, wherein the second protease is neutrophil elastase; and the first protease is different from the second protease. In this aspect, the first protease can comprise a

4

protease selected from the group consisting of trypsin, Asp-N, and Glu-C; furthermore, the polypeptide can be denatured, and/or reduced, and/or alkylated before digesting the polypeptide with the first protease. The neutrophil elastase can be, for example, human neutrophil elastase (EC 3.4.21.37). When analyzing the sample, the analysis can include at least one technique selected from the group consisting of chromatography, electrophoresis, spectroscopy, spectrometry, and combinations thereof. For example, when the technique for analyzing the sample comprises chromatography, the chromatographic technique can be selected from the group consisting of gas chromatography, liquid chromatography, high performance liquid chromatography, ultra-performance liquid chromatography, size-exclusion chromatography, ion-exchange chromatography, affinity chromatography, expanded bed adsorption chromatography, reverse-phase chromatography, hydrophobic interaction chromatography, and combinations thereof; when the technique is electrophoresis, the electrophoretic technique can be selected from the group consisting of gel electrophoresis, free-flow electrophoresis, electrofocusing, isotachopheresis, affinity electrophoresis, immunoelectrophoresis, counterelectrophoresis, and capillary electrophoresis, capillary zone electrophoresis, and combinations thereof; and when the technique is spectroscopy or spectrometry, the spectroscopic (or spectrometric) technique can be selected from the group consisting of mass spectrometry, ultraviolet spectroscopy, visible light spectroscopy, fluorescent spectroscopy, and ultraviolet-visible light spectroscopy, and combinations thereof. In some particular sub-aspects, the technique comprises liquid chromatography-mass spectrometry; in other cases, the technique comprises capillary zone electrophoresis coupled to mass spectrometry. The filter can have, for example, a molecular weight cut-off of 30 kDa.

In yet a fifth aspect, disclosed herein are methods of preparing a polypeptide for analysis by liquid chromatography, capillary zone electrophoresis, or mass spectrometry, comprising providing a sample comprising the polypeptide; applying the sample to a 30 kDa MWCO filter; capturing the sample on the filter to produce a retentate; denaturing the sample retentate on the filter; reducing the sample retentate on the filter; alkylating the sample on the filter; digesting the sample on the filter with a first protease; filtering the sample through the same filter and retaining the filtrate; digesting the sample on the filter with a second protease; filtering the sample through the same filter into the previously retained filtrate; quenching the active proteases in the filtrate with guanidine; and analyzing the filtrate by liquid chromatography, capillary zone electrophoresis, or mass spectrometry; wherein the second protease is neutrophil elastase; and the first protease is different from the second protease. In such aspect, the neutrophil elastase can be human neutrophil elastase (EC 3.4.21.37).

In these five aspects, the methods can be accomplished at least in part by automation. In some sub-aspects, the addition and exchange of solutions is handled by an automated liquid handling device or robot.

Furthermore, in these five aspects, the disclosed methods can be used to prepare and analyze therapeutic polypeptides. Therapeutic polypeptides can be selected from the group consisting of an antibody or antigen-binding fragment thereof, a derivative of an antibody or antibody fragment, and a fusion polypeptide. Examples of antibodies and antibody construct include those selected from the group consisting of infliximab, bevacizumab, cetuximab, ranibizumab, palivizumab, abagovomab, abciximab, actoxumab, adalim-

5

umab, afelimomab, afutuzumab, alacizumab, alacizumab pegol, ald518, alemtuzumab, alirocumab, altumomab, amatuximab, anatumomab mafenatox, anrukinzumab, apolizumab, arcitumomab, aselizumab, altinumab, atlizumab, atorolimumab, tocilizumab, bapineuzumab, basiliximab, bavituximab, bectumomab, belimumab, benralizumab, bertilimumab, besilesomab, bevacizumab, bezlotoxumab, biciromab, bivatuzumab, bivatuzumab mertansine, blinatumomab, blosozumab, brentuximab vedotin, briakinumab, brodalumab, canakinumab, cantuzumab mertansine, cantuzumab mertansine, caplacizumab, capromab pendetide, carlumab, catumaxomab, cc49, cedelizumab, certolizumab pegol, cetuximab, citatuzumab bogatox, cixutumumab, clazakizumab, clenoliximab, clivatuzumab tetraxetan, conatumumab, crenezumab, cr6261, dacetuzumab, daclizumab, dalotuzumab, daratumumab, demcizumab, denosumab, detumomab, dorlimomab aritox, drozitumab, duligotumab, dupilumab, ecomeximab, eculizumab, edobacomab, edrecolomab, efalizumab, efungumab, elotuzumab, elsilimumab, enavatuzumab, enlimomab pegol, enokizumab, enoticumab, ensituximab, epitumomab cituxetan, epratuzumab, erenumab, erlizumab, ertumaxomab, etaracizumab, etrolizumab, evolocumab, exbivirumab, fanolesomab, faralimomab, farletuzumab, fasinumab, fbta05, felvizumab, fezakinumab, ficlatuzumab, figitumumab, flantvotumab, fontolizumab, foralumab, foravirumab, fresolimumab, fulranumab, futuximab, galiximab, ganitumab, gantenerumab, gavilimumab, gentuzumab ozogamicin, gevokizumab, girentuximab, glembatumumab vedotin, golimumab, gomiliximab, gs6624, ibalizumab, ibritumomab tiuxetan, icrucumab, igovomab, inciromab, imgatuzumab, inclacumab, indatuximab ravtansine, infliximab, intetumumab, inolimumab, inotuzumab ozogamicin, ipilimumab, iratumumab, itolizumab, ixekizumab, keliximab, labetuzumab, lebrikizumab, lemalesomab, lerdelimomab, lexatumumab, libivirumab, ligelizumab, lintuzumab, lirlumab, lorvotuzumab mertansine, lucatumumab, lumiliximab, mapatumumab, maslimomab, mavrilimumab, matuzumab, mepolizumab, metelimomab, milatuzumab, minretumomab, mitumomab, mogamulizumab, morolimumab, motavizumab, moxetumomab pasudotox, muromonab-cd3, nacolomab tafenatox, namilumab, naptumomab estafenatox, narnatumab, natalizumab, nebacomab, necitumumab, nerelimomab, nesvacumab, nimotuzumab, nivolumab, nofetumomab merpentin, ocaratuzumab, ocrelizumab, odulimumab, ofatumumab, olaratumab, olokizumab, omalizumab, onartuzumab, oportuzumab monatox, oregovomab, orticumab, orelizumab, oxelumab, ozanomezumab, ozoralizumab, pagibaximab, palivizumab, panitumumab, panobacomab, parsatuzumab, pascolizumab, pateclizumab, patritumab, pemtumomab, perakizumab, pertuzumab, pexelizumab, pidilizumab, pintumomab, placulumab, ponezumab, priliximab, pritumumab, PRO 140, quilizumab, racotumomab, radretumab, rafivirumab, ramucirumab, ranibizumab, raxibacomab, regavirumab, reslizumab, rilotumumab, rituximab, robatumumab, roledumab, romosozumab, rontalizumab, rovelizumab, ruplizumab, samalizumab, sarilumab, satumomab pendetide, secukinumab, sevirumab, sibrotuzumab, sifalimumab, sil-tuximab, simtuzumab, sipilizumab, sirukumab, solanezumab, solitomab, sonpecizumab, sontuzumab, stamulumab, sulesomab, suvizumab, tabalumab, tacatuzumab tetraxetan, tadocizumab, talizumab, tanezumab, taplitumomab paptox, tefibazumab, telimomab aritox, tenatumomab, tefibazumab, teneliximab, teplizumab, teprotumumab, tezepelumab, TGN1412, tremelimomab, ticilimumab, tild-rakizumab, tigatuzumab, TNX-650, tocilizumab, torali-

6

zumab, tositumomab, tralokinumab, trastuzumab, TRBS07, tregalizumab, tucotuzumab celmoleukin, tuvirumab, ublituximab, urelumab, urtoxazumab, ustekinumab, vapaliximab, vatelizumab, vedolizumab, veltuzumab, vepalimumab, vesencumab, visilizumab, volociximab, vorsetuzumab mafodotin, votumumab, zalutumumab, zanolimumab, zatuximab, ziralimumab, zolimomab aritox, and those antibodies shown in Table H. In yet other sub-aspects, the therapeutic polypeptide is a polypeptide selected from the group consisting of a glycoprotein, CD polypeptide, a HER receptor polypeptide, a cell adhesion polypeptide, a growth factor polypeptide, an insulin polypeptide, an insulin-related polypeptide, a coagulation polypeptide, a coagulation-related polypeptide, albumin, IgE, a blood group antigen, a colony stimulating factor, a receptor, a neurotrophic factor, an interferon, an interleukin, a viral antigen, a lipoprotein, calcitonin, glucagon, atrial natriuretic factor, lung surfactant, tumor necrosis factor-alpha and -beta, enkephalinase, mouse gonadotropin-associated peptide, DNase, inhibin, activating, an integrin, protein A, protein D, a rheumatoid factor, an immunotoxin, a bone morphogenetic protein, a superoxide dismutase, a surface membrane polypeptide, a decay accelerating factor, an AIDS envelope, a transport polypeptide, a homing receptor, an addressin, a regulatory polypeptide, an immunoadhesin, a myostatin, a TALL polypeptide, an amyloid polypeptide, a thymic stromal lymphopoietin, a RANK ligand, a c-kit polypeptide, a TNF receptor, and an angiopoietin, and biologically active fragments, analogs or variants thereof.

In some cases, the therapeutic polypeptide is a BiTE® (bi-specific T-cell engager).

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 shows α -CD3 domain conserved among many BiTE® molecules (canonical and half-life extension). Tryptic digestion yields an 8 kDa peptide containing two complementarity determining regions (CDRs) with multiple hot spots.

FIG. 2 shows enzyme specificity for human neutrophil elastase. P1 corresponds to the C-terminus of one amino acid, and P1' corresponds to the N-terminus of the subsequent residue. Relative font size corresponds to the likelihood of cleavage at that particular residue (adapted from MEROPS database; (Rawlings et al., 2014))

FIG. 3 shows extracted ion chromatograms (EIC) of BiTE®-1 unstressed and forced deamidation (pH 8.5, 50° C., t=3 d) samples, showing unmodified and deamidated species after forced deamidation.

FIG. 4 shows tandem mass spectrometry (MSMS) of the three deamidated species observed in FIG. 3. The diagnostic b3 ion is shifted by +0.9838 Da in peak three relative to peaks 1 and 2. This mass shift corresponds to deamidation.

FIG. 5 shows a schematic highlighting differences between the single digest, conventional MAM and one MAM approach for BiTE® molecules, which incorporates 1:1 mixing of sample digested with trypsin and sample digested with trypsin followed by neutrophil elastase digestion.

FIG. 6 shows the experimental design used to evaluate effectiveness and robustness of Protocol 3 (see Examples, below, for further details).

FIG. 7 shows a schematic over-sample preparation outlined in Protocol 4 (see Examples, below, for further details).

FIG. 8 shows trypsin specificity of Protocol 4 for BiTE®-1, BiTE®-2, and BiTE®-3. Specificity was calculated with MassAnalyzer initially by specifying strict trypsin cleavage

in search parameters. Data were then researched with cleavage at the C-terminus of residues KRVITAL (low specificity). Trypsin specificity was the total area of peptides identified with the strict tryptic search divided by total area of peptides identified with KRVITAL cleavage search. This value was expressed as a percent.

FIG. 9 shows the percent of total area corresponding to the 8 kDa linker peptide for BiTE®-1, BiTE®-2, and BiTE®-3. Data were searched with MassAnalyzer with cleavage at the C-terminus of residues KRVITAL (low specificity). Total area of peptides corresponding to the 8 kDa linker peptide was divided by total area of all peptides identified with the KRVITAL cleavage search. This value was expressed as a percent. The percent of total area corresponding to the 8 kDa linker peptide is higher for BiTE®-2 because as a canonical BiTE® molecule, it does not contain the Fc domain added for half-life extension.

FIG. 10 shows total ion chromatogram (TIC) overlays of replicate injections of BiTE®-3 samples prepped with Protocol 3. Injections were from the same vial and were separated by ~4 d.

FIG. 11 shows TIC overlays of replicate injections of BiTE®-3 samples prepped with Protocol 4. Injections were from the same vial and were separated by ~4 d.

FIG. 12 shows TIC overlays comparing BiTE®-2 sample prepped in triplicate with Protocols 4 and 5. Two peaks (arrow) observed in all three Protocol 4 samples correspond to non-specific alkylation.

DETAILED DESCRIPTION

In order to address the issue of cleavage-resistant polypeptides for MAM analysis, cleavage sites and specificity for a number of commercially available enzymes were examined. Human neutrophil elastase (HNE; EC 3.4.21.37, also known as elastase-2, human leukocyte elastase (HLE), bone marrow serine protease, medullasin, and PMN elastase) has been reported to cleave at the C-terminus of valine, isoleucine, threonine, alanine, and leucine residues (FIG. 2; (Doucet and Overall, 2011; Rawlings et al., 2014)); however, the enzyme also cleaves non-specifically, a detriment which frequency decreases as substrate length decreases (Stein et al., 1987). Given an amino acid sequence of a target polypeptide, digestion with HNE can serve as a means to monitor desired regions in the polypeptide but for HNE's propensity to non-specifically cleave polypeptides.

For example, the inventors observed that when bi-specific T-cell engager (BiTE®) molecules were digested only with HNE, resulting peptides were very numerous (due to non-specific cleavage) and low-abundance. For example, FIG. 1 shows α -CD3 domain conserved among many BiTE® molecules (canonical and half-life extension). Tryptic digestion yields an 8 kDa peptide containing two complementarity determining regions (CDRs) with multiple hot spots that are difficult to identify and characterize using conventional mass spectrometry (MS)-based experiments. This peptide also lacks residues susceptible to cleavage by other widely-used enzymes such as Glu-C and Asp-N.

The inventors also unexpectedly observed that, following on-filter trypsin digestion, both the conserved 8 kDa linker peptide and molecule-specific linker peptides of BiTE® molecules were retained by 30 kDa molecular weight cut-off (MWCO) filters regardless of whether the filter membranes were composed of regenerated cellulose or polyether-sulfone. Enhancing specificity of HNE by first digesting with a first protease that is not HNE and exploiting the unanticipated retention of the linker peptides resulting from

the first protease digestion by 30 kDa MWCO (or other cut-off) filters allowed for the development of a novel, robust method for characterization of BiTE® molecules (and other polypeptides) and quantitation of critical attributes.

A description of a novel sample preparation method that uses a first digestion step followed by digestion with HNE is disclosed.

Definitions

Both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive. The use of the singular includes the plural unless specifically stated otherwise. The use of "or" means "and/or" unless stated otherwise. The use of the term "including", as well as other forms, such as "includes" and "included," is not limiting. Terms such as "element" or "component" encompass both elements and components comprising one unit and elements and components that comprise more than one subunit unless specifically stated otherwise. The use of the term "portion" can include part of a moiety or the entire moiety. When a numerical range is mentioned, e.g., 1-5, all intervening values are explicitly included, such as 1, 2, 3, 4, and 5, as well as fractions thereof, such as 1.5, 2.2, 3.4, and 4.1.

"About" or "approximately" means, when modifying a quantity (e.g., "about" 3 mM), that variation around the modified quantity can occur. These variations can occur by a variety of means, such as typical measuring and handling procedures, inadvertent errors, ingredient purity, and the like.

"Comprising" and "comprises" are intended to mean that methods include the listed elements but do not exclude other unlisted elements. The terms "consisting essentially of" and "consists essentially of," when used in the disclosed methods include the listed elements, exclude unlisted elements that alter the basic nature of the method, but do not exclude other unlisted elements. The terms "consisting of" and "consists of" when used to define methods exclude substantial method steps. Embodiments defined by each of these transition terms are within the scope of this disclosure.

"Coupled" means associated directly as well as indirectly. For example, a device or process can be directly associated with another device or process, or these devices and/or processes can be indirectly associated with each other, e.g., via another device or process.

"Protein", "peptide", and "polypeptide" are used interchangeably to mean a chain of amino acids wherein each amino acid is connected to the next by a peptide bond.

"Denaturation," "denaturing," "denature," and the like means a process in which polypeptides lose at least partially the quaternary structure, tertiary structure, and secondary structure that are found in the polypeptides in their native states by applying an external stress or reagent (i.e., a denaturant).

"Denaturant" means any substance, composition, energy, or condition that can denature a polypeptide. Examples of denaturants include a strong acid or base, an inorganic salt, an organic solvent, radiation, a chaotropic agent, or heat, or combinations of these.

"Denatured polypeptide," "denatured protein," and the like means a polypeptide that has its secondary, tertiary, and/or quaternary structure changed from the native polypeptide. A polypeptide can be fully denatured or partially denatured. A "non-denatured polypeptide" or "non-denatured protein" (and similar terms) means a polypeptide that has maintained its secondary, tertiary, and as applicable, quaternary structure. A "native polypeptide" or "native protein" (and the like) refers to a polypeptide as found in nature

and has any primary reference sequence described in the UniProt Knowledgebase database (The UniProt, 2017) or the UniGene database (Pontius et al., 2003).

“Reduced polypeptide” or “reduced protein” (and similar terms) means a polypeptide in which at least one of its interchain or intrachain disulfide bonds is broken. Such bonds can form between reduced thiol groups, such as those available on cysteine residues.

Threonine endopeptidases, EC 3.4.99 Endopeptidases of unknown catalytic mechanism. Specific examples of proteases include trypsin (EC 3.4.21.4), endoproteinase Asp-N (EC 3.4.24.33), and endoproteinase Glu-C (EC 3.4.21.19).

Elastases are proteases that (except for those classified as EC 3.4.21.70) hydrolyze elastin, an elastic fiber polypeptide of the extracellular matrix. There are eight human genes that encode elastases (Table A):

TABLE A

The human elastases				
Family	Gene	Name	EC number	Preferential cleavage
Neutrophil	ELANE (ELA2)	Neutrophil elastase (elastase 2)	3.4.21.37	Val-I-Xaa; Ala-I-Xaa; Thr-I-Xaa; Ile-I-Xaa; Leu-I-Xaa
Chymotrypsin	CTRC (ELA4)	Chymotrypsin C (cadlecrin) (elastase 4)	3.4.21.2	Leu-I-Xaa; Tyr-I-Xaa; Phe-I-Xaa; Met-I-Xaa; Trp-I-Xaa; Gln-I-Xaa; Asn-I-Xaa
Chymotrypsin-like	CELA1 (ELA1)	Chymotrypsin-like elastase family, member 1 (pancreatic elastase 1)	3.4.21.36	Ala-I-Xaa
	CELA2A (ELA2A)	Chymotrypsin-like elastase family, member 2A (pancreatic elastase 2A)	3.4.21.71	Leu-I-Xaa Met-I-Xaa Phe-I-Xaa
	CELA2B (ELA2B)	Chymotrypsin-like elastase family, member 2B (pancreatic elastase 2B)	3.4.21.71	Leu-I-Xaa Met-I-Xaa Phe-I-Xaa
	CELA3A (ELA3A)	Chymotrypsin-like elastase family, member 3A (pancreatic elastase 3A)	3.4.21.70	Ala-I-Xaa (Does not hydrolyze elastin)
	CELA3B (ELA3B)	Chymotrypsin-like elastase family, member 3B (pancreatic elastase 3B)	3.4.21.70	Ala-I-Xaa (Does not hydrolyze elastin)
Macrophage	MMP12 (HME)	Macrophage metalloelastase (macrophage elastase)	3.4.24.65	Hydrolysis of soluble and insoluble elastin. Specific cleavages are also produced at 14-Ala-I-Leu-15 and 16-Tyr-I-Leu-17 in the B chain of insulin

“Alkylated polypeptide” or “alkylated protein” (and the like) means a polypeptide to which an alkyl group has been transferred. Pragmatically, polypeptides are often alkylated on thiol groups (such as available on cysteine residues) to prevent reduced thiols from forming, or reforming after reduction, disulfide bonds or bridges.

“Digestion” and the like, in the context of polypeptides, means the fragmenting of a polypeptide into two or more fragments, which fragmentation is mediated by another substance, chemical, or enzyme.

“Proteolytic cleavage,” “proteolytic digestion,” and the like means the cleaving of a polypeptide by breaking the peptide bonds in a polypeptide, thus producing fragments. Proteolytic cleavage can be mediated by enzymes.

“Protease,” “peptidase,” “proteolytic cleavage enzyme,” “endoproteinaise,” “proteinase,” and the like means an enzyme and means a polypeptide or fragment thereof that catalyzes hydrolysis of peptide bonds. Proteases include variants of an amino acid sequences of known proteases that catalyze peptide bond hydrolysis, even wherein such catalyzing activity is reduced in the variant.

Enzyme catalyzed reactions are classified according to the Enzyme Commission numbering system. Because proteases catalyze peptide bond hydrolysis, they are classified in class EC 3.4, which includes, EC 3.4.11 Aminopeptidases, EC 3.4.13 Dipeptidases, EC 3.4.14 Dipeptidyl peptidases and tripeptidyl peptidases, EC 3.4.15 Peptidyl dipeptidases, EC 3.4.16 Serine type carboxypeptidases, EC 3.4.17 Metallo-carboxypeptidases, EC 3.4.18 Cysteine type carboxypeptidases, EC 3.4.19 Omega peptidases, EC 3.4.21: Serine proteases, EC 3.4.22 Cysteine proteases, EC 3.4.23 Aspartic endopeptidases, EC 3.4.24: Metallopeptidases, EC 3.4.25

“Neutrophil elastase” (sometimes referred to as elastase-2, leukocyte elastase, bone marrow serine protease, medullasin, or PMN elastase) is an elastase that neutrophils and macrophages secrete during inflammation to destroy bacteria. Human neutrophil elastase (HNE) was first described in 1968 (Janoff and Scherer, 1968). HNE was shown to differ from pancreatic elastases in that soybean trypsin inhibitor and salivary kallikrein inhibitor inhibit HNE, but do not inhibit pancreatic elastase activity. HNE maintains activity at physiological salt concentrations, while pancreatic elastase activity decreases. HNE is at least ten-fold more resistant to serum elastase inhibitor (as defined by Janoff and Scherer, 1968) than pancreatic elastase. Finally, HNE is also more low pH-resistant than pancreatic elastase (Janoff and Scherer, 1968).

Sinha et al. (Sinha et al., 1987) first described the amino acid sequence of HNE and showed that the polypeptide had only 43% sequence homology to pig pancreatic elastase. The UniProt database for UniProtKB-P08246 (ELNE_HUMAN) (The UniProt, 2017) describes ELANE as encoding HNE as a 267 amino acid precursor polypeptide consisting of a signal peptide (amino acid residues 1-27), a pro-peptide domain (residues 28-29), and the mature polypeptide (residues 30-267). Within the mature polypeptide resides the peptidase 51 domain (residues 30-247). Disulfide bonds can be found forming between residues 55 and 71, 151 and 208, 181 and 187, and 198-223. Asparagine residues at positions 88, 124, and 174 can be glycosylated (N-linked). The sequence for HNE is shown in Table B. In Table B, the signal peptides are boldfaced, the pro-peptide domain is single-underlined, the mature polypeptide is italicized, and the S1 peptidase domain is double-underlined, italicized, and boldfaced.

TABLE B

Amino acid sequence of human neutrophil elastase (HNE) (SEQ ID NO: 1)	
MTLGRRRLACL FLACVLPALL LGGTALASR <u>I VGGRRARPHA WPFMVSLQLR GGHFCGATLI</u>	60
<u>APNFVMSAAH CVANVNVRAV RVVLGAHNLS RREPTROVFA VORIFENGYD PVNLLNDIVI</u>	120
<u>LQNGSATIN ANVOVAOLPA QGRRIGNGVO CLAMGWLLIG RNRGIASVLO ELNVTVTSL</u>	180
<u>CRRSNVCTLV RGROAGVCFG DSGSPLVCNG LIHGIASFVR GGCASGLYPD AFAPVAQFVN</u>	240
<u>WIDSIIQRSE DNPCPHRDP DPASRTH</u>	267

HNE is capable of hydrolyzing almost every polypeptide member of the extracellular matrix, including several types of collagen, fibronectin, proteoglycans, heparin, and cross-linked fibrin.

HNE cleaves peptide bonds in which the P1 residue is a small alkyl group. Substrate specificity is summarized in FIG. 2.

“Chemical cleavage” in the context of producing fragments of a polypeptide, means the fragmenting of the polypeptide by a chemical. A “chemical” is a non-proteinaceous substance or compound. Chemicals can be organic or inorganic.

“Molecular weight cut-off” or “MWCO” means A membrane’s MWCO is a representation of membrane selectivity for solute molecules of different molecular weights (MWs), where the MW value (expressed in Daltons (Da)) is obtained from the solute molecule that gives a 90% rejection when a range of different MW solutes are filtered in the target solvent (which for most liquid based, pressure driven membrane applications is water), where rejection is defined as in Eq. (1):

$$\text{Rejection, } R(\%) = \left(1 - \frac{C_p}{C_f}\right) \times 100 \quad (1)$$

where C_p and C_f are the concentration of permeate and feed, respectively.

MWCO is determined experimentally using dextran, polyethylene glycol, and proteins of various molecular weights to rate the MWCO of membranes or filters. The rejection depends on many solute and process parameters like the type of solute, concentration, hydrodynamics, pressure, temperature and pH. MWCO measurements usually are carried out in separate experiments using different solutes, each with a certain MW.

For low range MW (approximately under 10 kDa), polyethylene glycols (PEGs), n-alkanes, and oligostyrenes can be used in solute rejection assays. For high range MW (approximately greater than 10 kDa), dextrans and sugars are often used (Rohani et al., 2011).

Methods

Disclosed herein are methods directed to digesting polypeptide in a sample with a first protease and a second protease, wherein the first protease produces at least two fragments of the polypeptide, which the at least two fragments of the polypeptide are subsequently digested by the second protease; and analyzing the sample after digesting with the second protease, wherein the second protease is neutrophil elastase; and the first protease is different from the second protease. Other steps in the method can include dividing the sample into two aliquots after digesting the polypeptide in the sample with the first protease, and then

digesting the polypeptide in one of the two aliquots with the second protease. These aliquots can then be combined before analysis. The polypeptide in the sample can be denatured and/or reduced and/or alkylated before digesting with the first protease. Analysis can include chromatography, electrophoresis, spectrometry, and combinations thereof.

In some embodiments, the methods include applying the sample to a filter having a MWCO before digesting the polypeptide in the sample on the filter with the first protease, digesting the polypeptide in the sample with the second protease, and analyzing the sample. Such embodiments can further incorporate filtering steps. The MWCO filter, in some embodiments, retains a significant proportion of the polypeptide and polypeptide fragments, even though the size of these polypeptides or polypeptide fragments have a MW that is significantly smaller than the MWCO of the filter.

Method Steps

Polypeptide Denaturing

In some embodiments, the polypeptide that is prepared and analyzed according to the disclosed methods is denatured.

Polypeptides can be denatured using a variety of art-accepted techniques and denaturants. In some embodiments, multiple denaturants are used together, either simultaneously or in sequence. For example, the denaturants of SDS and heat can be combined to denature polypeptides.

Protein denaturation can be accomplished by any means that disrupts quaternary, tertiary, or secondary polypeptide structure. For example, the use of chaotropes, such as urea, and denaturing detergents (e.g., sodium dodecyl sulfate (SDS)), heat, reducing agents, and agents that inactivate reactive thiol groups to block disulfide reformation. The pH of polypeptide-containing samples can also be manipulated to encourage denaturation. These components are often used together to effectively unfold polypeptides.

Additional examples of chaotropes include, in addition to urea, n-butanol, ethanol, guanidinium chloride, lithium perchlorate, lithium acetate, magnesium chloride, phenol, 2-propanol, and thiourea. Urea is preferred in most instances.

Detergents are classified in the form of the hydrophilic group: anionic, cationic, non-ionic, and zwitterionic. Anionic and cationic detergents are more likely to be denaturing, examples of which include: SDS, sodium cholate, sodium deoxycholate, sodium glycocholate, sodium taurocholate, sodium taurodeoxycholate, N-lauroylsarcosine, lithium dodecyl sulfate (anionic) and hexadecyltrimethyl ammonium bromide (CTAB) and trimethyl(tetradecyl) ammonium bromide (TTAB) (cationic). In some cases, a zwitterionic detergent can be useful, examples include amidosulfobetaine-14 (ASB-14), amidosulfobetaine-16 (ASB-16), C7Bz0, CHAPS, CHAPSO, EMPIGEN® BB, 3-(N,N-dimethyloctylammonio)propanesulfonate inner salt (SB3-

8), d (decyldimethylammonio) propanesulfonate inner salt (SB3-10), etc. Anionic detergents are preferred, with SDS being particularly preferred.

A denaturant can be heat, such as an elevated temperature at or greater than 30° C. (for most polypeptides). Denaturants include agitation. In some embodiments, low salt, including essentially or substantially or no salt can denature polypeptides.

A denaturant can be a solvent, such as ethanol or other alcohols.

Denaturing of polypeptides has been extensively studied and described; for example, see (Tanford, 1968) for further details. A person of ordinary skill in the art understands how to denature polypeptides given the nature of the polypeptide and the many denaturants from which to choose.

Reduction of Polypeptides

A reduced polypeptide is a polypeptide that is exposed to reducing conditions sufficient to reduce a reducible residue in the polypeptide structure, such as a cysteine. If the reduced polypeptide contains a thiol group, or sulfur-containing residue, then the thiol group in the reduced polypeptide is reduced. A reduced polypeptide comprising a cysteine residue has the sulfur atom of the cysteine residue reduced, which can be indicated as “—SH.” A reduced polypeptide can be a disulfide bond-containing polypeptide. A disulfide bond-containing polypeptide can become a reduced polypeptide by exposure to reducing conditions that cause one or more disulfide bonds (disulfide bridges) in the disulfide bond-containing polypeptide to break.

A “reducing agent”, “reductant” or “reducer” is an element or compound that loses (or donates) an electron to another chemical species in a redox chemical reaction. A reducing agent allows disulfide groups to become reactive by generating thiol (—SH) groups. Common polypeptide reducing reagents are shown in Table C.

TABLE C

Reducing reagents	
Product	Notes (including alternative names and CAS entries)
2-Mercaptoethanol	β-mercaptoethanol (BME, 2BME, 2-ME, b-mer, CAS 60-24-2)
2-Mercaptoethylamine-HCl	2-aminoethanethiol (2-MEA-HCl, cysteamine-HCl, CAS 156-57-0), selectively reduces antibody hinge-region disulfide bonds
Dithiothreitol	Dithiothreitol (DTT, CAS 3483-12-3)
TCEP-HCl	Tris (2-carboxyethyl) phosphine hydrochloride (TCEP, CAS 5961-85-3) is a thiol-free reductant for polypeptide disulfide bonds

In some embodiments, polypeptide denaturation and reduction are carried out simultaneously. In other embodiments, the polypeptide denaturation and reduction are performed in discrete steps.

A person of ordinary skill in the art understands how to reduce polypeptides given the nature of the polypeptide and the reducing reagents from which to choose.

Alkylating a Polypeptide Comprised in a Sample

“Inactivating reactive thiol groups” means blocking free thiol groups in a polypeptide to prevent unwanted thiol-disulfide exchange reactions. Alkylating agents are substances that cause the replacement of hydrogen by an alkyl group.

Alkylation of free cysteines, often following their reduction, prevents formation and reformation of disulfide bonds that might otherwise form between free thiols of cysteine residues. Commonly used alkylating agents include n-ethylmaleimide (NEM), iodoacetamide (IAA) and iodoacetic acid. Examples of other suitable alkylating agents include

dithiobis(2-nitro)benzoic acid; acrylamide; 4-vinylpyridine; nitrogen mustards, such as chlorambucil and cyclophosphamide; cisplatin; nitrosoureas, such as carmustine, lomustine and semustine; alkyl sulfonates, such as busulfan; ethyleneimines, such as thiotepa; and triazines, such as dacarbazine. The person skilled in the art is aware of the reagents that can be used to protect sulfhydryl groups, as well as how to use such reagents.

Polypeptide Digestion

The methods disclosed herein comprise cleaving the polypeptide to be analyzed in a sample in a first digestion, wherein the cleaving produces at least two fragments of the polypeptide. Any method of fragmenting the polypeptide can be used, provided that at least two fragments of the polypeptide are produced; furthermore, complete degradation of the polypeptide into, for example, single amino acids, is undesirable.

In such digestion step, convenient means of cleavage include using a protease (wherein the protease is not neutrophil elastase). Any protease can be used, as long as such protease cleaves the polypeptide into at least two fragments. The rationale supporting this first digestion before digesting with a neutrophil elastase is to provide polypeptide fragments to the neutrophil elastase (such as human neutrophil elastase), wherein the neutrophil elastase has an increased specificity, as compared to when presented with the intact polypeptide.

In some embodiments, mixtures of two or more proteases can be used. In other embodiments, the first digestion can include multiple sequential digestions. For example, a first digestion is performed with a first protease, and then in a subsequent reaction, a second digestion is performed with a second protease (wherein neither protease is neutrophil elastase). A second (or more) digestions are completed to

improve the specificity of the neutrophil elastase used in the step of cleaving the polypeptide further with the neutrophil elastase.

Examples of useful proteases (but not at all inclusive) include trypsin, endoproteinase Glu-C, endoproteinase Arg-C, pepsin, chymotrypsin, chymotrypsin B, Lys-N protease, Lys-C protease, Glu-C protease, Asp-N protease, pancrato-peptidase, carboxypeptidase A, carboxypeptidase B, proteinase K, and thermolysin. In some embodiments, combinations of these proteases are used. In some embodiments, trypsin alone is used.

These and other proteases, including peptide bond selectivity and E.C. numbers, are shown in Table D, which is adapted from (Unspecified, 2007). The sources shown for each protease are exemplary only; many of these proteases are commercially available.

In some embodiments, a protein:protease ratio (w/w) of 10:1, 20:1, 25:1, 50:1, or 100:1 can be used. In some embodiments, the ratio is 20:1. In some embodiments, the

15

neutrophil elastase used is at a concentration of about 100 ng/ml-1 mg/ml, or about 100 ng/ml-500 µg/ml, or about 100 ng/ml-100 µg/ml, or about 1 µg/ml-1 mg/ml, or about 1 µg/ml-500 µg/ml, or about 1 µg/ml-100 µg/ml, or about 10 µg/mg-1 mg/ml, or about 10 µg/mg-500 µg/ml, or about 10 µg/mg-100 µg/ml. In some embodiments, the digestion step is for 10 minutes to 48 hours, or 30 minutes to 48 hours, or 30 minutes to 24 hours, or 30 minutes to 16 hours, or 1 hour to 48 hours, or 1 hour to 24 hours, or 1 hour to 16 hours, or 1 to 8 hours, or 1 to 6 hours, or 1 to 4 hours. In some embodiments, the digestion step is incubated at a temperature between 20° C. and 45° C., or between 20° C. and 40° C., or between 22° C. and 40° C., or between 25° C. and 37° C. In some embodiments, the digestion step is incubated at 37° C. One of skill in the art can choose appropriate conditions (buffers, incubation times, amount of protease, volumes, etc.), as in vitro protease digestion is well understood in the art.

16

chemical that cleaves the polypeptide into at least two fragments, and then in a subsequent reaction, a second digestion is performed with a protease (wherein the protease is not neutrophil elastase), or vice versa. A second (or more) digestions are completed to improve the specificity of the neutrophil elastase used in the step of cleaving the polypeptide further with the neutrophil elastase.

In the disclosed methods, a subsequent digestion is performed with a neutrophil elastase. In some embodiments, the neutrophil elastase is human.

In some embodiments, a protein: (human) neutrophil elastase ratio (w/w) of 10:1, 20:1, 25:1, 50:1, or 100:1 can be used. In some embodiments, the ratio is 20:1. In some embodiments, the neutrophil elastase used is at a concentration of about 100 ng/ml-1 mg/ml, or about 100 ng/ml-500 µg/ml, or about 100 ng/ml-100 µg/ml, or about 1 µg/ml-1 mg/ml, or about 1 µg/ml-500 µg/ml, or about 1 µg/ml-100 µg/ml, or about 10 µg/mg-1 mg/ml, or about 10 µg/mg-500

TABLE D

Proteases commonly used for protein fragmentation			
Protease ^a	EC no.	Peptide bond selectivity	Exemplary Accession no. ^b
Trypsin (bovine)	3.4.21.4	P ₁ -P ₁ ¹ - (P ₁ = Lys, Arg)	P00760 ^S
Chymotrypsin (bovine)	3.4.21.1	P ₁ -P ₁ ¹ - (P ₁ = aromatic, P ₁ ¹ = nonspecific)	P00766 ^S
Endoproteinase Asp-N (<i>Pseudomonas fragi</i>)	3.4.24.33	P ₁ -Asp- (and -P ₁ -cysteic acid)	φ
Endoproteinase Arg-C (mouse submaxillary gland)	φ	-Arg-P ₁ -	—
Endoproteinase Glu-C (V8 protease) (<i>Staphylococcus aureus</i>)	3.4.21.19	-Glu-P ₁ ¹ - (and -Asp-P ₁ ¹ -) (2)	P04188 ^S
Endoproteinase Lys-C (<i>Lysobacter enzymogenes</i>)	3.4.21.50	-Lys-P ₁ ¹ -	S77957 ^P
Pepsin (porcine)	3.4.23.1	P ₁ -P ₁ ¹ - (P ₁ = hydrophob pref.)	P00791 ^S
Thermolysin (<i>Bacillus thermo-proteolyticus</i>)	3.4.24.27	P ₁ -P ₁ ¹ - (P ₁ = Leu, Phe, Ile, Val, Met, Ala)	P00800 ^S
Elastase (porcine) (not neutrophil elastase)	3.4.21.36	P ₁ -P ₁ ¹ - (P ₁ = uncharged, nonaromatic)	P00772 ^S
Papain (<i>Carica papaya</i>)	3.4.22.2	P ₁ -P ₁ ¹ - (P ₁ = Arg, Lys pref.)	P00784 ^S
Proteinase K (<i>Tritirachium album</i>)	3.4.21.64	P ₁ -P ₁ ¹ - (P ₁ = aromatic, hydrophob pref.)	P06873 ^S
Subtilisin (<i>Bacillus subtilis</i>)	3.4.21.62	P ₁ -P ₁ ¹ - (P ₁ = neutral/acidic pref.)	P04189 ^S
Clostripain (endoproteinase-Arg-C) (<i>Clostridium histolyticum</i>)	3.4.22.8	-Arg-P ₁ - (P ₁ = Pro pref.)	P09870 ^S
Carboxypeptidase A (bovine)	3.4.17.1	P ₁ -P ₁ ¹ - (P ₁ cannot be Arg, Lys, Pro)	P00730 ^S
Carboxypeptidase B (porcine)	3.4.17.2	P ₁ -P ₁ ¹ - (P ₁ = Lys, Arg)	P00732 ^S
Carboxypeptidase P (<i>Penicillium janthinellum</i>)	φ	P ₁ -P ₁ ¹ - (nonspecific)	—
Carboxypeptidase Y (yeast)	3.4.16.5	P ₁ -P ₁ ¹ - (nonspecific)	P00729 ^S
Cathepsin C	3.4.14.1	χ-P ₁ -P ₁ ¹ - (removes amino-terminal dipeptide)	—
Acylamino-acid-releasing enzyme (porcine)	3.4.19.1	Ac-P ₁ -P ₁ ¹ - (P ₁ = Ser, Ala, Met pref.)	P19205 ^S +
Pyroglutamate aminopeptidase (bovine)	3.4.19.3	P ₁ -P ₁ ¹ - (P ₁ = 5-oxoproline or pyroglutamate)	—

^aExemplary source shown in parentheses

^bS = SwissProt; P = PIR; + = porcine sequence; φ = partial sequences of Asp-N; accession numbers: AAB35279, AAB35280, AAB35281, AAB35282

In some embodiments, the first digestion is accomplished using a chemical. Especially useful chemicals include those that cleave polypeptides in a site-specific manner. Such chemicals include cyanogen bromide (CNBr; carbononitridic bromide), which cleaves C-terminal of methionine residues; 2-nitro-5-thiocyanobenzoate (NTCB), which cleaves N-terminally of cysteine residues; asparagine-glycine dipeptides can be cleaved using hydroxylamine; formic acid, which cleaves at aspartic acid-proline (Asp-Pro) peptide bonds, and BNPS-skatole (3-bromo-3-methyl-2-(2-nitrophenyl)sulfanylindole), which cleaves C-terminal of tryptophan residues. One of skill in the art understands how to select appropriate variables, including polypeptide concentration, chemical concentration, incubation time and temperature, etc. See also, for example, (Crimmins et al., 2001; Li et al., 2001; Tanabe et al., 2014).

In some embodiments, the first digestion can include multiple sequential digestions, wherein at least one of such sequential digestions comprises using a chemical, such as CNBr, NTCB, hydroxylamine, formic acid, and BNPS-skatole. For example, a first digestion is performed with

45

µg/ml, or about 10 µg/mg-100 µg/ml. In some embodiments, the digestion step is for 10 minutes to 48 hours, or 30 minutes to 48 hours, or 30 minutes to 24 hours, or 30 minutes to 16 hours, or 1 hour to 48 hours, or 1 hour to 24 hours, or 1 hour to 16 hours, or 1 to 8 hours, or 1 to 6 hours, or 1 to 4 hours. In some embodiments, the digestion step is incubated for 35 minutes. In some embodiments, the digestion step is incubated at a temperature between 20° C. and 45° C., or between 20° C. and 40° C., or between 22° C. and 40° C., or between 25° C. and 37° C. In some embodiments, the digestion step is incubated at 37° C. One of skill in the art is capable of understanding the various parameters and can select appropriate conditions for the digestion with neutrophil elastase.

Dividing Sample into Aliquots

60

In embodiments, before the sample comprising the polypeptide to be analyzed is digested with neutrophil elastase, the sample is divided into aliquots. In some cases, dividing the sample into aliquots is performed by simply taking half the volume of the sample before digesting with neutrophil elastase. In some cases, the aliquots can be equalized for polypeptide concentration using, for example, UV spectrophotometry.

65

Filter Selection

When choosing the appropriate MWCO for specific applications, many factors are usually considered, including sample concentration, composition, molecular shape, and operating conditions such as temperature, pressure, and cross-flow velocity. Other variables regarding the flow of molecule passage are also factored in. For example, linear molecules, high transmembrane pressure (TMP) and low sample concentration can increase molecule passage, while low temperature and membrane fouling can decrease molecule passage. Qualification methods for MWCO are not always comparable, as they vary across manufacturers. In the art, it is commonly advised to select a MWCO that is at least two times smaller than the molecular weight of the solute that is being retained.

For the disclosed methods, filter shape is also considered. For 1-2 mL volumes, filter units where the filter is slanted when the filter unit is perpendicular to the ground are less desirable than those where the filter is horizontal when the filter unit is perpendicular to the ground.

Suitable materials for the MWCO membrane are hydrophilic. Suitable hydrophilic materials include polyether-sulfone, polyvinylidene difluoride, and regenerated cellulose.

Suitable cut-off filters can be obtained commercially, such as from Pall Corporation (Port Washington, NY) or Millipore, Inc (Burlington, MA, such as Microcon® filters). For the disclosed methods, filters from these particular manufacturers have MWCO specifications that are suitable. Comparable filters from other manufacturers can also be used. The MWCO specifications provided by manufacturers can be relied upon because even accounting for differences between MWCO ratings between manufacturers, the disclosed methods take advantage in part of the surprising observation that small molecular weight molecules are retained by the filter even though these molecules have a

molecular weight that is equal to or less than (and often substantially less than) the rated MWCO of the filter (see, for example, Example 11).

Filtering the sample using MWCO filters can use any suitable filter method, using any suitable filter device. Filtering can take place by gravity, capillary force, or commonly centrifugation, including ultracentrifugation.

Analyzing the Sample

After the polypeptide has been digested with neutrophil elastase, the resulting polypeptide fragment can be analyzed by any suitable method. The proceeding discussion is not meant to limit in any way the methods that can be used to analyze the prepared polypeptides.

In general, suitable analytical methods can be chromatographic, electrophoretic, and spectrometric. Some of these methods can be combined.

One of skill in the art has access to, for example, handbooks, that facilitate the selection of appropriate analytical methods, as well as appropriate conditions to conduct those methods, including for example, (Gunzler and Williams, 2001).

Chromatographic methods are those methods that separate polypeptide fragments in a mobile phase, which phase is processed through a structure holding a stationary phase. Because the polypeptide fragments are of different sizes and compositions, each fragment has its own partition coefficient. Because of the different partition coefficients, the polypeptides are differentially retained on the stationary phase. Examples of such methods known in the art include gas chromatography, liquid chromatography, high performance liquid chromatography, ultra-performance liquid chromatography, size-exclusion chromatography, ion-exchange chromatography, affinity chromatography, expanded bed adsorption chromatography, reverse-phase chromatography, and hydrophobic interaction chromatography.

A summary of some of the known chromatographic methods is shown in Table E.

TABLE E

Examples of chromatographic methods	
Chromatographic type	Description
Adsorption	Adsorbent stationary phase
Affinity	Based on a highly specific interaction such as that between antigen and antibody or receptor and ligand, one such substance being immobilized and acting as the sorbent
Column	The various solutes of a solution travel down an absorptive column where the individual components are absorbed by the stationary phase. The most strongly adsorbed component will remain near the top of the column; the other components will pass to positions farther down the column according to their affinity for the adsorbent
Exclusion (including gel-filtration, gel-permeation, molecular exclusion, molecular sieve gel-filtration)	Stationary phase is a gel having a closely controlled pore size. Molecules are separated based on molecular size and shape; smaller molecules being temporarily retained in the pores
Expanded bed adsorption (EBA)	Useful for viscous and particulate solutions. Uses for the solid phase particles that are in a fluidized state, wherein a gradient of particle size is created.
Gas (GC)	An inert gas moves the vapors of the materials to be separated through a column of inert material
Gas-liquid (GLC)	Gas chromatography where the sorbent is a nonvolatile liquid coated on a solid support
Gas-solid (GSC)	Gas chromatography where the sorbent is an inert porous solid
High-performance liquid, high-pressure liquid (HPLC).	Mobile phase is a liquid which is forced under high pressure through a column packed with a sorbent
Hydrophobic interaction chromatography	Matrix is substituted with hydrophobic groups (such as methyl, ethyl, propyl, octyl, or phenyl). At high salt concentrations, non-polar sidechains on polypeptide surfaces interact with the hydrophobic groups; that is, both types of groups are excluded by a polar solvent; elution accomplished with decreasing salt, increasing concentrations of detergent, and/or changes in pH.
Ion exchange	Stationary phase is an ion exchange resin to which are coupled either cations or anions that exchange with other cations or anions in the material passed through.
Paper	Paper is used for adsorption
Partition	The partition of the solutes occurs between two liquid phases (the original solvent and the film of solvent on an adsorption column)

TABLE E-continued

Examples of chromatographic methods	
Chromatographic type	Description
Reverse-phase	Any liquid chromatography in which the mobile phase is significantly more polar than the stationary phase. Hydrophobic molecules in the mobile phase adsorb to the hydrophobic stationary phase; hydrophilic molecules in the mobile phase tend to elute first.
Thin-layer (TLC)	Chromatography through a thin layer of inert material, such as cellulose
Ultra-performance liquid (UPLC)	A liquid chromatographic technique that uses a solid phase with particles less than 2.5 μm (smaller than in HPLC) and has higher flow rates; pressure used is 2-3 times more than in HPLC.

Prepared polypeptides can be analyzed also using electrophoretic methods—gel electrophoresis, free-flow electrophoresis, electrofocusing, isotachopheresis, affinity electrophoresis, immunoelectrophoresis, counterelectrophoresis, capillary electrophoresis, and capillary zone electrophoresis. Overviews and handbooks are available to one of skill in the art, such as (Kurien and Scofield, 2012; Lord, 2004).

Electrophoresis can be used to analyze charged molecules, such as polypeptides which are not at their isoelectric point, which are transported through a solvent by an electrical field. The polypeptides migrate at a rate proportional to their charge density. A polypeptide's mobility through an

electric field depends on: field strength, net charge on the polypeptide, size and shape of the polypeptide, ionic strength, and properties of the matrix through which the polypeptide migrates (e.g., viscosity, pore size). Polyacrylamide and agarose are two common support matrices. These matrices serve as porous media and behave like a molecular sieve. Polyacrylamide forms supports with smaller pore sizes and is especially useful in the disclosed methods, being ideal for separating most polypeptide fragments.

Table F presents examples of polypeptide electrophoretic techniques.

TABLE F

Examples of electrophoretic methods	
Technique	Description
Gel electrophoresis	Refers to electrophoretic techniques that use a gel as a matrix through which polypeptides travel. Many electrophoretic techniques use gels, including those based on polyacrylamide (polyacrylamide gel electrophoresis (PAGE)), including denaturing and non-denaturing PAGE). Pore size of polyacrylamide gels is controlled by modulating the concentrations of acrylamide and bis-acrylamide (which cross-links the acrylamide monomers)
Free-flow electrophoresis (Carrier-free electrophoresis)	No matrices are used; instead, polypeptides migrate through a solution; fast, high reproducibility, compatible with downstream detection techniques; can be run under native or denaturing conditions; only small sample volumes required (although can be used as a preparative technique)
Electrofocusing (Isoelectrofocusing)	Polypeptides are separated by differences in their isoelectric point (pI), usually performed in gels and based on the principle that overall charge on the polypeptide is a function of pH. An ampholyte solution is used to make immobilized pH gradient (IPG) gels. The immobilized pH gradient is obtained by the continuous change in the ratio of immobilines (weak acid or base defined by pK). Polypeptides migrate through the pH gradient until its charge is 0. Very high resolution, separating polypeptides differing by a single charge
Isotachopheresis (ITP)	Orders and concentrates polypeptides of intermediate effective mobilities between an ion of high effective mobility and one of much lower effective mobility, followed by their migration at a uniform speed. A multianalyte sample is introduced between the leading electrolyte (LE, containing leading ion) and the terminating electrolyte (TE, containing terminating ion) where the leading ion, the terminating ion, and the sample components have the same charge polarity, and the sample ions must have lower electrophoretic mobilities than the leading ion but larger than the terminating ion. After electrophoresis, the polypeptides move forward behind the leading ion and in front of the terminating ion, forming discrete, contiguous zones in order of their electrophoretic mobilities. Transient ITP includes an additional step of separating after ITP with zone electrophoresis.
Affinity electrophoresis	Based on changes in the electrophoretic pattern of molecules through specific interactions with other molecules or complex formation; examples include mobility shift, charge shift and affinity capillary electrophoresis. Various types are known, including those using agarose gel, rapid agarose gel, boronate affinity, affinity-trap polyacrylamide, and phosphate affinity electrophoresis
Immunoelectrophoresis	Separates polypeptides based on electrophoresis and reaction with antibodies. Includes immunoelectrophoretic analysis (one-dimensional immunoelectrophoresis), crossed immunoelectrophoresis (two-dimensional quantitative immunoelectrophoresis), rocket-immunoelectrophoresis (one-dimensional quantitative immunoelectrophoresis), fused rocket immunoelectrophoresis, and affinity immunoelectrophoresis. Often uses agarose gels buffered at high pH
Counterelectrophoresis (counterimmunoelectrophoresis)	Antibody and antigen migrate through a buffered diffusion medium. Antigens in a gel with a controlled pH are strongly negatively charged and migrate rapidly across the electric field toward the anode. The antibody in such a medium is less negatively charged and migrates in the opposite direction toward the cathode. If the antigen and antibody are specific for each other, they combine and form a distinct precipitin line.
Capillary electrophoresis	Refers to electrokinetic separation methods performed in submillimeter diameter capillaries and in micro- and nanofluidic channels. Examples include capillary zone electrophoresis (CZE), capillary gel electrophoresis (CGE), capillary isoelectric focusing (CIEF), capillary isotachopheresis and micellar electrokinetic chromatography (MEKC).

TABLE F-continued

Examples of electrophoretic methods	
Technique	Description
Capillary zone electrophoresis	A type of capillary electrophoresis, CZE separates ions based on their charge and frictional forces within a fine bore capillary. Sensitive in the picomolar range

Prepared polypeptides can be analyzed also using spectrometric methods—mass spectrometry (Rubakhin and Sweedler, 2010), ultraviolet spectrometry, visible light spectrometry, fluorescent spectrometry, and ultraviolet-visible light spectrometry (Nowicka-Jankowska, 1986).

Table G presents examples of polypeptide electrophoretic techniques.

TABLE G

Examples of spectrometric methods	
Technique	Description
Mass Spectrometry (MS)	Sample molecules are ionized by high energy electrons. The mass to charge ratio of these ions is measured by electrostatic acceleration and magnetic field perturbation, providing a precise molecular weight. Ion fragmentation patterns may be related to the structure of the molecular ion. Mass spectrum is a plot of the ion signal as a function of the mass-to-charge ratio. Analyzers include sector field mass, time-of-flight (TOF), and quadrupole mass analyzers. Ion traps include three-dimensional quadrupole, cylindrical, linear quadrupole, and Orbitrap ion traps. Detectors include electron multipliers, Faraday cups, and ion-to-photon detectors. Variations of MS include tandem MS. Mass spectrometers can be configured in a variety of ways, including matrix-assisted laser desorption/ionization source configured with a TOF analyzer (MALDI-TOF); electrospray ionization-mass spectrometry (ESI-MS), inductively coupled plasma-mass spectrometry (ICP-MS), accelerator mass spectrometry (AMS), thermal ionization-mass spectrometry (TIMS), and spark source mass spectrometry (SSMS).
Ultraviolet-Visible Spectroscopy	Absorption of high-energy UV light causes electronic excitation. Wavelengths of 200 to 800 nm show absorption if conjugated pi-electron systems are present
Infrared Spectroscopy	Absorption of infrared radiation causes vibrational and rotational excitation of groups of atoms within the polypeptide. Because of their characteristic absorptions, functional groups are identified

The principle enabling mass spectrometry (MS) consists of ionizing chemical compounds to generate charged molecules or molecule fragments, and then measuring their mass-to-charge ratios. In an illustrative MS procedure, a sample is loaded onto the MS instrument and undergoes vaporization, the components of the sample are ionized by one of a variety of methods (e.g., by impacting them with an electron beam), which results in the formation of positively charged particles, the positive ions are then accelerated by a magnetic field, computations are performed on the mass-to-charge ratio (m/z) of the particles based on the details of motion of the ions as they transit through electromagnetic fields, and, detection of the ions, which have been sorted according to their m/z ratios.

An illustrative MS instrument has three modules: an ion source, which converts gas phase sample molecules into ions (or, in the case of electrospray ionization, move ions that exist in solution into the gas phase); a mass analyzer, which sorts the ions by their mass-to-charge ratios by applying electromagnetic fields; and a detector, which measures the value of an indicator quantity and thus provides data for calculating the abundances of each ion present.

The MS technique has both qualitative and quantitative uses, including identifying unknown compounds, determining the isotopic composition of elements in a molecule, and determining the structure of a compound by observing its fragmentation. Included are gas chromatography-mass spec-

trometry (GC/MS or GC-MS), liquid chromatography mass spectrometry (LC/MS or LC-MS), and ion mobility spectrometry/mass spectrometry (IMS/MS or IMMS).

The analytical methods (chromatographic, electrophoretic, and spectrometric) can be combined. For example, combinations such as liquid chromatography-mass spec-

trometry, capillary zone electrophoresis coupled to mass spectrometry, and ion mobility spectrometry-mass spectrometry.

Automation

The various steps of the disclosed methods can be accomplished using liquid handling robots. Such robots dispense reagents, samples or other liquids to a designated container. Robots are controlled by software, either integrated directly into the robot itself, or by a connected computer. By automating liquid handling, the disclosed methods can be accomplished with high through-put, fewer errors, and reduced analyst hands-on time.

Liquid handling robots can be configured to use various laboratory instruments, such as centrifuges, PCR machines, colony pickers, shaking devices, heating devices, etc. Such customization permits adapting these machines to a particular method.

In some cases, such robots replace the use of pipettes and/or syringes by using sound to move liquids (acoustic liquid handling).

Currently, Agilent Technologies (Santa Clara, CA), Beckman Coulter, Inc. (Indianapolis, IN), Eppendorf North America (Hauppauge, NY), Hamilton Robotics (Reno, NV), Hudson Robotics, Inc. (Springfield, NJ), and Tecan AG (Männedorf, Switzerland) are some of the manufacturers of such robots.

Therapeutic Polypeptides

Polypeptides, including those that bind to one or more of the following, can be prepared and analyzed in the disclosed methods. These include CD proteins, including CD3, CD4, CD8, CD19, CD20, CD22, CD30, and CD34; including those that interfere with receptor binding. HER receptor family proteins, including HER2, HER3, HER4, and the EGF receptor. Cell adhesion molecules, for example, LFA-I, Mol, p150, 95, VLA-4, ICAM-1, VCAM, and alpha v/beta 3 integrin. Growth factors, such as vascular endothelial growth factor ("VEGF"), growth hormone, thyroid stimulating hormone, follicle stimulating hormone, luteinizing hormone, growth hormone releasing factor, parathyroid hormone, Mullerian-inhibiting substance, human macrophage inflammatory protein (MIP-1-alpha), erythropoietin (EPO), nerve growth factor, such as NGF-beta, platelet-derived growth factor (PDGF), fibroblast growth factors, including, for instance, aFGF and bFGF, epidermal growth factor (EGF), transforming growth factors (TGF), including, among others, TGF- α and TGF- β , including TGF- β 1, TGF- β 2, TGF- β 3, TGF- β 4, or TGF- β 5, insulin-like growth factors-I and -II (IGF-I and IGF-II), des(I-3)-IGF-I (brain IGF-I), and osteoinductive factors. Insulins and insulin-related proteins, including insulin, insulin A-chain, insulin B-chain, proinsulin, and insulin-like growth factor binding proteins. Coagulation and coagulation-related proteins, such as, among others, factor VIII, tissue factor, von Willebrand factor, protein C, alpha-1-antitrypsin, plasminogen activators, such as urokinase and tissue plasminogen activator ("t-PA"), bombazine, thrombin, and thrombopoietin; (vii) other blood and serum proteins, including but not limited to albumin, IgE, and blood group antigens. Colony stimulating factors and receptors thereof, including the following, among others, M-CSF, GM-CSF, and G-CSF, and receptors thereof, such as CSF-1 receptor (c-fms). Receptors and receptor-associated proteins, including, for example, flk2/flt3 receptor, obesity (OB) receptor, LDL receptor, growth hormone receptors, thrombopoietin receptors ("TPO-R," "c-mpl"), glucagon receptors, interleukin receptors, interferon receptors, T-cell receptors, stem cell factor receptors, such as c-Kit, and other receptors. Receptor ligands, including, for example, OX40L, the ligand for the OX40 receptor. Neurotrophic factors, including bone-derived neurotrophic factor (BDNF) and neurotrophin-3, -4, -5, or -6 (NT-3, NT-4, NT-5, or NT-6). Relaxin A-chain, relaxin B-chain, and prorelaxin; interferons and interferon receptors, including for example, interferon- α , - β , and - γ , and their receptors. Interleukins and interleukin receptors, including IL-1 to IL-33 and IL-1 to IL-33 receptors, such as the IL-8 receptor, among others. Viral antigens, including an AIDS envelope viral antigen. Lipoproteins, calcitonin, glucagon, atrial natriuretic factor, lung surfactant, tumor necrosis factor-alpha and -beta, enkephalinase, RANTES (regulated on activation normally T-cell expressed and secreted), mouse gonadotropin-associated peptide, DNase, inhibin, and activin. Integrin, protein A or D, rheumatoid factors, immunotoxins, bone morphogenetic protein (BMP), superoxide dismutase, surface membrane proteins, decay accelerating factor (DAF), AIDS envelope, transport proteins, homing receptors, addressins, regulatory proteins, immunoadhesins, antibodies. Myostatins, TALL proteins, including TALL-I, amyloid proteins, including but not limited to amyloid-beta proteins, thymic stromal lymphopoietins ("TSLP"), RANK ligand ("OPGL"), c-kit, TNF receptors, including TNF Receptor Type 1, TRAIL-R2, angiopoietins, and biologically active fragments or analogs or variants of any of the foregoing.

Exemplary polypeptides and antibodies include Acti-vase® (Alteplase); alirocumab, Aranesp® (Darbepoetin-alfa), Epogen® (Epoetin alfa, or erythropoietin); Avonex® (Interferon β -1a); Bexxar® (Tositumomab); Betaseron® (Interferon β); bococizumab (anti-PCSK9 monoclonal antibody designated as L1L3, see U.S. Pat. No. 8,080,243); Campath® (Alemtuzumab); Dynepo® (Epoetin delta); Velcade® (bortezomib); MLN0002 (anti- α 4 β 7 mAb); MLN1202 (anti-CCR2 chemokine receptor mAb); Enbrel® (etanercept); Eprex® (Epoetin alfa); Erbitux® (Cetuximab); evolocumab; Genotropin® (Somatropin); Herceptin® (Trastuzumab); Humatrope® (somatropin [rDNA origin] for injection); Humira® (Adalimumab); Infergen® (Interferon Alfacon-1); Natrecor® (nesiritide); Kineret® (Anakinra), Leukine® (Sargamostim); LymphoCide® (Epratuzumab); Benlysta™ (Belimumab); Metalyse® (Tenecteplase); Mircera® (methoxy polyethylene glycol-epoetin beta); Mylotarg® (Gemtuzumab ozogamicin); Raptiva® (efalizumab); Cimzia® (certolizumab pegol); Soliris™ (Eculizumab); Pexelizumab (Anti-C5 Complement); MEDI-524 (Numax®); Lucentis® (Ranibizumab); Edrecolomab (Panorex®); Trabio® (lerdelimumab); TheraCim hR3 (Nimotuzumab); Omnitarg (Pertuzumab, 2C4); Osidem® (IDM-I); OvaRex® (B43.13); Nuvion® (visilizumab); Cantuzumab mertansine (huC242-DMI); NeoRecormon® (Epoetin beta); Neumega® (Oprelvekin); Neulasta® (Pegylated filgrastim, pegylated G-CSF, pegylated hu-Met-G-CSF); Neupogen® (Filgrastim); Orthoclone OKT3® (Muromonab-CD3), Procrit® (Epoetin alfa); Remicade® (Infliximab), Reopro® (Abciximab), Actemra® (anti-IL6 Receptor mAb), Avastin® (Bevacizumab), HuMax-CD4 (zanolimimumab), Rituxan® (Rituximab); Tarceva® (Erlotinib); Roferon-A® (Interferon alfa-2a); Simulect® (Basiliximab); Stelara™ (Ustekinumab); Prexige® (lumiracoxib); Synagis® (Palivizumab); 14667-CHO (anti-IL15 antibody, see U.S. Pat. No. 7,153,507), Tysabri® (Natalizumab); Valortim® (MDX-1303, anti-*B. anthracis* Protective Antigen mAb); ABthrax™; Vectibix® (Panitumumab); Xolair® (Omalizumab), ETI211 (anti-MRSA mAb), IL-1 Trap (the Fc portion of human IgG1 and the extracellular domains of both IL-1 receptor components (the Type I receptor and receptor accessory protein)), VEGF Trap (Ig domains of VEGFR1 fused to IgG1 Fc), Zenapax® (Daclizumab); Zenapax® (Daclizumab), Zevalin® (Ibritumomab tiuxetan), Zetia (ezetimibe), Atacicept (TACI-Ig), anti- α 4 β 7 mAb (vedolizumab); galiximab (anti-CD80 monoclonal antibody), anti-CD23 mAb (lumiliximab); BR2-Fc (huBR3/huFc fusion protein, soluble BAFF antagonist); Simponi™ (Golimumab); Mapatumumab (human anti-TRAIL Receptor-1 mAb); Ocrelizumab (anti-CD20 human mAb); HuMax-EGFR (zalutumumab); M200 (Volociximab, anti- α 5 β 1 integrin mAb); MDX-010 (Ipilimumab, anti-CTLA-4 mAb and VEGFR-I (IMC-18F1); anti-BR3 mAb; anti-*C. difficile* Toxin A and Toxin B C mAbs MDX-066 (CDA-I) and MDX-1388); anti-CD22 dsFv-PE38 conjugates (CAT-3888 and CAT-8015); anti-CD25 mAb (HuMax-TAC); anti-TSLP antibodies; anti-TSLP receptor antibody (U.S. Pat. No. 8,101,182); anti-TSLP antibody designated as A5 (U.S. Pat. No. 7,982,016); (anti-CD3 mAb (NI-0401); Adecatumumab (MT201, anti-EpCAM-CD326 mAb); MDX-060, SGN-30, SGN-35 (anti-CD30 mAbs); MDX-1333 (anti-IFNAR); HuMax CD38 (anti-CD38 mAb); anti-CD40L mAb; anti-Cripto mAb; anti-CTGF Idiopathic Pulmonary Fibrosis Phase I Fibrogen (FG-3019); anti-CTLA4 mAb; anti-eotaxin1 mAb (CAT-213); anti-FGF8 mAb; anti-ganglioside GD2 mAb; anti-sclerostin antibodies (see, U.S. Pat. No. 8,715,663 or 7,592,429) anti-sclerostin antibody designated

as Ab-5 (U.S. Pat. No. 8,715,663 or 7,592,429); anti-ganglioside GM2 mAb; anti-GDF-8 human mAb (MYO-029); anti-GM-CSF Receptor mAb (CAM-3001); anti-HepC mAb (HuMax HepC); MEDI-545, MDX-1103 (anti-IFN α mAb); anti-IGFIR mAb; anti-IGF-IR mAb (HuMax-Inflam); anti-IL12/IL23p40 mAb (Briakinumab); anti-IL-23p19 mAb (LY2525623); anti-IL13 mAb (CAT-354); anti-IL-17 mAb (AIN457); anti-IL2Ra mAb (HuMax-TAC); anti-IL5 Receptor mAb; anti-integrin receptors mAb (MDX-018, CNTO 95); anti-IPIO Ulcerative Colitis mAb (MDX-1100); anti-LLY antibody; BMS-66513; anti-Mannose Receptor/hCG β mAb (MDX-1307); anti-mesothelin dsFv-PE38 conjugate (CAT-5001); anti-PDImAb (MDX-1 106 (ONO-4538)); anti-PDGFR α antibody (IMC-3G3); anti-TGF β mAb (GC-1008); anti-TRAIL Receptor-2 human mAb (HGS-ETR2); anti-TWEAK mAb; anti-VEGFR/Fit-1 mAb; anti-ZP3 mAb (HuMax-ZP3); NVS Antibody #1; NVS Antibody #2; and an amyloid-beta monoclonal antibody.

Examples of antibodies suitable for the methods and pharmaceutical formulations include the antibodies shown in Table H. Other examples of suitable antibodies include infliximab, bevacizumab, cetuximab, ranibizumab, palivizumab, abagovomab, abciximab, actoxumab, adalimumab, afelimomab, afutuzumab, alacizumab, alacizumab pegol, ald518, alemtuzumab, aliocumab, altumomab, amatuximab, anatumomab mafenatox, anrukinzumab, apolizumab, arcitumomab, aselizumab, altinumab, atlizumab, atorolimumab, tocilizumab, bapineuzumab, basiliximab, bavituximab, bectumomab, belimumab, benralizumab, bertilimumab, besilesomab, bevacizumab, bezlotoxumab, biciromab, bivatuzumab, bivatuzumab mertansine, blinatumomab, blosozumab, brentuximab vedotin, briakinumab, brodalumab, canakinumab, cantuzumab mertansine, cantuzumab mertansine, caplacizumab, capromab pendetide, carlumab, catumaxomab, cc49, cedelizumab, certolizumab pegol, cetuximab, citatuzumab bogatox, cixutumumab, clazakizumab, clenoliximab, clivatuzumab tetraxetan, conatumumab, crenezumab, cr6261, dacetuzumab, daclizumab, dalotuzumab, daratumumab, demcizumab, denosumab, detumomab, dorlimomab aritox, drozitumab, duligotumab, dupilumab, ecromeximab, eculizumab, edobacomab, edrecolomab, efalizumab, efungumab, elotuzumab, elsilimumab, enavatuzumab, enlimomab pegol, enokizumab, enoticumab, ensituximab, epitumomab cituxetan, epratuzumab, erenumab, erlizumab, ertumaxomab, etaracizumab, etrolizumab, evolocumab, exbivirumab, fanolesomab, faralimomab, farletuzumab, fasinumab, fbta05, felvizumab, fezakinumab, ficlatuzumab, figitumumab, flanvotumab, fontolizumab, foralumab, foravirumab, fresolimumab, fulranumab, futuximab, galiximab, ganitumab, gantenerumab, gavilimumab, gemtuzumab ozogamicin, gevokizumab, girentuximab, glembatumumab vedotin, golimumab, gomiliximab, gs6624, ibalizumab, ibritumomab tiuxetan, icrucumab, igovomab, imciromab, imgatuzumab, inclacumab, indatuximab ravtansine, infliximab, intetumumab, inolimumab, inotuzumab ozogamicin, ipilimumab, iratumumab, itolizumab, ixekizumab, keliximab, labetuzumab, lebrikizumab, lemalesomab, lerdelimomab, lexatumumab, libivirumab, ligelizumab, lintuzumab, lirlumab, lorvotuzumab mertansine, lucatumumab, lumiliximab, mapatumumab, maslimomab, mavrilimumab, matuzumab, mepolizumab, metelimomab, milatuzumab, minretumomab, mitumomab, mogamulizumab, morolimumab, motavizumab, moxetumomab pasudotox, muromonab-cd3, nacolomab tafenatox, namilumab, naptumomab estafenatox, narnatumab, natalizumab, nebacomab,

necitumumab, nerelimomab, nesvacumab, nimotuzumab, nivolumab, nofetumomab merpentan, ocaratuzumab, ocrelizumab, odulimomab, ofatumumab, olaratumab, olokizumab, omalizumab, onartuzumab, oportuzumab monatox, oregovomab, orticumab, otelixizumab, oxelumab, ozanvezumab, ozoralizumab, pagibaximab, palivizumab, panitumumab, panobacumab, parsatuzumab, pascolizumab, pateclizumab, patritumab, pentumomab, perakizumab, pertuzumab, pexelizumab, pidilizumab, pintumomab, placulumab, ponezumab, priliximab, primumab, PRO 140, quilizumab, racotumomab, radretumab, rafivirumab, ramucirumab, ranibizumab, raxibacumab, regavirumab, reslizumab, rilotumumab, rituximab, robatumumab, roledumab, romosozumab, rontalizumab, rovelizumab, ruplizumab, samalizumab, sarilumab, satumomab pendetide, secukinumab, sevirumab, sibrotuzumab, sifalimumab, sil-tuximab, simtuzumab, siplizumab, sirukumab, solanvezumab, solitumab, sonepcizumab, sontuzumab, stamulumab, sulesomab, suvizumab, tabalumab, tacatuzumab tetraxetan, tadocizumab, talizumab, tanezumab, taplitumomab paptox, tefibazumab, telimomab aritox, tenatumomab, tefibazumab, teneliximab, teplizumab, teprotumumab, TGN1412, tremelimomab, ticilimumab, tildrakizumab, tigatuzumab, TNX-650, tocilizumab, toralizumab, tositumomab, tralokinumab, trastuzumab, TRBS07, tregalizumab, tucotuzumab celmoleukin, tuvirumab, ublituximab, urelumab, urtoxazumab, ustekinumab, vapaliximab, vatelizumab, vedolizumab, veltuzumab, vepalimumab, vesencumab, visilizumab, volociximab, vorsetuzumab mafodotin, votumumab, zalutumumab, zanolimumab, zatuximab, ziralimumab, zolimomab aritox.

Antibodies also include adalimumab, bevacizumab, blinatumomab, cetuximab, conatumumab, denosumab, eculizumab, erenumab, evolocumab, infliximab, natalizumab, panitumumab, rilotumumab, rituximab, romosozumab, tezepelumab, and trastuzumab, and antibodies selected from Table H.

TABLE H

Examples of therapeutic antibodies					
Target (informal name)	HC* Type (including allotypes)	LC* Type	LC* pI	SEQ ID NO:	HC* SEQ ID NO:
anti-amyloid	IgG1 (f) (R; EM)	Kappa	9.0	2	3
GMCSF (247)	IgG2	Kappa	8.7	4	5
CGRPR	IgG2	Lambda	8.6	6	7
RANKL	IgG2	Kappa	8.6	8	9
Sclerostin (27H6)	IgG2	Kappa	6.6	10	11
IL-1R1	IgG2	Kappa	7.4	12	13
Myostatin	IgG1 (z) (K; EM)	Kappa	8.7	14	15
B7RP1	IgG2	Kappa	7.7	16	17
Amyloid	IgG1 (za) (K; DL)	Kappa	8.7	18	19
GMCSF (3.112)	IgG2	Kappa	8.8	20	21
CGRP (32H7)	IgG2	Kappa	8.7	22	23
CGRP (3B6.2)	IgG2	Lambda	8.6	24	25
PCSK9 (8A3.1)	IgG2	Kappa	6.7	26	27
PCSK9 (492)	IgG2	Kappa	6.9	28	29
CGRP	IgG2	Lambda	8.8	30	31
Hepcidin	IgG2	Lambda	7.3	32	33
TNFR p55)	IgG2	Kappa	8.2	34	35
OX40L	IgG2	Kappa	8.7	36	37
HGF	IgG2	Kappa	8.1	38	39
GMCSF	IgG2	Kappa	8.1	40	41
Glucagon R	IgG2	Kappa	8.4	42	43
GMCSF (4.381)	IgG2	Kappa	8.4	44	45
Sclerostin (13F3)	IgG2	Kappa	7.8	46	47
CD-22	IgG1 (f) (R; EM)	Kappa	8.8	48	49
INF γ R	IgG1 (za) (K; DL)	Kappa	8.8	50	51
Ang2	IgG2	Kappa	7.4	52	53

TABLE H-continued

Examples of therapeutic antibodies					
Target (informal name)	HC* Type (including allotypes)	LC* Type	pI	LC* SEQ ID NO:	HC* SEQ ID NO:
TRAILR2	IgG1 (f) (R; EM)	Kappa	8.7	54	55
EGFR	IgG2	Kappa	6.8	56	57
IL-4R	IgG2	Kappa	8.6	58	59
IL-15	IgG1 (f) (R; EM)	Kappa	8.8	60	61
IGF1R	IgG1 (za) (K; DL)	Kappa	8.6	62	63
IL-17R	IgG2	Kappa	8.6	64	65
Dkk1 (6.37.5)	IgG2	Kappa	8.2	66	67
Sclerostin	IgG2	Kappa	7.4	68	69
TSLP	IgG2	Lambda	7.2	70	71
Dkk1 (11H10)	IgG2	Kappa	8.2	72	73
PCSK9	IgG2	Lambda	8.1	74	75
GIPR (2G10.006)	IgG1 (z) (K; EM)	Kappa	8.1	76	77
Activin	IgG2	Lambda	7.0	78	79
Sclerostin (2B8)	IgG2	Lambda	6.7	80	81
Sclerostin	IgG2	Kappa	6.8	82	83
c-fms	IgG2	Kappa	6.6	84	85
$\alpha 4\beta 7$	IgG2	Kappa	6.5	86	87

*HC—antibody heavy chain; LC—antibody light chain.

In some embodiments, the therapeutic polypeptide is a BiTE® molecule. BiTE® molecules are engineered bispecific monoclonal antibodies which direct the cytotoxic activity of T cells against cancer cells. They are the fusion of two single-chain variable fragments (scFvs) of different antibodies, or amino acid sequences from four different genes, on a single peptide chain of about 55 kilodaltons. One of the scFvs binds to T cells via the CD3 receptor, and the other to a tumor cell via a tumor specific molecule. Blinatumomab (BLINCYTO®) is an example of a BiTE® molecule, specific for CD19. BiTE® molecules that are modified, such as those modified to extend their half-lives, can also be used in the disclosed methods.

EXAMPLES

The following Examples section is given solely by way of example and are not set forth to limit the disclosure or claims in any way.

Example 1—Reagent Preparation for Examples 2-4

Denaturing buffer (7M guanidine HCl, 100 mM tris, 20 mM methionine, pH 8.3) was prepared by adding 10 mL 1 M (hydroxymethyl) aminomethane hydrochloride (tris), pH 7.8 (Teknova, Hollister, CA, P/N T1078) to 87.5 mL 8 M guanidine HCl (Pierce, Rockford, IL, P/N 24115) followed by addition of 299 mg L-methionine (J. T. Baker, P/N 2085-05). The pH of the solution was adjusted to pH 8.3 with 6 N hydrochloric acid (HCl) (Sigma, St. Louis, MO, P/N 84429). Volume was adjusted to 100 mL with HPLC-grade water. Reduction solution (500 mM DTT) was prepared by dissolving 7.7 mg pre-weighed dithiothreitol (DTT) (Pierce, Rockford, IL, P/N 20291) in 100 μ L denaturing buffer. Alkylation solution (500 mM NaIAA) was prepared by dissolving 15-65 mg sodium iodoacetate (NaIAA) (Sigma, St. Louis, MO, P/N I-9148) in a volume of denaturing buffer sufficient to yield 500 mM NaIAA. Digestion buffer (100 mM tris, 20 mM methionine, pH 7.8) was prepared by dissolving 299 mg L-methionine in 10 mL 1 M tris, pH 7.8 and adding 100 mL HPLC-grade water. The pH of the solution was adjusted to pH 7.8 with HCl, and the volume was adjusted to 100 mL with HPLC-grade water. Enzyme solutions (1 mg/mL trypsin, 1 mg/mL HNE) were

prepared by adding 100 μ L digestion buffer to 100 μ g trypsin (Roche, Basel, Switzerland, P/N 03708969001) or 100 μ g HNE (Elastin Products Company, Owensville, MO, P/N SE563). Digest quenching solution (10% TFA) was prepared by adding 1.0 mL 100% trifluoroacetic acid (TFA) (Pierce, Rockford, IL, P/N 28904) to 9.0 mL HPLC-grade water

Example 2—Trypsin and HNE Independent Digests

Samples were denatured and reduced by adding 57.1 μ L denaturing buffer and 2 μ L reducing solution to 42.9 μ L sample (1-2 mg/mL in formulation buffer) followed by incubation at 37° C. for 35 min. Following reduction, 9 μ L alkylation solution was added to each sample followed by incubation at room temperature in dark for 20 min. Alkylation was quenching by addition of 7 μ L reducing solution. Samples were desalted with Zeba Spin Desalting Columns (Thermo Scientific, Waltham, MA, P/N 89883) in accordance with manufacturer's instructions using an Eppendorf 5430 centrifuge (Hamburg, Germany) with spins at 1,500xg. Sample concentration was measured followed desalting using either an Implen NanoPhotometer Pearl (München, Germany) or a Thermo Scientific Nanodrop 2000c (Waltham, MA). Enzyme solution (either trypsin or HNE) was added to each sample at an enzyme:substrate ratio of 1:20. Samples were incubated in a water bath at 37° C. for 35 min. Digest was quenched by the addition of 6 μ L digestion quenching solution.

Example 3—Trypsin and HNE Sequential Digests

Samples were denatured and reduced by adding 57.1 μ L denaturing buffer and 2 μ L reducing solution to 42.9 μ L sample (1-2 mg/mL in formulation buffer) followed by incubation at 37° C. for 35 min. Following reduction, 9 μ L alkylation solution was added to each sample followed by incubation at room temperature in dark for 20 min. Alkylation was quenching by addition of 7 μ L reducing solution. Samples were desalted with Zeba Spin Desalting Columns (Thermo Scientific, Waltham, MA, P/N 89883) in accordance with manufacturer's instructions using an Eppendorf 5430 centrifuge (Hamburg, Germany) with spins at 1,500xg. Sample concentration was measured followed desalting using either an Implen NanoPhotometer Pearl (München, Germany) or a Thermo Scientific Nanodrop 2000c (Waltham, MA). Trypsin solution was added to each sample at an enzyme:substrate ratio of 1:20. Samples were incubated in a water bath at 37° C. for 35 min. HNE solution was added to each sample at an enzyme:substrate ratio of 1:20. Samples were incubated in a water bath at 37° C. for 30 min. Digest was quenched by the addition of 6 μ L digestion quenching solution.

Example 4—Mixture of Trypsin Digest and Trypsin-HNE Sequential Digest

Samples were denatured and reduced by adding 57.1 μ L denaturing buffer and 2 μ L reducing solution to 42.9 μ L sample (1 mg/mL in formulation buffer) followed by incubation at 37° C. for 35 min. Following reduction, 9 μ L alkylation solution was added to each sample followed by incubation at room temperature in dark for 20 min. Alkylation was quenching by addition of 7 μ L reducing solution. Samples were desalted with Zeba Desalting Columns in accordance with manufacturer's instructions using an Eppendorf 5430 centrifuge with spins at 1,500xg. Sample concentration was measured followed desalting using either

a NanoPhotometer Pearl or a Nanodrop 2000c. Trypsin solution was added to each sample at an enzyme:substrate ratio of 1:20. Samples were incubated in a water bath at 37° C. for 35 min. Following trypsin digestion, each sample was split into two equal aliquots (~55 µL each). 3 µL digestion quenching solution was added to the first aliquot, which was then set aside. To the second aliquot, HNE solution was added at an enzyme:substrate ratio of 1:20. This aliquot was incubated in a water bath at 37° C. for 30 min followed by addition of 3 µL digestion quenching solution. The two aliquots were then combined at a 1:1 ratio followed by gentle mixing.

Example 5—Reagent Preparation for Examples 6 and 7

Denaturing buffer (6M guanidine HCl, 200 mM tris, 20 mM methionine, pH 8.3) was prepared by adding 20 mL 1 M (hydroxymethyl) aminomethane hydrochloride (tris), pH 8.3 (Teknova, St. Louis, MO, P/N T1083) to 87.5 mL 8 M guanidine HCl (Pierce, Rockford, IL, P/N 24115) followed by addition of 299 mg L-methionine (J. T. Baker, P/N 2085-05). The pH of the solution was adjusted to pH 8.3 with 1 N hydrochloric acid (HCl) (Ricca, Arlington, TX, P/N R3700100-120A) or 1 N Sodium hydroxide (NaOH) (Merck, Kenilworth, NJ, P/N 1.09137.100). Volume was adjusted to 100 mL with HPLC-grade water. Reduction solution (500 mM DTT) was prepared by dissolving 7.7 mg pre-weighed dithiothreitol (DTT) (Pierce, Rockford, IL, P/N 20291) in 100 µL denaturing buffer. Alkylation solution (500 mM NaIAA) was prepared by dissolving 15-65 mg sodium iodoacetate (NaIAA) (Sigma, St. Louis, MO, P/N 1-9148) in a volume of denaturing buffer sufficient to yield 500 mM NaIAA. Digestion buffer (50 mM tris, 20 mM methionine, pH 7.8) was prepared by dissolving 299 mg L-methionine in 10 mL 1 M tris, pH 7.8 and adding 100 mL HPLC-grade water. The pH of the solution was adjusted to pH 7.8 with 1 N hydrochloric acid (HCl) (Ricca, Arlington, TX, P/N R3700100-120A) or 1 N Sodium hydroxide (NaOH) (Merck, Kenilworth, NJ, P/N 1.09137.100), and the volume was adjusted to 100 mL with HPLC-grade water. Enzyme solutions (1 mg/mL trypsin, 1 mg/mL HNE) were prepared by adding 100 µL digestion buffer to 100 µg trypsin (Roche, Basel, Switzerland, P/N 03708969001) or 100 µg HNE (Elastin Products Company, Owensville, MO, P/N SE563). Digest quenching solution (8 M guanidine HCl, 250 mM acetate, pH 4.7) was prepared by dissolving 76.4 g guanidine HCl (Sigma, St. Louis, MO, P/N 50933) and 1.0 g sodium acetate (Sigma, P/N 32319) in 95 mL HPLC grade water. 716 µL glacial acetic acid (Sigma, St. Louis, MO, P/N 320099) was then added, and the pH was adjusted to pH 4.7 with either HCl or NaOH. Volume was then adjusted to 100 mL with HPLC grade water.

Example 6—MWCO Spin Filter-Aided Sequential Digest

100 µg sample in formulation buffer was added to a 30 kDa molecular weight cut-off spin unit consisting of a membrane unit positioned inside a centrifuge tube for filtrate collection. (Millipore, Billerica, MA, P/N MRCF0R030 or Pall, Port Washington, NY, P/N OD030C34). This unit was spun for 15 min at 14,000×g using an Eppendorf 5430 centrifuge. Filtrate was discarded. 200 µL denaturing buffer was added to the sample, the filter unit was spun for 15 min at 14,000×g, and the filtrate was discarded; this was repeated two additional times. For each sample, 3 µL reducing

solution was added to 37 µL denaturing buffer, and 40 µL of this solution was added to the filter unit. Samples were denatured and reduced by incubating at 37° C. water bath for 45 min. Samples were then spun for 15 min at 14,000×g, and the filtrate was discarded. For each sample 7 µL alkylation solution was added to 33 µL denaturing buffer, and 40 µL of this solution was added to the filter unit. Samples were alkylated by incubating at room temperature in dark for 20 min. Samples were then spun for 15 min at 14,000×g, and the filtrate was discarded. For each sample 4 µL denaturing solution was added to 36 µL denaturing buffer, and 40 µL of this solution was added to the filter unit to quench alkylation. Samples were then spun for 15 min at 14,000×g, and the filtrate was discarded. 200 µL digest buffer was added to the sample, the filter unit was spun for 15 min at 14,000×g, and the filtrate was discarded; this was repeated two additional times to remove denaturing, reducing, and alkylating agents. For each sample, 5 µL trypsin solution was added to 35 µL digest buffer, and 40 µL of this solution was added to the filter unit (1:20 enzyme:substrate ratio). Sample was incubated in a 37° C. water bath for 60 min. The filter unit was transferred to a new collection tube (Collection Tube 2). The initial collection tube (Collection Tube 1) was set aside. The filter unit was centrifuged for 15 min at 14,000×g. Filtrate, which contained tryptic peptides, was retained in Collection Tube 2. 20 µL digest buffer was added to the filter unit, the filter unit (in Collection Tube 2) was spun for 15 min at 14,000×g, and the filtrate was retained in Collection Tube 2; this was repeated one additional time. The filter unit was transferred back to Collection Tube 1, and Collection Tube 2 was set aside. For each sample, 5 µL HNE solution was added to 35 µL digest buffer, and 40 µL of this solution was added to the filter unit (now in Collection Tube 1, 1:20 enzyme:substrate ratio based on starting material). Sample was incubated in a 37° C. water bath for 30 min. The filter unit was transferred to Collection Tube 2. Collection Tube 1 was discarded. The filter unit was centrifuged for 15 min at 14,000×g. Filtrate, which contained peptides resulting from HNE digestion, was retained in Collection Tube 2 (along with tryptic peptides from previous steps). 20 µL digest buffer was added to the filter unit, the filter unit (in Collection Tube 2) was spun for 15 min at 14,000×g, and the filtrate was retained in Collection Tube 2; this was repeated one additional time. Digest was quenched by the addition of 160 µL digest quenching buffer to Collection Tube 2.

Example 7—MWCO Spin Filter-Aided Sequential Digest, Shortened

200 µL denaturing buffer was added to a 30 kDa molecular weight cut-off spin unit consisting of a membrane unit positioned inside a centrifuge tube for filtrate collection. (Millipore, Billerica, MA, P/N MRCF0R030 or Pall, Port Washington, NY, P/N OD030C34). This unit was spun for 10 min at 14,000×g using an Eppendorf 5430 centrifuge. Filtrate was discarded. 100 µg sample in formulation buffer was added to the filter unit and spun for 10 min at 14,000×g. Filtrate was discarded. For each sample, 3 µL reducing solution was added to 37 µL denaturing buffer, and 40 µL of this solution was added to the filter unit. Samples were denatured and reduced by incubating at 37° C. water bath for 30 min. For each sample 7 µL alkylation solution was added to 33 µL denaturing buffer, and 40 µL of this solution was added to the filter unit. Samples were alkylated by incubating at room temperature in dark for 20 min. For each sample 4 µL denaturing solution was added to 36 µL denaturing buffer, and 40 µL of this solution was added to the filter unit

31

to quench alkylation. Samples were then spun for 15 min at 14,000×g, and the filtrate was discarded. 200 µL digest buffer was added to the sample, the filter unit was spun for 15 min at 14,000×g, and the filtrate was discarded; this was repeated two additional times to remove denaturing, reducing, and alkylating agents. For each sample, 5 µL trypsin solution was added to 35 µL digest buffer, and 40 µL of this solution was added to the filter unit (1:20 enzyme:substrate ratio). Sample was incubated in a 37° C. water bath for 60 min. The filter unit was transferred to a new collection tube (Collection Tube 2). The initial collection tube (Collection Tube 1) was set aside. The filter unit was centrifuged for 10 min at 14,000×g. Filtrate, which contained tryptic peptides, was retained in Collection Tube 2. 20 µL digest buffer was added to the filter unit, the filter unit (in Collection Tube 2) was spun for 10 min at 14,000×g, and the filtrate was retained in Collection Tube 2; this was repeated one additional time. The filter unit was transferred back to Collection Tube 1, and Collection Tube 2 was set aside. For each sample, 5 µL HNE solution was added to 35 µL digest buffer, and 40 µL of this solution was added to the filter unit (now in Collection Tube 1, 1:20 enzyme:substrate ratio based on starting material). Sample was incubated in a 37° C. water bath for 30 min. The filter unit was transferred to Collection Tube 2. Collection Tube 1 was discarded. The filter unit was centrifuged for 10 min at 14,000×g. Filtrate, which contained peptides resulting from HNE digestion, was retained in Collection Tube 2 (along with tryptic peptides from previous steps). 20 µL digest buffer was added to the filter unit, the filter unit (in Collection Tube 2) was spun for 10 min at 14,000×g, and the filtrate was retained in Collection Tube 2; this was repeated one additional time. Digest was quenched by the addition of 160 µL digest quenching buffer to Collection Tube 2. A comparison highlighting differences between Protocols 4 and 5 is collected in Table 9.

Example 8—Ultra-Performance Liquid Chromatography (UPLC) Conditions

For all samples, Mobile Phase A consisted of 0.1% formic acid in water, and Mobile Phase B consisted of 0.1% formic acid in acetonitrile. For initial experiments using Protocols 1 and 2, peptides were separated using a CSH C18 1.7 µm, 2.1×150 mm UPLC column (Waters, Milford, MA, P/N 186005298). Following acquisition of bridging data, remaining experiments using the methods from Examples 2-4 were performed with a BEH C18 1.7 µm, 2.1×150 mm UPLC column (Waters, Milford, MA, P/N 186003556). UPLC separations were performed using either a Thermo Scientific U-3000 system (Waltham, MA), a Waters Acquity H-Class system (Milford, MA), or an Agilent 1290 system (Santa Clara, CA) utilizing gradients outlined in Tables 1-3 (depending on the experiment). Based on starting material, ~3-4 µg of sample was loaded on the column.

Example 9—Mass Spectrometry Conditions

Peptides resulting from digestion were analyzed using a Thermo Scientific Q Exactive (Waltham, MA), a Thermo Scientific Q Exactive Plus (Waltham, MA), or a Thermo Scientific Q Exactive BioPharma (Waltham, MA). Because multiple instruments were used, data collection parameters varied slightly depending on the instrument. Instruments were operated in data-dependent mode (top 4-8) over a scan range of 200-2,000 m/z. The AGC target was set to 1E6 for MS1 scans and 5E5 for tandem mass spectrometry (MSMS) scans. MS1 scans were collected at a resolution of either

32

35,000 or 140,000, and MS2 scans were collected at a resolution of 17,500. An isolation window of 2-4 m/z was specified for MSMS scans. Peaks with unassigned charge states and charge states greater than 8 were excluded from MSMS. Dynamic exclusion was set to 10 s. Lock Mass of m/z 391.28430 was enabled.

Example 10—Data Analysis

MS data were searched with MassAnalyzer (data were collected over several months, so multiple versions of MassAnalyzer were used). Carboxymethylation was specified as a static modification. Depending on the experiment, cleavage was specified as either nonspecific or the C-terminus of amino acids KRVITAL amino acids. For all searches, signal-to-noise ratio was set to 20, mass accuracy of 15 ppm was specified, and confidence was set to 0.95. For sequence coverage maps, minimum peak area was set to 1% of the base peak, relative peak area threshold was set to 17%, minimum confidence was set to 0.95, and maximum peptide mass was set to 15,000.

Example 11—Results

Initial experiments probing the efficacy of HNE for digestion of BiTE® molecules were direct comparisons of trypsin digestion to HNE digestion for BiTE®-3 using Protocol 1 and Gradient 1. Trypsin digestion did not yield peptides which could be used to characterize attributes in either CDR. On the other hand, digestion with HNE resulted in several peptides corresponding to the linker region. Many of these peptides were identified with low signal intensity (<10⁶), and this lack of specificity (14 peptides identified) could potentially compromise quantitation of attributes in this region. However, HNE specificity was thought to be related to length of the substrate being digested (Stein et al., 1987). This property of the enzyme was leveraged by first digesting BiTE®-3 with trypsin to produce potential substrates with shorter amino acid lengths followed by digestion with HNE (Protocol 2). This sequential digestion resulted in a single peptide with high signal intensity (>1.5×10⁷).

Having demonstrated feasibility of sample preparation utilizing trypsin-HNE digestion for monitoring attributes in the α-CD3 CDRs of interest, additional molecules (BiTE®-1, half-life extension BiTE®, BiTE®-2, canonical BiTE®, and CDH19, half-life extension BiTE®; in addition to BiTE®-3, half-life extension BiTE®) that had been subjected to photodegradation (1.2 million lux-h of cool white light, t=3 d) and forced deamidation (pH 8.5, 50, t=3 d) in addition to unstressed samples were prepared using Protocol 2 and Gradient 1. For unstressed samples, sequence coverages for α-CD3 CDR 1 were very reproducible (FIG. 4). One or two peptides corresponding to a potential asparagine deamidation sites in this CDR were consistently identified with high signal intensity (>10⁷). Likewise, for all molecules except CDH19, a single peptide corresponding to a potential tryptophan oxidation site was identified with high signal intensity. Forced deamidation conditions resulted in three peaks corresponding to deamidation in the α-CD3 CDR 1 for all four molecules that were not observed in the unstressed samples (FIG. 3 illustrates BiTE®-1 as a representative example). MSMS (FIG. 4) confirmed that the two peaks that elute first both corresponded to -GNS- motif deamidation while the peak that eluted last corresponded to -GNF- motif deamidation. A comparison of modification percentages for all four molecules (Table 11.1) highlights that both potential deamidation sites in this CDR were

susceptible to modification and should be monitored. CDR tryptophan oxidation was also observed followed photodegradation at significantly lower percentages (Table 11.2). Increased deamidation was not observed for the second CDR in close proximity to the large linker peptide (α -CD3 CDR2). These initial results established that not only are labile sites present in the α -CD3 CDRs that cannot be monitored using trypsin digestion but also that trypsin-HNE sequential digestion can be used to identify and monitor these modifications.

TABLE 11.1

Quantitation of asparagine deamidation observed for multiple BiTE® molecules		
Molecule	Unstressed	Forced Deamidation*
BiTE®-1	1.0%	39.6%
BiTE®-3	1.8%	39.8%
BiTE®-2	1.8%	5.7%
CDH19	1.0%	42.4%

*Stressed conditions were pH 8.5, 50° C. for 3 d. Deamidation levels account for peak areas of all three deamidated species relative to the unmodified peak area.

TABLE 11.2

Quantitation of tryptophan oxidation observed for multiple BiTE® molecules				
Molecule	CDR Tryptophan Oxidation*		Non-CDR Tryptophan Oxidation*	
	Unstressed	Photo-degradation~	Unstressed	Photo-degradation~
BiTE®-1	0.1%	1.9%	0.1%	1.0%
BiTE®-3	0.4%	2.9%	0.0%	0.6%
BiTE®-2	0.3%	1.2%	0.3%	0.5%
CDH19	0.1%	0.9%	0.2%	0.3%

*Oxidation levels account for peak areas of oxidation and kynurenine relative to the unmodified peak area.

~Stressed conditions were 1.2 million lux-h of cool white light for 3 d.

While the trypsin-HNE digestion yielded high quality peptides for the two α -CD3 CDRs of interest, sequence coverage and modification quantitation for the rest of the molecule still required the tryptic peptides utilized by conventional MAM.

A single analysis was performed on a 1:1 mixture of trypsin-digested sample combined with an aliquot of sample digested with trypsin followed by HNE (FIG. 5). An additional strength of this approach of analyzing a mixture of two digests lies in the fact that data for what is essentially two peptides maps is acquired in a single injection; not only does this allow for greater identified sequence coverage, but in the case of BiTE®-2, a peptide containing a critical attribute (CDR aspartic acid isomerization) resulting from trypsin-HNE digestion yielded better signal quality and reproducibility than the corresponding tryptic peptide. In addition, the UPLC gradient utilized for separation was switched to Gradient 2, and the column was switched from the Waters CSH C18 column to the Waters BEH C18 column. Robustness of this approach was evaluated by analyzing unstressed samples, photodegradation samples, pH jump samples, and thermal degradation samples corresponding to two molecules (BiTE®-2 and BiTE®-3). Trypsin digestion, trypsin-HNE sequential digestion, and a 1:1 mixture of trypsin digest and trypsin-HNE digest samples were prepared by two analysts and analyzed using two UPLC systems (Thermo U-3000 and Waters Acquity H-Class) connected to two different mass spectrometers (Thermo QExactive and Thermo QExactive Plus) (FIG. 8).

All samples were prepared in duplicate, so a total of four injections were acquired for each condition.

Table 11.3 collects quantitation results α -CD3 CDR 1 deamidation from both BiTE®-2 and BiTE®-3 as a representative peptide with asparagine deamidation. For the data presented in this table, photodegradation was performed with 192,000 lux-h for 2 d; pH jump degradation was performed at pH 8.4 at 37° C. for 7 d; and thermal degradation was performed at 40° C. for 4 weeks. These results indicate that mixing the trypsin digest with the trypsin-HNE digest had minimal impact on quantitation results. Reducing the gradient still resulted in clear separation of the three deamidation species (unmodified separated from -GNS- peak 1 by ~1.5 min, -GNS- peak 1 separated from -GNS- peak 2 by ~0.5 min, -GNS- peak 2 separated from -GNF- by ~1.3 min, data not shown). In addition, precision of the deamidation levels for both molecules was comparable regardless of whether trypsin-HNE digestion was run separately or mixed with the trypsin digest. Because deamidation and aspartic acid isomerization were typically the most challenging modifications to chromatographically separate from corresponding unmodified peptides, Table 11.4 collects quantitation results for BiTE®-3 D⁵¹⁰ isomerization as a representative peptide with aspartic acid isomerization. For the data presented in this table, photodegradation was performed with 192,000 lux-h for 2 d; pH jump degradation was performed at pH 8.4 at 37° C. for 7 d; and thermal degradation was performed at 40° C. for 4 weeks. Despite containing threonine and leucine residues, a portion of the sequence of BiTE®-3 undergoes minimal cleavage during sequential digestion with HNE, so it was possible to quantitate D⁵¹⁰ isomerization using a fully tryptic peptide regardless of whether the trypsin digest, the trypsin-HNE sequential digest, or the 1:1 mixture of the two digests was analyzed. In these results, D⁵¹⁰ isomerization levels were consistent regardless of the digest conditions, again reinforcing the observation that mixing these digests had minimal impact on attribute quantitation. These results demonstrate that mixing the two parallel digests is a reasonable approach for BiTE® molecules (as well as other polypeptides) and that doing so has minimal impact on modification quantitation regardless of whether tryptic peptides or trypsin-HNE peptides are used.

TABLE 11.3

Quantitation of BiTE®-2 and BiTE®-3 asparagine deamidation			
Condition	Digest	BiTE®-2 N ³⁶⁰ + N ³⁶³	BiTE®-3 N ³⁵⁹ + N ³⁶²
		Deamidation**	Deamidation**
Control	Trypsin → Neutrophil Elastase*	1.9% ± 0.2%	1.9% ± 0.1%
	Mix~	2.2% ± 0.6%	1.6% ± 0.3%
Photo Degradation	Trypsin → Neutrophil Elastase*	2.1% ± 0.4%	2.0% ± 0.3%
	Mix~	2.2% ± 0.7%	1.8% ± 0.2%
pH Jump Degradation	Trypsin → Neutrophil Elastase*	54.7% ± 0.3%	42.2% ± 1.1%
	Mix~	54.5% ± 1.0%	41.5% ± 1.2%
Thermal Degradation	Trypsin → Neutrophil Elastase	13.5% ± 0.4%	4.9% ± 0.3%
	Mix	13.6% ± 0.8%	4.5% ± 0.3%

*"Trypsin → Neutrophil Elastase" corresponds to BiTE®-2/BiTE®-3 digested with trypsin followed by sequential digestion with neutrophil elastase.

~"Mix" corresponds to "Trypsin → Neutrophil Elastase" mixed (1:1) with BiTE®-2/BiTE®-3 digested with trypsin.

**Deamidation levels are total and account for peak areas of all three deamidated species (-GNS- and -GNF-) relative to the unmodified peak area.

TABLE 11.4

Quantitation of BiTE®-3 aspartic acid isomerization		
Condition	Digest	D ⁵¹⁰ Isomerization
Control	Trypsin*	0.2% ± 0.1%
	Trypsin → Neutrophil Elastase [^]	0.4% ± 0.1%
	Mix**	0.3% ± 0.1%
Photo Degradation	Trypsin*	0.3% ± 0.1%
	Trypsin → Neutrophil Elastase [^]	0.4% ± 0.1%
	Mix**	0.3% ± 0.1%
pH Jump Degradation	Trypsin*	0.8% ± 0.1%
	Trypsin → Neutrophil Elastase [^]	1.0% ± 0.2%
	Mix**	1.0% ± 0.2%
Thermal Degradation	Trypsin*	4.1% ± 0.4%
	Trypsin → Neutrophil Elastase [^]	4.1% ± 0.4%
	Mix**	3.9% ± 0.5%

*"Trypsin" indicates BiTE®-3 digested with trypsin.

[^]"Trypsin → Neutrophil Elastase" corresponds to BiTE®-2/BiTE®-3 digested with trypsin followed by sequential digestion with neutrophil elastase.

**"Mix" corresponds to "Trypsin → Neutrophil Elastase" mixed (1:1) with BiTE®-2/BiTE®-3 digested with trypsin

MAM utilizing the method described above (Protocol 3, Gradient 2) was qualified for BiTE®-2 (ATM-000401) and BiTE®-3 (ATM-000391). However, during analysis, the new peak detection procedure (Sieve) incorporated into the method identified a number of new peaks. This resulted in system suitability failure of the sequences. After further evaluation, a number of the peaks corresponded to missed tryptic cleavages. However, some of these peaks were observed in final system suitability injection (but not in the initial pre-run injection). These findings suggested that trypsin efficiency was being somehow impeded, but they also suggested that one of the enzymes was still active despite acidification of the sample following digestion. However, attribute quantitation was not affected by the new peaks (only the system suitability checks). Troubleshooting indicated that the desalting columns utilized in Protocols 1-3 were not completely removing guanidine from the sample prior to digestion. Trypsin efficiency can be inhibited by low levels of guanidine, so it was likely that variable removal of guanidine by the desalting column resulted in inconsistent tryptic digestion. In addition, because acid quenching of tryptic digestion has been reported, it was speculated that acidification was not sufficiently quenching HNE digestion. Enzymatic activity, though reduced due to acidification, would result in the observation of new peaks if a replicate injection was performed from a single vial after a period of time; this, in turn, would result in failure of system suitability assessment.

To address these issues, a thorough exploration of multiple preparation methods was conducted. These included evaluating multiple desalting columns (Zeba, Biospin, NAP-5), preparation methods that do not require guanidine for denaturation (Rapigest), and MWCO filter-based methods using multiple manufacturers (Pall and Microcon® (available from Millipore (Waltham, MA)) as well as multiple molecular weight cut-offs (30 kDa, 5 kDa, as specified by the manufacturers). In addition, the digest quenching buffer was updated to include a high concentration of guanidine to inactivate both enzymes. In Protocols 1-3, acid quenching typically resulted in a final sample pH of ~2. The updated digest quenching buffer was low enough to inhibit trypsin and HNE activities but high enough to minimize artefactual modifications (e.g., deamidation and isomerization) that may result due to extended exposure to low pH. Because this quenching buffer contained guanidine, the initial wash step in the UPLC gradient was extended; all experiments utiliz-

ing the guanidine quenching buffer required the use of Gradient 3 to thoroughly desalt the samples prior to MS analysis.

The conclusion of this investigation was that a MWCO filter-based method (Protocol 4 above, adapted from the previously reported filter-aided sample preparation (FASP) method (Wisniewski et al., 2009)) outperformed other preparation methods based on several metrics (e.g., repeatability, recovery, quantitation, sequence coverage, concentration range, etc.). A schematic of Protocol 4 is presented in FIG. 7. In this method, the molecule is captured on top of the 30 kDa MWCO spin filter, and denaturation, reduction, alkylation, desalting, and digestion all occur on filter. Following digestion, resulting peptides are collected through centrifugation. In sequential digest outlined in Protocol 4, following trypsin digestion, the tryptic peptides are collected, and species that do not pass through the filter are then subjected to digestion with HNE. One of the surprising observations of this method is that the 8 kDa linker peptide of interest was retained by the 30 kDa filter, despite the recommendations made by manufacturers Microcon® and Pall, respectively, of "using a membrane with a MWCO at least two times smaller than the molecular weight of the protein solute that one intends to concentrate" and "a MWCO be selected that is three to six times smaller than the molecular weight of the solute being retained." While manufacturers recommend using a MWCO two to six times smaller than the species targeted for retention, surprisingly the 8 kDa peptide is being retained by a MWCO nearly four times larger than the peptide itself.

Retention of the 8 kDa peptide by 30 kDa MWCO filters with membranes consisting of different materials (Microcon® membranes are regenerated cellulose, Pall membranes are modified polyethersulfone) emphasizes that this is not a manufacturer-specific artefact. Not only did this novel use of a MWCO filter make a sequential digestion possible, but direct comparison of this method utilizing the 30 kDa MWCO filter to a different method using a 5 kDa MWCO filter showed that the 30 kDa filter method significantly outperformed the 5 kDa filter method in terms of trypsin digest efficiency, repeatability, precision of attribute quantitation, and recovery. The primary advantage of this method is that enzyme specificity afforded by trypsin digestion is preserved for the bulk of the molecule, and only the large peptides difficult to characterize by MS are further digested into smaller peptides because they are retained on-filter. This is illustrated by FIG. 8, which highlights trypsin specificity of peptides identified for BiTE®-1, BiTE®-2, and BiTE®-3 (triplicate analysis for each molecule) using filters manufactured by both Microcon® and Pall. In these results, over 95% of identified peptides typically corresponded to trypsin digestion, while the remaining identified peptides primarily corresponded to the large 8 kDa linker peptide of interest (FIG. 9). Attribute quantitation for BiTE®-2 using filters from both manufacturers was comparable to results for the parallel digestion (Protocol 3) (Table 11.5). Similar results were also observed for BiTE®-1 and BiTE®-3. This method resolved the new peak issue by removing guanidine more completely compared to the other method discussed above, and the enzymes were also inactivated either by the updated quenching buffer or by retention on the MWCO filter. In FIG. 10, a BiTE®-3 sample prepared using Protocol 3 was reinjected after ~4 d. Several peaks were only present in the initial injection, which were absent in the second injection. In contrast, samples prepared using Protocol 4 were consistent even after 4 d had passed between injections (FIG. 11). These results suggest that the MWCO filter-based prepara-

tion outline in Protocol 4 is robust, yields quantitation results consistent to those previously reported, and eliminates many of the root causes of new peaks observed with Protocol 3.

Protocol 3, these modifications reduced sample preparation time to a point where it is comparable to solution-based methods.

TABLE 11.6

Comparison of Protocols 4 and 5					
Protocol 4			Protocol 5		
Step	Description	Time (min)	Step	Description	Time (min)
—	—	—	1	Equilibrate filter with denaturing buffer. Spin	10
1	Spin (remove formulation buffer)	15	2	Spin (remove formulation buffer)	10
2	Add denaturing buffer, spin	15	—	—	—
3	Add denaturing buffer, spin	15	—	—	—
4	Add denaturing/reducing solution. Incubate	45	3	Add denaturing/reducing solution. Incubate	30
5	Spin	15	—	—	—
6	Add alkylation solution. Incubate	20	4	Add alkylation solution. Incubate	20
7	Spin	15	—	—	—
8	Add reducing solution. Spin	15	5	Add reducing solution. Spin	15
9	Add digest buffer. Spin	15	6	Add digest buffer. Spin	15
10	Add digest buffer. Spin	15	7	Add digest buffer. Spin	15
11	Add digest buffer. Spin	15	8	Add digest buffer. Spin	15
12	Add trypsin solution. Incubate	60	9	Add trypsin solution. Incubate	60
13	Spin	15	10	Spin	10
14	Add digest buffer. Spin	15	11	Add digest buffer. Spin	10
15	Add digest buffer. Spin	15	12	Add digest buffer. Spin	10
16	Add HNE solution. Incubate	30	13	Add HNE solution. Incubate	30
17	Spin	15	14	Spin	10
18	Add digest buffer. Spin	15	15	Add digest buffer. Spin	10
19	Add digest buffer. Spin	15	—	—	—
Total Time		380	Total Time		270
Spin Steps		15	Spin Steps		9

TABLE 11.5

Comparison of attribute quantitation for BiTE®-2 using 30 kDa MWCO filters manufactured by Microcon® and Pall*			
Attribute	Reference Protocol	Microcon® 30 kDa	Pall 30 kDa
M ²³³ Oxidation	0.254% ± 0.046%	0.14% ± 0.02%	0.22% ± 0.03%
D ⁵⁴ + D ⁵⁷ + D ⁶²	1.256% ± 0.298%	1.01% ± 0.04%	0.89% ± 0.00%
Isomerization [†]			
N ³⁶³ Deamidation	0.946% ± 0.375%	0.96% ± 0.02%	0.88% ± 0.02%

*The Microcon® prep was performed in triplicate, but due to instrument failure, only two preps for the Pall filters were analyzed.

[†]D⁵⁴ + D⁵⁷ + D⁶² isomerization was quantified using I⁵¹-R⁶⁵ for the Reference Protocol. C⁴⁴-R⁶⁵ was used for quantitation in filter-based experiments due to more efficient trypsin digestion.

One of the primary concerns about Protocol 4 was the length of time necessary to prepare the samples; a sequential digest utilizing Protocol 4 could take more than 8 h. In addition to analyst fatigue and other factors, the amount of time necessary to prepare samples using this method could introduce artefactual modifications resulting from extended exposure to ambient laboratory conditions. To address these concerns, a shortened filter-based preparation (Protocol 5) was developed. Table 11.6 highlights differences between Protocols 4 and 5. The bulk of time savings introduced by Protocol 5 were made possible by eliminating superfluous centrifugation steps, reducing centrifugation times, and reducing incubation times. These reductions theoretically yield roughly 2 h in time savings, but practically, time savings were typically greater than 3 h. While still slightly longer than solution-based preparation methods such as

Overlays comparing BiTE®-2 samples prepped with Protocols 4 and 5 are presented in FIG. 12 (each preparation performed in triplicate). The only noticeable peak differences observed were a shoulder at ~24 min and a peak at ~47 min that were present in all samples prepped with Protocol 4 but none of the samples prepped with Protocol 5. These peaks were attributed to nonspecific carboxymethylation resulting from overalkylation. Similarly, because denaturing/reduction incubation time was reduced and spin steps between reduction and alkylation were eliminated, alkylation levels for all cysteine residues were compared (Table 11.7). For the data in this table, alkylation percentages were determined by searching data with MassAnalyzer, specifying carboxymethylation as a variable modification (rather than a static modification, which is typically specified). Alkylation levels for all BiTE®-1 samples, regardless of whether samples were prepped with Protocol 4 or 5, were above 99.7%. Similar results were observed for BiTE®-2 and BiTE®-3, suggesting that time reductions introduced by Protocol 5 had no negative impact on alkylation levels. Attribute quantitation was comparable when samples were prepped with both Protocols 4 and 5, which in turn were consistent with modification levels historically observed (Table 11.8). For the data in this table, each preparation was performed in triplicate. D⁵⁴+D⁵⁷+D⁶² isomerization was quantified using I⁵¹-R⁶⁵ for MDR-001581. C⁴⁴-R⁶⁵ was used for quantitation in filter-based experiments due to more efficient trypsin digestion. These initial results for Protocol 5 were encouraging, so a limited robustness evaluation was performed (two analysts, BiTE®-2 samples prepped in triplicate). New peak detection passed for all samples (e.g.,

no new peaks were detected), and attribute quantitation for all 6 injections was consistent with MDR-001581 (Table 11.9). Based on these results, along with comparison of other metrics such as trypsin specificity, total identified area, recovery of the 8 kDa linker peptide, and percent of identified area corresponding to enzymes used for digestion, Protocol 5 introduces significant time savings to preparation time with no significant negative impact on data quality.

TABLE 11.7

Comparison of BiTE @-1 alkylation levels resulting from Protocols 4 and 5						
Cys Residue	Protocol 4			Protocol 5		
	Prep 1	Prep 2	Prep 3	Prep 1	Prep 2	Prep 3
C22	99.95%	99.97%	99.96%	100.00%	100.00%	100.00%
C44	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%
C96	99.93%	99.95%	99.94%	99.99%	99.99%	99.99%
C162	99.93%	99.94%	99.95%	99.97%	99.98%	99.98%
C232	99.75%	99.73%	99.79%	99.94%	99.93%	99.94%
C244	99.95%	99.95%	99.96%	99.98%	99.98%	99.99%
C279	99.96%	99.97%	99.97%	99.99%	99.99%	100.00%
C355	99.92%	99.93%	99.94%	99.99%	99.98%	99.98%
C419	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%
C487	99.98%	99.99%	99.99%	100.00%	100.00%	100.00%

TABLE 11.8

Comparison of attribute quantitation for AMG596 using Protocols 4 and 5			
Attribute	Reference Protocol	Protocol 4	Protocol 5
M ²³³ Oxidation	0.254% ± 0.046%	0.14% ± 0.02%	0.13% ± 0.01%
D ⁵⁴ + D ⁵⁷ + D ⁶²	1.256% ± 0.298%	1.01% ± 0.04%	0.89% ± 0.02%
Isomerization			
N ³⁶³ Deamidation	0.946% ± 0.375%	0.96% ± 0.02%	0.75% ± 0.04%

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SEQUENCE LISTING

<160> NUMBER OF SEQ ID NOS: 87

<210> SEQ ID NO 1
 <211> LENGTH: 267
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 <213> ORGANISM: Homo sapiens
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 <220> FEATURE:
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 <222> LOCATION: (30)..(247)
 <223> OTHER INFORMATION: peptidase S1 domain

<400> SEQUENCE: 1

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Met Thr Leu Gly Arg Arg Leu Ala Cys Leu Phe Leu Ala Cys Val Leu
      -25                      -20                      -15

Pro Ala Leu Leu Leu Gly Gly Thr Ala Leu Ala Ser Glu Ile Val Gly
      -10                      -5                      -1  1

Gly Arg Arg Ala Arg Pro His Ala Trp Pro Phe Met Val Ser Leu Gln
      5                      10                      15

Leu Arg Gly Gly His Phe Cys Gly Ala Thr Leu Ile Ala Pro Asn Phe
      20                      25                      30                      35

Val Met Ser Ala Ala His Cys Val Ala Asn Val Asn Val Arg Ala Val
      40                      45                      50

Arg Val Val Leu Gly Ala His Asn Leu Ser Arg Arg Glu Pro Thr Arg
      55                      60                      65

Gln Val Phe Ala Val Gln Arg Ile Phe Glu Asn Gly Tyr Asp Pro Val
      70                      75                      80

Asn Leu Leu Asn Asp Ile Val Ile Leu Gln Leu Asn Gly Ser Ala Thr
      85                      90                      95

Ile Asn Ala Asn Val Gln Val Ala Gln Leu Pro Ala Gln Gly Arg Arg
      100                     105                     110                     115

Leu Gly Asn Gly Val Gln Cys Leu Ala Met Gly Trp Gly Leu Leu Gly
      120                     125                     130

Arg Asn Arg Gly Ile Ala Ser Val Leu Gln Glu Leu Asn Val Thr Val
      135                     140                     145

Val Thr Ser Leu Cys Arg Arg Ser Asn Val Cys Thr Leu Val Arg Gly
      150                     155                     160

Arg Gln Ala Gly Val Cys Phe Gly Asp Ser Gly Ser Pro Leu Val Cys
      165                     170                     175

Asn Gly Leu Ile His Gly Ile Ala Ser Phe Val Arg Gly Gly Cys Ala
      180                     185                     190                     195

Ser Gly Leu Tyr Pro Asp Ala Phe Ala Pro Val Ala Gln Phe Val Asn
      200                     205                     210

Trp Ile Asp Ser Ile Ile Gln Arg Ser Glu Asp Asn Pro Cys Pro His
      215                     220                     225

Pro Arg Asp Pro Asp Pro Ala Ser Arg Thr His
      230                     235

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<210> SEQ ID NO 2
 <211> LENGTH: 239
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

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<400> SEQUENCE: 2

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Met Arg Leu Pro Ala Gln Leu Leu Gly Leu Leu Met Leu Trp Val Pro
1           5           10           15

Gly Ser Ser Gly Asp Ile Val Met Thr Gln Thr Pro Leu Ser Ser Pro
20           25           30

Val Thr Leu Gly Gln Pro Ala Ser Ile Ser Cys Arg Ser Ser Gln Ser
35           40           45

Leu Val His Ser Asp Gly Asn Thr Tyr Leu Asn Trp Leu Gln Gln Arg
50           55           60

Pro Gly Gln Pro Pro Arg Leu Leu Ile Tyr Lys Ile Ser Asn Arg Phe
65           70           75           80

Ser Gly Val Pro Asp Arg Phe Thr Gly Ser Gly Ala Gly Thr Asp Phe
85           90           95

Thr Leu Lys Ile Ser Arg Val Glu Ala Glu Asp Val Gly Val Tyr Thr
100          105          110

Cys Met Gln Val Thr Gln Phe Pro Leu Thr Phe Gly Gln Gly Thr Arg
115          120          125

Leu Glu Ile Lys Arg Thr Val Ala Ala Pro Ser Val Phe Ile Phe Pro
130          135          140

Pro Ser Asp Glu Gln Leu Lys Ser Gly Thr Ala Ser Val Val Cys Leu
145          150          155          160

Leu Asn Asn Phe Tyr Pro Arg Glu Ala Lys Val Gln Trp Lys Val Asp
165          170          175

Asn Ala Leu Gln Ser Gly Asn Ser Gln Glu Ser Val Thr Glu Gln Asp
180          185          190

Ser Lys Asp Ser Thr Tyr Ser Leu Ser Ser Thr Leu Thr Leu Ser Lys
195          200          205

Ala Asp Tyr Glu Lys His Lys Val Tyr Ala Cys Glu Val Thr His Gln
210          215          220

Gly Leu Ser Ser Pro Val Thr Lys Ser Phe Asn Arg Gly Glu Cys
225          230          235

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<210> SEQ ID NO 3

<211> LENGTH: 470

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 3

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Met Asp Thr Leu Cys Tyr Thr Leu Leu Leu Leu Thr Thr Pro Ser Trp
1           5           10           15

Val Leu Ser Gln Val Thr Leu Lys Glu Ser Gly Pro Val Leu Leu Lys
20           25           30

Pro Thr Glu Thr Leu Thr Leu Thr Cys Thr Val Ser Gly Phe Ser Leu
35           40           45

Ser Asn Ala Arg Met Gly Val Ser Trp Ile Arg Gln Pro Pro Gly Lys
50           55           60

Ala Leu Glu Trp Leu Ala His Ile Phe Ser Asn Asp Glu Lys Ser Tyr
65           70           75           80

Ile Thr Ser Leu Lys Ser Arg Leu Thr Ile Ser Lys Asp Thr Ser Lys
85           90           95

Ser Gln Val Val Leu Thr Met Thr Asn Met Asp Pro Val Asp Thr Ala
100          105          110

Thr Tyr Tyr Cys Ala Arg Ile Pro Leu Arg Ser Pro Gly Ala Phe Asp
115          120          125

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Ile Trp Gly Gln Gly Thr Met Val Thr Val Ser Ser Ala Ser Thr Lys
 130                      135                      140

Gly Pro Ser Val Phe Pro Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly
145                      150                      155                      160

Gly Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro
                      165                      170                      175

Val Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr
                      180                      185                      190

Phe Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val
                      195                      200                      205

Val Thr Val Pro Ser Ser Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn
210                      215                      220

Val Asn His Lys Pro Ser Asn Thr Lys Val Asp Lys Arg Val Glu Pro
225                      230                      235                      240

Lys Ser Cys Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu
                      245                      250                      255

Leu Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp
260                      265                      270

Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp
275                      280                      285

Val Ser His Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly
290                      295                      300

Val Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn
305                      310                      315                      320

Ser Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp
                      325                      330                      335

Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro
                      340                      345                      350

Ala Pro Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu
                      355                      360                      365

Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn
370                      375                      380

Gln Val Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile
385                      390                      395                      400

Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr
405                      410                      415

Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys
420                      425                      430

Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys
435                      440                      445

Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu
450                      455                      460

Ser Leu Ser Pro Gly Lys
465                      470

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<210> SEQ ID NO 4
<211> LENGTH: 235
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

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<400> SEQUENCE: 4

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Met Glu Thr Pro Ala Gln Leu Leu Phe Leu Leu Leu Trp Leu Pro
 1              5              10              15

Asp Thr Thr Gly Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser

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Ala	Leu	Gly	Cys	Leu	Val	Lys	Asp	Tyr	Phe	Pro	Glu	Pro	Val	Thr	Val	
				165					170					175		
Ser	Trp	Asn	Ser	Gly	Ala	Leu	Thr	Ser	Gly	Val	His	Thr	Phe	Pro	Ala	
		180						185					190			
Val	Leu	Gln	Ser	Ser	Gly	Leu	Tyr	Ser	Leu	Ser	Ser	Val	Val	Thr	Val	
	195						200					205				
Pro	Ser	Ser	Asn	Phe	Gly	Thr	Gln	Thr	Tyr	Thr	Cys	Asn	Val	Asp	His	
	210					215					220					
Lys	Pro	Ser	Asn	Thr	Lys	Val	Asp	Lys	Thr	Val	Glu	Arg	Lys	Cys	Cys	
225					230					235					240	
Val	Glu	Cys	Pro	Pro	Cys	Pro	Ala	Pro	Pro	Val	Ala	Gly	Pro	Ser	Val	
			245						250					255		
Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg	Thr	
		260						265					270			
Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	His	Glu	Asp	Pro	Glu	
	275						280					285				
Val	Gln	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys	
	290					295					300					
Thr	Lys	Pro	Arg	Glu	Glu	Gln	Phe	Asn	Ser	Thr	Phe	Arg	Val	Val	Ser	
305					310					315					320	
Val	Leu	Thr	Val	Val	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys	
			325					330						335		
Cys	Lys	Val	Ser	Asn	Lys	Gly	Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile	
		340						345					350			
Ser	Lys	Thr	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro	
		355					360					365				
Pro	Ser	Arg	Glu	Glu	Met	Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Leu	
	370					375					380					
Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Asn	
385					390					395					400	
Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Met	Leu	Asp	Ser	
			405						410					415		
Asp	Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Lys	Leu	Thr	Val	Asp	Lys	Ser	Arg	
			420					425				430				
Trp	Gln	Gln	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val	Met	His	Glu	Ala	Leu	
	435						440					445				
His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser	Leu	Ser	Pro	Gly	Lys		
	450					455					460					

<210> SEQ ID NO 6

<211> LENGTH: 235

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 6

Met	Ala	Trp	Ala	Leu	Leu	Leu	Leu	Thr	Leu	Leu	Thr	Gln	Asp	Thr	Gly	
1				5					10					15		
Ser	Trp	Ala	Gln	Ser	Val	Leu	Thr	Gln	Ser	Pro	Ser	Ala	Ser	Gly	Thr	
		20						25					30			
Pro	Gly	Gln	Arg	Val	Thr	Ile	Ser	Cys	Ser	Gly	Ser	Ser	Ser	Asn	Ile	
		35					40					45				
Gly	Ser	Asn	Tyr	Val	Tyr	Trp	Tyr	Gln	Gln	Leu	Pro	Gly	Ala	Ala	Pro	
	50					55					60					
Lys	Leu	Leu	Ile	Leu	Arg	Asn	Asn	Gln	Arg	Pro	Ser	Gly	Val	Pro	Asp	
65					70					75					80	

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Arg Phe Ser Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Thr Ile Ser
      85                      90                      95
Gly Leu Arg Ser Glu Asp Glu Ala Asp Tyr Tyr Cys Ala Ala Trp Asp
      100                    105                    110
Asp Ser Leu Ser Gly Trp Val Phe Gly Gly Gly Thr Lys Leu Thr Val
      115                    120                    125
Leu Gly Gln Pro Lys Ala Asn Pro Thr Val Thr Leu Phe Pro Pro Ser
      130                    135                    140
Ser Glu Glu Leu Gln Ala Asn Lys Ala Thr Leu Val Cys Leu Ile Ser
      145                    150                    155                    160
Asp Phe Tyr Pro Gly Ala Val Thr Val Ala Trp Lys Ala Asp Gly Ser
      165                    170                    175
Pro Val Lys Ala Gly Val Glu Thr Thr Lys Pro Ser Lys Gln Ser Asn
      180                    185                    190
Asn Lys Tyr Ala Ala Ser Ser Tyr Leu Ser Leu Thr Pro Glu Gln Trp
      195                    200                    205
Lys Ser His Arg Ser Tyr Ser Cys Gln Val Thr His Glu Gly Ser Thr
      210                    215                    220
Val Glu Lys Thr Val Ala Pro Thr Glu Cys Ser
      225                    230                    235

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<210> SEQ ID NO 7

<211> LENGTH: 479

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 7

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Met Asp Met Arg Val Pro Ala Gln Leu Leu Gly Leu Leu Leu Trp
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Leu Arg Gly Ala Arg Cys Glu Val Gln Leu Val Glu Ser Gly Gly Gly
      20                    25                    30
Leu Val Lys Pro Gly Gly Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly
      35                    40                    45
Phe Thr Phe Ser Asn Ala Trp Met Ser Trp Val Arg Gln Ala Pro Gly
      50                    55                    60
Lys Gly Leu Glu Trp Val Gly Arg Ile Lys Ser Lys Thr Asp Gly Gly
      65                    70                    75                    80
Thr Thr Asp Tyr Thr Ala Pro Val Lys Gly Arg Phe Thr Ile Ser Arg
      85                    90                    95
Asp Asp Ser Lys Asn Thr Leu Tyr Leu Gln Met Asn Ser Leu Lys Ala
      100                    105                    110
Glu Asp Thr Ala Val Tyr Tyr Cys Thr Thr Asp Arg Thr Gly Tyr Ser
      115                    120                    125
Ile Ser Trp Ser Ser Tyr Tyr Tyr Tyr Tyr Gly Met Asp Val Trp Gly
      130                    135                    140
Gln Gly Thr Thr Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro Ser
      145                    150                    155                    160
Val Phe Pro Leu Ala Pro Cys Ser Arg Ser Thr Ser Glu Ser Thr Ala
      165                    170                    175
Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val
      180                    185                    190
Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala
      195                    200                    205
Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val

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210	215	220
Pro Ser Ser Asn Phe Gly Thr Gln Thr Tyr Thr Cys Asn Val Asp His 225 230 235 240		
Lys Pro Ser Asn Thr Lys Val Asp Lys Thr Val Glu Arg Lys Cys Cys 245 250 255		
Val Glu Cys Pro Pro Cys Pro Ala Pro Pro Val Ala Gly Pro Ser Val 260 265 270		
Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser Arg Thr 275 280 285		
Pro Glu Val Thr Cys Val Val Val Asp Val Ser His Glu Asp Pro Glu 290 295 300		
Val Gln Phe Asn Trp Tyr Val Asp Gly Val Glu Val His Asn Ala Lys 305 310 315 320		
Thr Lys Pro Arg Glu Glu Gln Phe Asn Ser Thr Phe Arg Val Val Ser 325 330 335		
Val Leu Thr Val Val His Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys 340 345 350		
Cys Lys Val Ser Asn Lys Gly Leu Pro Ala Pro Ile Glu Lys Thr Ile 355 360 365		
Ser Lys Thr Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro 370 375 380		
Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser Leu Thr Cys Leu 385 390 395 400		
Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn 405 410 415		
Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Met Leu Asp Ser 420 425 430		
Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val Asp Lys Ser Arg 435 440 445		
Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu 450 455 460		
His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly Lys 465 470 475		
<210> SEQ ID NO 8		
<211> LENGTH: 235		
<212> TYPE: PRT		
<213> ORGANISM: Homo sapiens		
<400> SEQUENCE: 8		
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Asp Thr Thr Gly Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser 20 25 30		
Leu Ser Pro Gly Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser 35 40 45		
Val Arg Gly Arg Tyr Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala 50 55 60		
Pro Arg Leu Leu Ile Tyr Gly Ala Ser Ser Arg Ala Thr Gly Ile Pro 65 70 75 80		
Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile 85 90 95		
Ser Arg Leu Glu Pro Glu Asp Phe Ala Val Phe Tyr Cys Gln Gln Tyr 100 105 110		

-continued

Gly	Ser	Ser	Pro	Arg	Thr	Phe	Gly	Gln	Gly	Thr	Lys	Val	Glu	Ile	Lys
	115						120					125			
Arg	Thr	Val	Ala	Ala	Pro	Ser	Val	Phe	Ile	Phe	Pro	Pro	Ser	Asp	Glu
130						135					140				
Gln	Leu	Lys	Ser	Gly	Thr	Ala	Ser	Val	Val	Cys	Leu	Leu	Asn	Asn	Phe
145					150					155					160
Tyr	Pro	Arg	Glu	Ala	Lys	Val	Gln	Trp	Lys	Val	Asp	Asn	Ala	Leu	Gln
			165						170					175	
Ser	Gly	Asn	Ser	Gln	Glu	Ser	Val	Thr	Glu	Gln	Asp	Ser	Lys	Asp	Ser
		180						185					190		
Thr	Tyr	Ser	Leu	Ser	Ser	Thr	Leu	Thr	Leu	Ser	Lys	Ala	Asp	Tyr	Glu
	195						200					205			
Lys	His	Lys	Val	Tyr	Ala	Cys	Glu	Val	Thr	His	Gln	Gly	Leu	Ser	Ser
	210					215					220				
Pro	Val	Thr	Lys	Ser	Phe	Asn	Arg	Gly	Glu	Cys					
225					230					235					

<210> SEQ ID NO 9

<211> LENGTH: 467

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 9

Met	Glu	Phe	Gly	Leu	Ser	Trp	Leu	Phe	Leu	Val	Ala	Ile	Leu	Lys	Gly
1				5					10					15	
Val	Gln	Cys	Glu	Val	Gln	Leu	Leu	Glu	Ser	Gly	Gly	Gly	Leu	Val	Gln
			20					25					30		
Pro	Gly	Gly	Ser	Leu	Arg	Leu	Ser	Cys	Ala	Ala	Ser	Gly	Phe	Thr	Phe
	35					40						45			
Ser	Ser	Tyr	Ala	Met	Ser	Trp	Val	Arg	Gln	Ala	Pro	Gly	Lys	Gly	Leu
	50				55					60					
Glu	Trp	Val	Ser	Gly	Ile	Thr	Gly	Ser	Gly	Gly	Ser	Thr	Tyr	Tyr	Ala
65				70					75						80
Asp	Ser	Val	Lys	Gly	Arg	Phe	Thr	Ile	Ser	Arg	Asp	Asn	Ser	Lys	Asn
			85					90					95		
Thr	Leu	Tyr	Leu	Gln	Met	Asn	Ser	Leu	Arg	Ala	Glu	Asp	Thr	Ala	Val
	100						105						110		
Tyr	Tyr	Cys	Ala	Lys	Asp	Pro	Gly	Thr	Thr	Val	Ile	Met	Ser	Trp	Phe
	115					120						125			
Asp	Pro	Trp	Gly	Gln	Gly	Thr	Leu	Val	Thr	Val	Ser	Ser	Ala	Ser	Thr
	130				135					140					
Lys	Gly	Pro	Ser	Val	Phe	Pro	Leu	Ala	Pro	Cys	Ser	Arg	Ser	Thr	Ser
145				150						155					160
Glu	Ser	Thr	Ala	Ala	Leu	Gly	Cys	Leu	Val	Lys	Asp	Tyr	Phe	Pro	Glu
			165					170						175	
Pro	Val	Thr	Val	Ser	Trp	Asn	Ser	Gly	Ala	Leu	Thr	Ser	Gly	Val	His
		180						185					190		
Thr	Phe	Pro	Ala	Val	Leu	Gln	Ser	Ser	Gly	Leu	Tyr	Ser	Leu	Ser	Ser
	195					200						205			
Val	Val	Thr	Val	Pro	Ser	Ser	Asn	Phe	Gly	Thr	Gln	Thr	Tyr	Thr	Cys
	210					215					220				
Asn	Val	Asp	His	Lys	Pro	Ser	Asn	Thr	Lys	Val	Asp	Lys	Thr	Val	Glu
225					230					235					240
Arg	Lys	Cys	Cys	Val	Glu	Cys	Pro	Pro	Cys	Pro	Ala	Pro	Pro	Val	Ala
			245						250						255

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Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met
 260 265 270
 Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His
 275 280 285
 Glu Asp Pro Glu Val Gln Phe Asn Trp Tyr Val Asp Gly Val Glu Val
 290 295 300
 His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Phe Asn Ser Thr Phe
 305 310 315 320
 Arg Val Val Ser Val Leu Thr Val Val His Gln Asp Trp Leu Asn Gly
 325 330 335
 Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Gly Leu Pro Ala Pro Ile
 340 345 350
 Glu Lys Thr Ile Ser Lys Thr Lys Gly Gln Pro Arg Glu Pro Gln Val
 355 360 365
 Tyr Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser
 370 375 380
 Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu
 385 390 395 400
 Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro
 405 410 415
 Met Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val
 420 425 430
 Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met
 435 440 445
 His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser
 450 455 460
 Pro Gly Lys
 465

<210> SEQ ID NO 10
 <211> LENGTH: 242
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 10

Met Asp Met Arg Val Pro Ala Gln Leu Leu Gly Leu Leu Leu Trp
 1 5 10 15
 Leu Arg Gly Ala Arg Cys Asp Ile Val Met Thr Gln Thr Pro Leu Ser
 20 25 30
 Leu Pro Val Thr Pro Gly Glu Pro Ala Ser Ile Ser Cys Arg Ser Ser
 35 40 45
 Gln Ser Leu Leu Asn Ser Val Asp Gly Ser Thr Asn Leu Asp Trp Tyr
 50 55 60
 Leu Gln Lys Pro Gly Gln Ser Pro Gln Leu Leu Ile Tyr Thr Leu Ser
 65 70 75 80
 Tyr Arg Ala Ser Gly Val Pro Asp Arg Phe Ser Gly Ser Gly Ser Gly
 85 90 95
 Thr Asp Phe Thr Leu Lys Ile Ser Arg Val Glu Ala Glu Asp Val Gly
 100 105 110
 Val Tyr Tyr Cys Met Gln Arg Ile Glu Phe Pro Leu Thr Phe Gly Gly
 115 120 125
 Gly Thr Lys Val Glu Ile Lys Arg Thr Val Ala Ala Pro Ser Val Phe
 130 135 140
 Ile Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser Gly Thr Ala Ser Val

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145	150	155	160
Val Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu Ala Lys Val Gln Trp			
	165	170	175
Lys Val Asp Asn Ala Leu Gln Ser Gly Asn Ser Gln Glu Ser Val Thr			
	180	185	190
Glu Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu Ser Ser Thr Leu Thr			
	195	200	205
Leu Ser Lys Ala Asp Tyr Glu Lys His Lys Val Tyr Ala Cys Glu Val			
	210	215	220
Thr His Gln Gly Leu Ser Ser Pro Val Thr Lys Ser Phe Asn Arg Gly			
225	230	235	240
Glu Cys			
<210> SEQ ID NO 11			
<211> LENGTH: 465			
<212> TYPE: PRT			
<213> ORGANISM: Homo sapiens			
<400> SEQUENCE: 11			
Met Glu Leu Gly Leu Cys Trp Val Phe Leu Val Ala Ile Leu Glu Gly			
1	5	10	15
Val Gln Cys Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln			
	20	25	30
Pro Gly Gly Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe			
	35	40	45
Ser Ser Tyr Ser Met Asn Trp Val Arg Gln Ala Pro Gly Lys Gly Leu			
	50	55	60
Glu Trp Val Ser Tyr Ile Ser Ser Ser Gly Ser Ser Ile Tyr Tyr Ala			
	65	70	75
Asp Ser Val Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Asn			
	85	90	95
Ser Leu Tyr Leu Gln Met Asn Ser Leu Arg Asp Glu Asp Thr Ala Val			
	100	105	110
Tyr Tyr Cys Ala Arg Glu Arg Tyr Tyr Gly Asp Thr Pro Phe Asp Tyr			
	115	120	125
Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly			
	130	135	140
Pro Ser Val Phe Pro Leu Ala Pro Cys Ser Arg Ser Thr Ser Glu Ser			
	145	150	155
Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val			
	165	170	175
Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe			
	180	185	190
Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val			
	195	200	205
Thr Val Pro Ser Ser Asn Phe Gly Thr Gln Thr Tyr Thr Cys Asn Val			
	210	215	220
Asp His Lys Pro Ser Asn Thr Lys Val Asp Lys Thr Val Glu Arg Lys			
	225	230	235
Cys Cys Val Glu Cys Pro Pro Cys Pro Ala Pro Pro Val Ala Gly Pro			
	245	250	255
Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser			
	260	265	270
Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His Glu Asp			

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275	280	285
Pro Glu Val Gln Phe Asn Trp	Tyr Val Asp Gly Val Glu Val His Asn	
290	295	300
Ala Lys Thr Lys Pro Arg Glu Glu Gln Phe Asn Ser Thr Phe Arg Val		
305	310	315 320
Val Ser Val Leu Thr Val Val His Gln Asp Trp Leu Asn Gly Lys Glu		
	325	330 335
Tyr Lys Cys Lys Val Ser Asn Lys Gly Leu Pro Ala Pro Ile Glu Lys		
	340	345 350
Thr Ile Ser Lys Thr Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr		
	355	360 365
Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser Leu Thr		
	370	375 380
Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu		
385	390	395 400
Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Met Leu		
	405	410 415
Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val Asp Lys		
	420	425 430
Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His Glu		
	435	440 445
Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly		
	450	455 460

Lys
465

<210> SEQ ID NO 12
 <211> LENGTH: 233
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 12

Met Ser Pro Ser Gln Leu Ile Gly Phe Leu Leu Leu Trp Val Pro Ala		
1	5	10 15
Ser Arg Gly Glu Ile Val Leu Thr Gln Ser Pro Asp Phe Gln Ser Val		
	20	25 30
Thr Pro Lys Glu Lys Val Thr Ile Thr Cys Arg Ala Ser Gln Ser Ile		
	35	40 45
Gly Ser Ser Leu His Trp Tyr Gln Gln Lys Pro Asp Gln Ser Pro Lys		
	50	55 60
Leu Leu Ile Lys Tyr Ala Ser Gln Ser Phe Ser Gly Val Pro Ser Arg		
65	70	75 80
Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Asn Ser		
	85	90 95
Leu Glu Ala Glu Asp Ala Ala Ala Tyr Tyr Cys His Gln Ser Ser Ser		
	100	105 110
Leu Pro Leu Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys Arg Thr		
	115	120 125
Val Ala Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu		
	130	135 140
Lys Ser Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro		
145	150	155 160
Arg Glu Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly		
	165	170 175

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Asn Ser Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr
 180 185 190

Ser Leu Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His
 195 200 205

Lys Val Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val
 210 215 220

Thr Lys Ser Phe Asn Arg Gly Glu Cys
 225 230

<210> SEQ ID NO 13
 <211> LENGTH: 463
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 13

Met Gly Ser Thr Ala Ile Leu Ala Leu Leu Ala Val Leu Gln Gly
 1 5 10 15

Val Cys Ala Glu Val Gln Leu Met Gln Ser Gly Ala Glu Val Lys Lys
 20 25 30

Pro Gly Glu Ser Leu Lys Ile Ser Cys Lys Gly Ser Gly Tyr Ser Phe
 35 40 45

Ser Phe His Trp Ile Ala Trp Val Arg Gln Met Pro Gly Lys Gly Leu
 50 55 60

Glu Trp Met Gly Ile Ile His Pro Gly Ala Ser Asp Thr Arg Tyr Ser
 65 70 75 80

Pro Ser Phe Gln Gly Gln Val Thr Ile Ser Ala Asp Asn Ser Asn Ser
 85 90 95

Ala Thr Tyr Leu Gln Trp Ser Ser Leu Lys Ala Ser Asp Thr Ala Met
 100 105 110

Tyr Phe Cys Ala Arg Gln Arg Glu Leu Asp Tyr Phe Asp Tyr Trp Gly
 115 120 125

Gln Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro Ser
 130 135 140

Val Phe Pro Leu Ala Pro Cys Ser Arg Ser Thr Ser Glu Ser Thr Ala
 145 150 155 160

Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val
 165 170 175

Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala
 180 185 190

Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val
 195 200 205

Pro Ser Ser Asn Phe Gly Thr Gln Thr Tyr Thr Cys Asn Val Asp His
 210 215 220

Lys Pro Ser Asn Thr Lys Val Asp Lys Thr Val Glu Arg Lys Cys Cys
 225 230 235 240

Val Glu Cys Pro Pro Cys Pro Ala Pro Pro Val Ala Gly Pro Ser Val
 245 250 255

Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser Arg Thr
 260 265 270

Pro Glu Val Thr Cys Val Val Val Asp Val Ser His Glu Asp Pro Glu
 275 280 285

Val Gln Phe Asn Trp Tyr Val Asp Gly Val Glu Val His Asn Ala Lys
 290 295 300

Thr Lys Pro Arg Glu Glu Gln Phe Asn Ser Thr Phe Arg Val Val Ser
 305 310 315 320

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Val Leu Thr Val Val His Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys
 325 330 335
 Cys Lys Val Ser Asn Lys Gly Leu Pro Ala Pro Ile Glu Lys Thr Ile
 340 345 350
 Ser Lys Thr Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro
 355 360 365
 Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser Leu Thr Cys Leu
 370 375 380
 Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn
 385 390 395 400
 Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Met Leu Asp Ser
 405 410 415
 Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val Asp Lys Ser Arg
 420 425 430
 Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu
 435 440 445
 His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly Lys
 450 455 460

<210> SEQ ID NO 14
 <211> LENGTH: 236
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 14

Met Asp Met Arg Val Pro Ala Gln Leu Leu Gly Leu Leu Leu Trp
 1 5 10 15
 Leu Arg Gly Ala Arg Cys Asp Ile Gln Met Thr Gln Ser Pro Ser Ser
 20 25 30
 Leu Ser Ala Ser Val Gly Asp Arg Val Thr Ile Thr Cys Lys Ala Ser
 35 40 45
 Gln Asp Ile Asn Lys Tyr Val Ala Trp Tyr Gln Gln Lys Pro Gly Lys
 50 55 60
 Ala Pro Lys Leu Leu Ile Tyr Tyr Thr Ser Trp Leu Gln Pro Gly Val
 65 70 75 80
 Pro Ser Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Phe Thr
 85 90 95
 Ile Ser Ser Leu Gln Pro Glu Asp Ile Ala Thr Tyr Tyr Cys Leu Gln
 100 105 110
 Tyr Asp Asn Leu Leu Tyr Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile
 115 120 125
 Lys Arg Thr Val Ala Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp
 130 135 140
 Glu Gln Leu Lys Ser Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn
 145 150 155 160
 Phe Tyr Pro Arg Glu Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu
 165 170 175
 Gln Ser Gly Asn Ser Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp
 180 185 190
 Ser Thr Tyr Ser Leu Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr
 195 200 205
 Glu Lys His Lys Val Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser
 210 215 220
 Ser Pro Val Thr Lys Ser Phe Asn Arg Gly Glu Cys

225						230							235							
<210> SEQ ID NO 15																				
<211> LENGTH: 467																				
<212> TYPE: PRT																				
<213> ORGANISM: Homo sapiens																				
<400> SEQUENCE: 15																				
Met 1	Asp	Met	Arg	Val 5	Pro	Ala	Gln	Leu	Leu 10	Gly	Leu	Leu	Leu	Trp 15						
Leu	Arg	Gly	Ala 20	Arg	Cys	Glu	Val	Gln 25	Leu	Val	Glu	Ser	Gly 30	Gly						
Leu	Val	Gln 35	Pro	Gly	Gly	Ser	Leu 40	Arg	Leu	Ser	Cys	Ala 45	Ala	Ser	Gly					
Phe 50	Thr	Phe	Ser	Arg	Tyr 55	Trp	Met	Asn	Trp	Val	Arg 60	Gln	Ala	Pro	Gly					
Lys 65	Gly	Leu	Glu	Trp 70	Val	Ala	Gln	Ile	Arg 75	Leu	Lys	Ser	Asp	Asn 80	Tyr					
Ala	Thr	His	Tyr 85	Ala	Glu	Ser	Val	Lys 90	Gly	Arg	Phe	Thr	Ile 95	Ser	Arg					
Asp	Asn	Ala	Lys 100	Asn	Ser	Leu	Tyr 105	Leu	Gln	Met	Asn	Ser 110	Leu	Arg	Ala					
Glu	Asp	Thr 115	Ala	Val	Tyr	Tyr	Cys 120	Thr	Glu	Gly	Leu 125	Asp	Tyr	Trp	Gly					
Gln 130	Gly	Thr	Thr	Val	Thr 135	Val	Ser	Ser	Ala	Ser	Thr 140	Lys	Gly	Pro	Ser					
Val 145	Phe	Pro	Leu	Ala 150	Pro	Ser	Ser	Lys	Ser 155	Thr	Ser	Gly	Gly	Thr	Ala					
Ala	Leu	Gly	Cys 165	Leu	Val	Lys	Asp	Tyr 170	Phe	Pro	Glu	Pro	Val 175	Thr	Val					
Ser	Trp	Asn 180	Ser	Gly	Ala	Leu	Thr 185	Ser	Gly	Val	His 190	Thr	Phe	Pro	Ala					
Val 195	Leu	Gln	Ser	Ser	Gly	Leu	Tyr 200	Ser	Leu	Ser	Ser 205	Val	Val	Thr	Val					
Pro 210	Ser	Ser	Ser	Leu	Gly 215	Thr	Gln	Thr	Tyr	Ile	Cys 220	Asn	Val	Asn	His					
Lys 225	Pro	Ser	Asn 230	Thr	Lys	Val	Asp	Lys	Lys 235	Val	Glu	Pro	Lys	Ser	Cys					
Asp	Lys	Thr	His 245	Thr	Cys	Pro	Pro	Cys	Pro 250	Ala	Pro	Glu	Leu 255	Leu	Gly					
Gly	Pro	Ser	Val 260	Phe	Leu	Phe	Pro	Pro 265	Lys	Pro	Lys	Asp	Thr 270	Leu	Met					
Ile	Ser	Arg 275	Thr	Pro	Glu	Val	Thr 280	Cys	Val	Val	Val 285	Asp	Val	Ser	His					
Glu 290	Asp	Pro	Glu	Val	Lys	Phe 295	Asn	Trp	Tyr	Val	Asp 300	Gly	Val	Glu	Val					
His 305	Asn	Ala	Lys	Thr 310	Lys	Pro	Arg	Glu	Glu 315	Gln	Tyr	Asn	Ser	Thr	Tyr					
Arg	Val	Val	Ser 325	Val	Leu	Thr	Val	Leu 330	His	Gln	Asp	Trp	Leu 335	Asn	Gly					
Lys	Glu	Tyr 340	Lys	Cys	Lys	Val	Ser 345	Asn	Lys	Ala	Leu	Pro 350	Ala	Pro	Ile					
Glu 355	Lys	Thr	Ile	Ser	Lys	Ala 360	Lys	Gly	Gln	Pro	Arg 365	Glu	Pro	Gln	Val					

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Tyr Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser
 370 375 380
 Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu
 385 390 395 400
 Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro
 405 410 415
 Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val
 420 425 430
 Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met
 435 440 445
 His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser
 450 455 460
 Pro Gly Lys
 465

<210> SEQ ID NO 16
 <211> LENGTH: 236
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 16

Met Asp Met Arg Val Leu Ala Gln Leu Leu Gly Leu Leu Leu Leu Cys
 1 5 10 15
 Phe Pro Gly Ala Arg Cys Asp Ile Gln Met Thr Gln Ser Pro Ser Ser
 20 25 30
 Leu Ser Ala Ser Val Gly Asp Arg Val Thr Ile Thr Cys Arg Ala Ser
 35 40 45
 Gln Gly Ile Ser Asn Trp Leu Ala Trp Tyr Gln Gln Lys Pro Glu Lys
 50 55 60
 Ala Pro Lys Ser Leu Ile Tyr Ala Ala Ser Ser Leu Gln Ser Gly Val
 65 70 75 80
 Pro Ser Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr
 85 90 95
 Ile Ser Ser Leu Gln Pro Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln
 100 105 110
 Tyr Asp Ser Tyr Pro Arg Thr Phe Gly Gln Gly Thr Lys Val Glu Ile
 115 120 125
 Lys Arg Thr Val Ala Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp
 130 135 140
 Glu Gln Leu Lys Ser Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn
 145 150 155 160
 Phe Tyr Pro Arg Glu Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu
 165 170 175
 Gln Ser Gly Asn Ser Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp
 180 185 190
 Ser Thr Tyr Ser Leu Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr
 195 200 205
 Glu Lys His Lys Val Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser
 210 215 220
 Ser Pro Val Thr Lys Ser Phe Asn Arg Gly Glu Cys
 225 230 235

<210> SEQ ID NO 17
 <211> LENGTH: 466
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

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<400> SEQUENCE: 17

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Met Glu Leu Gly Leu Asn Trp Val Phe Leu Val Ala Ile Leu Glu Gly
1      5      10      15
Val His Cys Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln
20      25      30
Pro Gly Gly Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe
35      40      45
Ser Ser Tyr Trp Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu
50      55      60
Glu Trp Val Ala Tyr Ile Lys Gln Asp Gly Asn Glu Lys Tyr Tyr Val
65      70      75      80
Asp Ser Val Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Asn
85      90      95
Ser Leu Tyr Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val
100     105     110
Tyr Tyr Cys Ala Arg Glu Gly Ile Leu Trp Phe Gly Asp Leu Pro Thr
115     120     125
Phe Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys
130     135     140
Gly Pro Ser Val Phe Pro Leu Ala Pro Cys Ser Arg Ser Thr Ser Glu
145     150     155     160
Ser Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro
165     170     175
Val Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr
180     185     190
Phe Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val
195     200     205
Val Thr Val Pro Ser Ser Asn Phe Gly Thr Gln Thr Tyr Thr Cys Asn
210     215     220
Val Asp His Lys Pro Ser Asn Thr Lys Val Asp Lys Thr Val Glu Arg
225     230     235     240
Lys Cys Cys Val Glu Cys Pro Pro Cys Pro Ala Pro Pro Val Ala Gly
245     250     255
Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile
260     265     270
Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His Glu
275     280     285
Asp Pro Glu Val Gln Phe Asn Trp Tyr Val Asp Gly Val Glu Val His
290     295     300
Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Phe Asn Ser Thr Phe Arg
305     310     315     320
Val Val Ser Val Leu Thr Val Val His Gln Asp Trp Leu Asn Gly Lys
325     330     335
Glu Tyr Lys Cys Lys Val Ser Asn Lys Gly Leu Pro Ala Pro Ile Glu
340     345     350
Lys Thr Ile Ser Lys Thr Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr
355     360     365
Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser Leu
370     375     380
Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp
385     390     395     400
Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Met

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405	410	415
Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val Asp		
420	425	430
Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His		
435	440	445
Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro		
450	455	460
Gly Lys		
465		

<210> SEQ ID NO 18
 <211> LENGTH: 241
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 18

Met Asp Met Arg Leu Pro Ala Gln Leu Leu Gly Leu Leu Met Leu Trp		
1	5	10
Val Pro Gly Ser Ser Gly Asp Val Leu Met Thr Gln Ser Pro Leu Ser		
20	25	30
Leu Pro Val Thr Leu Gly Gln Pro Ala Ser Ile Ser Cys Arg Ser Ser		
35	40	45
Gln Ser Ile Val His Ser Asn Gly Asn Thr Tyr Leu Glu Trp Tyr Leu		
50	55	60
Gln Arg Pro Gly Gln Ser Pro Lys Leu Leu Ile Tyr Lys Val Ser Asn		
65	70	75
Arg Phe Ser Gly Val Pro Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr		
85	90	95
Asp Phe Thr Leu Lys Ile Ser Arg Val Glu Ala Glu Asp Val Gly Val		
100	105	110
Tyr Tyr Cys Phe Gln Gly Ser His Val Pro Leu Thr Phe Gly Ala Gly		
115	120	125
Thr Lys Leu Glu Ile Lys Arg Thr Val Ala Ala Pro Ser Val Phe Ile		
130	135	140
Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser Gly Thr Ala Ser Val Val		
145	150	155
Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu Ala Lys Val Gln Trp Lys		
165	170	175
Val Asp Asn Ala Leu Gln Ser Gly Asn Ser Gln Glu Ser Val Thr Glu		
180	185	190
Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu Ser Ser Thr Leu Thr Leu		
195	200	205
Ser Lys Ala Asp Tyr Glu Lys His Lys Val Tyr Ala Cys Glu Val Thr		
210	215	220
His Gln Gly Leu Ser Ser Pro Val Thr Lys Ser Phe Asn Arg Gly Glu		
225	230	235
240		
Cys		

<210> SEQ ID NO 19
 <211> LENGTH: 470
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 19

Met Asp Thr Leu Cys Ser Thr Leu Leu Leu Leu Thr Ile Pro Ser Trp		
1	5	10
15		

Val	Leu	Ser	Gln	Val	Thr	Leu	Lys	Glu	Ser	Gly	Pro	Ala	Leu	Val	Lys	
			20				25							30		
Pro	Thr	Gln	Thr	Leu	Thr	Leu	Thr	Cys	Thr	Phe	Ser	Gly	Phe	Ser	Leu	
			35				40							45		
Arg	Thr	Ser	Gly	Met	Gly	Val	Gly	Trp	Ile	Arg	Gln	Pro	Pro	Gly	Lys	
			50				55							60		
Ala	Leu	Glu	Trp	Leu	Ala	His	Ile	Trp	Trp	Asp	Asp	Asp	Lys	Ser	Tyr	
			65				70							75		
Asn	Pro	Ser	Leu	Lys	Ser	Gln	Leu	Thr	Ile	Ser	Lys	Asp	Thr	Ser	Lys	
			85							90						
Asn	Gln	Val	Val	Leu	Thr	Met	Thr	Asn	Met	Asp	Pro	Val	Asp	Thr	Ala	
			100							105						
Thr	Tyr	Tyr	Cys	Ala	Arg	Arg	Asn	Tyr	Tyr	Tyr	Asp	Asp	Tyr	Phe	Ala	
			115							120						
Tyr	Trp	Gly	Gln	Gly	Thr	Leu	Val	Thr	Val	Ser	Ser	Ala	Ser	Thr	Lys	
			130													
Gly	Pro	Ser	Val	Phe	Pro	Leu	Ala	Pro	Ser	Ser	Lys	Ser	Thr	Ser	Gly	
			145													
Gly	Thr	Ala	Ala	Leu	Gly	Cys	Leu	Val	Lys	Asp	Tyr	Phe	Pro	Glu	Pro	
			165													
Val	Thr	Val	Ser	Trp	Asn	Ser	Gly	Ala	Leu	Thr	Ser	Gly	Val	His	Thr	
			180													
Phe	Pro	Ala	Val	Leu	Gln	Ser	Ser	Gly	Leu	Tyr	Ser	Leu	Ser	Ser	Val	
			195													
Val	Thr	Val	Pro	Ser	Ser	Ser	Leu	Gly	Thr	Gln	Thr	Tyr	Ile	Cys	Asn	
			210													
Val	Asn	His	Lys	Pro	Ser	Asn	Thr	Lys	Val	Asp	Lys	Lys	Val	Glu	Pro	
			225													
Lys	Ser	Cys	Asp	Lys	Thr	His	Thr	Cys	Pro	Pro	Cys	Pro	Ala	Pro	Glu	
			245													
Leu	Leu	Gly	Gly	Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	
			260													
Thr	Leu	Met	Ile	Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	
			275													
Val	Ser	His	Glu	Asp	Pro	Glu	Val	Lys	Phe	Asn	Trp	Tyr	Val	Asp	Gly	
			290													
Val	Glu	Val	His	Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu	Gln	Tyr	Asn	
			305													
Ser	Thr	Tyr	Arg	Val	Val	Ser	Val	Leu	Thr	Val	Leu	His	Gln	Asp	Trp	
			325													
Leu	Asn	Gly	Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	Ala	Leu	Pro	
			340													
Ala	Pro	Ile	Glu	Lys	Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	
			355													
Pro	Gln	Val	Tyr	Thr	Leu	Pro	Pro	Ser	Arg	Asp	Glu	Leu	Thr	Lys	Asn	
			370													
Gln	Val	Ser	Leu	Thr	Cys	Leu	Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	
			385													
Ala	Val	Glu	Trp	Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	
			405													
Thr	Pro	Pro	Val	Leu	Asp	Ser	Asp	Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Lys	
			420													

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Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys
 435 440 445

Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu
 450 455 460

Ser Leu Ser Pro Gly Lys
 465 470

<210> SEQ ID NO 20
 <211> LENGTH: 235
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 20

Met Glu Thr Pro Ala Gln Leu Leu Phe Leu Leu Leu Trp Leu Pro
 1 5 10 15

Asp Thr Thr Gly Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser
 20 25 30

Leu Ser Pro Gly Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser
 35 40 45

Val Ser Ser Ser Tyr Phe Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala
 50 55 60

Pro Arg Leu Leu Ile Tyr Gly Thr Ser Ser Arg Ala Thr Gly Ile Pro
 65 70 75 80

Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Val
 85 90 95

Ser Arg Leu Glu Pro Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr
 100 105 110

Asp Arg Ser Pro Arg Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys
 115 120 125

Arg Thr Val Ala Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu
 130 135 140

Gln Leu Lys Ser Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe
 145 150 155 160

Tyr Pro Arg Glu Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln
 165 170 175

Ser Gly Asn Ser Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser
 180 185 190

Thr Tyr Ser Leu Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu
 195 200 205

Lys His Lys Val Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser
 210 215 220

Pro Val Thr Lys Ser Phe Asn Arg Gly Glu Cys
 225 230 235

<210> SEQ ID NO 21
 <211> LENGTH: 463
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 21

Met Asp Trp Thr Trp Arg Ile Leu Phe Leu Val Ala Ala Ala Thr Gly
 1 5 10 15

Ala His Ser Gln Val Gln Leu Val Gln Ser Gly Ala Ala Val Lys Lys
 20 25 30

Pro Gly Ala Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Thr Phe
 35 40 45

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Thr	Gly	Tyr	Tyr	Ile	His	Trp	Val	Arg	Gln	Ala	Pro	Gly	Gln	Gly	Leu
50						55					60				
Glu	Trp	Met	Gly	Trp	Ile	Asn	Pro	Asn	Ser	Gly	Gly	Thr	Asn	Tyr	Ala
65					70					75					80
Gln	Lys	Phe	Gln	Gly	Arg	Val	Thr	Met	Thr	Arg	Asp	Thr	Ser	Ile	Ser
			85						90					95	
Thr	Ala	Ser	Met	Glu	Leu	Ser	Arg	Leu	Arg	Ser	Asp	Asp	Thr	Ala	Val
			100					105					110		
Tyr	Phe	Cys	Ala	Arg	Asp	Arg	Trp	Leu	Asp	Ala	Phe	Asp	Ile	Trp	Gly
		115					120					125			
Gln	Gly	Thr	Met	Val	Thr	Val	Ser	Ser	Ala	Ser	Thr	Lys	Gly	Pro	Ser
	130					135						140			
Val	Phe	Pro	Leu	Ala	Pro	Cys	Ser	Arg	Ser	Thr	Ser	Glu	Ser	Thr	Ala
145					150					155					160
Ala	Leu	Gly	Cys	Leu	Val	Lys	Asp	Tyr	Phe	Pro	Glu	Pro	Val	Thr	Val
			165						170					175	
Ser	Trp	Asn	Ser	Gly	Ala	Leu	Thr	Ser	Gly	Val	His	Thr	Phe	Pro	Ala
		180						185					190		
Val	Leu	Gln	Ser	Ser	Gly	Leu	Tyr	Ser	Leu	Ser	Ser	Val	Val	Thr	Val
	195						200					205			
Pro	Ser	Ser	Asn	Phe	Gly	Thr	Gln	Thr	Tyr	Thr	Cys	Asn	Val	Asp	His
	210					215					220				
Lys	Pro	Ser	Asn	Thr	Lys	Val	Asp	Lys	Thr	Val	Glu	Arg	Lys	Cys	Cys
225					230					235					240
Val	Glu	Cys	Pro	Pro	Cys	Pro	Ala	Pro	Pro	Val	Ala	Gly	Pro	Ser	Val
			245						250					255	
Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg	Thr
		260						265					270		
Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	His	Glu	Asp	Pro	Glu
	275						280					285			
Val	Gln	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys
	290					295					300				
Thr	Lys	Pro	Arg	Glu	Glu	Gln	Phe	Asn	Ser	Thr	Phe	Arg	Val	Val	Ser
305					310					315					320
Val	Leu	Thr	Val	Val	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys
			325					330						335	
Cys	Lys	Val	Ser	Asn	Lys	Gly	Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile
		340						345					350		
Ser	Lys	Thr	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro
		355					360					365			
Pro	Ser	Arg	Glu	Glu	Met	Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Leu
	370					375					380				
Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Asn
385					390					395					400
Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Met	Leu	Asp	Ser
			405					410						415	
Asp	Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Lys	Leu	Thr	Val	Asp	Lys	Ser	Arg
			420					425				430			
Trp	Gln	Gln	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val	Met	His	Glu	Ala	Leu
	435						440					445			
His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser	Leu	Ser	Pro	Gly	Lys	
	450					455					460				

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<210> SEQ ID NO 22
 <211> LENGTH: 235
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 22

```

Met Glu Thr Pro Ala Gln Leu Leu Phe Leu Leu Leu Leu Trp Leu Pro
 1          5          10          15

Asp Thr Thr Gly Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser
          20          25          30

Leu Ser Pro Gly Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser
          35          40          45

Val Ser Ser Gly Tyr Leu Thr Trp Tyr Gln Gln Lys Pro Gly Gln Ala
 50          55          60

Pro Arg Leu Leu Ile Tyr Gly Ala Ser Ser Arg Ala Thr Gly Ile Pro
65          70          75          80

Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile
          85          90          95

Ser Arg Leu Glu Pro Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr
          100          105          110

Gly Asn Ser Leu Ser Arg Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys
          115          120          125

Arg Thr Val Ala Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu
          130          135          140

Gln Leu Lys Ser Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe
          145          150          155          160

Tyr Pro Arg Glu Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln
          165          170          175

Ser Gly Asn Ser Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser
          180          185          190

Thr Tyr Ser Leu Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu
          195          200          205

Lys His Lys Val Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser
          210          215          220

Pro Val Thr Lys Ser Phe Asn Arg Gly Glu Cys
          225          230          235

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<210> SEQ ID NO 23
 <211> LENGTH: 474
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 23

```

Met Asp Met Arg Val Pro Ala Gln Leu Leu Gly Leu Leu Leu Trp
 1          5          10          15

Leu Arg Gly Ala Arg Cys Gln Val Gln Leu Val Glu Ser Gly Gly Gly
          20          25          30

Val Val Gln Pro Gly Arg Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly
          35          40          45

Phe Thr Phe Ser Ser Tyr Gly Met His Trp Val Arg Gln Ala Pro Gly
          50          55          60

Lys Gly Leu Glu Trp Val Ala Val Ile Trp Tyr Asp Gly Ser Asn Lys
          65          70          75          80

Tyr Tyr Ala Asp Ser Val Lys Gly Arg Phe Ile Ile Ser Arg Asp Lys
          85          90          95

Ser Lys Asn Thr Leu Tyr Leu Gln Met Asn Ser Leu Arg Ala Glu Asp

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100					105					110					
Thr	Ala	Val	Tyr	Tyr	Cys	Ala	Arg	Ala	Gly	Gly	Ile	Ala	Ala	Ala	Gly
	115						120					125			
Leu	Tyr	Tyr	Tyr	Tyr	Gly	Met	Asp	Val	Trp	Gly	Gln	Gly	Thr	Thr	Val
	130					135					140				
Thr	Val	Ser	Ser	Ala	Ser	Thr	Lys	Gly	Pro	Ser	Val	Phe	Pro	Leu	Ala
	145					150					155				160
Pro	Cys	Ser	Arg	Ser	Thr	Ser	Glu	Ser	Thr	Ala	Ala	Leu	Gly	Cys	Leu
			165						170					175	
Val	Lys	Asp	Tyr	Phe	Pro	Glu	Pro	Val	Thr	Val	Ser	Trp	Asn	Ser	Gly
		180						185					190		
Ala	Leu	Thr	Ser	Gly	Val	His	Thr	Phe	Pro	Ala	Val	Leu	Gln	Ser	Ser
	195						200					205			
Gly	Leu	Tyr	Ser	Leu	Ser	Ser	Val	Val	Thr	Val	Pro	Ser	Ser	Asn	Phe
	210					215					220				
Gly	Thr	Gln	Thr	Tyr	Thr	Cys	Asn	Val	Asp	His	Lys	Pro	Ser	Asn	Thr
	225					230					235				240
Lys	Val	Asp	Lys	Thr	Val	Glu	Arg	Lys	Cys	Cys	Val	Glu	Cys	Pro	Pro
			245						250					255	
Cys	Pro	Ala	Pro	Pro	Val	Ala	Gly	Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro
		260						265					270		
Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys
		275					280					285			
Val	Val	Val	Asp	Val	Ser	His	Glu	Asp	Pro	Glu	Val	Gln	Phe	Asn	Trp
	290					295						300			
Tyr	Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu
	305					310					315				320
Glu	Gln	Phe	Asn	Ser	Thr	Phe	Arg	Val	Val	Ser	Val	Leu	Thr	Val	Val
			325					330						335	
His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn
		340					345						350		
Lys	Gly	Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile	Ser	Lys	Thr	Lys	Gly
	355						360					365			
Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro	Pro	Ser	Arg	Glu	Glu
	370					375					380				
Met	Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Leu	Val	Lys	Gly	Phe	Tyr
	385					390					395				400
Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn
			405					410					415		
Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Met	Leu	Asp	Ser	Asp	Gly	Ser	Phe	Phe
		420					425						430		
Leu	Tyr	Ser	Lys	Leu	Thr	Val	Asp	Lys	Ser	Arg	Trp	Gln	Gln	Gly	Asn
	435					440						445			
Val	Phe	Ser	Cys	Ser	Val	Met	His	Glu	Ala	Leu	His	Asn	His	Tyr	Thr
	450					455					460				
Gln	Lys	Ser	Leu	Ser	Leu	Ser	Pro	Gly	Lys						
	465					470									

<210> SEQ ID NO 24

<211> LENGTH: 236

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 24

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Met Asp Met Arg Val Pro Ala Gln Leu Leu Gly Leu Leu Leu Trp
1      5      10      15
Leu Arg Gly Ala Arg Cys Ser Ser Glu Leu Thr Gln Asp Pro Thr Val
20      25      30
Ser Val Ala Leu Gly Gln Thr Val Lys Ile Thr Cys Gln Gly Asp Ser
35      40      45
Leu Arg Ser Phe Tyr Ala Ser Trp Tyr Gln Gln Lys Pro Gly Gln Ala
50      55      60
Pro Val Leu Val Phe Tyr Gly Lys Asn Asn Arg Pro Ser Gly Ile Pro
65      70      75      80
Asp Arg Phe Ser Gly Ser Ser Ser Gly Asn Thr Ala Ser Leu Thr Ile
85      90      95
Thr Gly Ala Gln Ala Glu Asp Glu Ala Asp Tyr Tyr Cys Asn Ser Arg
100     105     110
Asp Ser Ser Val Tyr His Leu Val Leu Gly Gly Gly Thr Lys Leu Thr
115     120     125
Val Leu Gly Gln Pro Lys Ala Asn Pro Thr Val Thr Leu Phe Pro Pro
130     135     140
Ser Ser Glu Glu Leu Gln Ala Asn Lys Ala Thr Leu Val Cys Leu Ile
145     150     155     160
Ser Asp Phe Tyr Pro Gly Ala Val Thr Val Ala Trp Lys Ala Asp Gly
165     170     175
Ser Pro Val Lys Ala Gly Val Glu Thr Thr Lys Pro Ser Lys Gln Ser
180     185     190
Asn Asn Lys Tyr Ala Ala Ser Ser Tyr Leu Ser Leu Thr Pro Glu Gln
195     200     205
Trp Lys Ser His Arg Ser Tyr Ser Cys Gln Val Thr His Glu Gly Ser
210     215     220
Thr Val Glu Lys Thr Val Ala Pro Thr Glu Cys Ser
225     230     235

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<210> SEQ ID NO 25

<211> LENGTH: 478

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 25

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Met Asp Met Arg Val Pro Ala Gln Leu Leu Gly Leu Leu Leu Trp
1      5      10      15
Leu Arg Gly Ala Arg Cys Gln Val Gln Leu Val Gln Ser Gly Ala Glu
20      25      30
Val Lys Lys Pro Gly Ala Ser Val Lys Val Ser Cys Lys Ala Ser Gly
35      40      45
Tyr Thr Phe Thr Gly Tyr Tyr Met His Trp Val Arg Gln Ala Pro Gly
50      55      60
Gln Gly Leu Glu Trp Met Gly Trp Ile Asn Pro Asn Ser Gly Gly Thr
65      70      75      80
Asn Tyr Ala Gln Lys Phe Gln Gly Arg Val Thr Met Thr Arg Asp Thr
85      90      95
Ser Ile Ser Thr Ala Tyr Met Glu Leu Ser Arg Leu Arg Ser Asp Asp
100     105     110
Thr Ala Val Tyr Phe Cys Ala Arg Asp Gln Met Ser Ile Ile Met Leu
115     120     125
Arg Gly Val Phe Pro Pro Tyr Tyr Tyr Gly Met Asp Val Trp Gly Gln
130     135     140

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Gly Thr Thr Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro Ser Val
 145 150 155 160
 Phe Pro Leu Ala Pro Cys Ser Arg Ser Thr Ser Glu Ser Thr Ala Ala
 165 170 175
 Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val Ser
 180 185 190
 Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala Val
 195 200 205
 Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val Pro
 210 215 220
 Ser Ser Asn Phe Gly Thr Gln Thr Tyr Thr Cys Asn Val Asp His Lys
 225 230 235 240
 Pro Ser Asn Thr Lys Val Asp Lys Thr Val Glu Arg Lys Cys Cys Val
 245 250 255
 Glu Cys Pro Pro Cys Pro Ala Pro Pro Val Ala Gly Pro Ser Val Phe
 260 265 270
 Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser Arg Thr Pro
 275 280 285
 Glu Val Thr Cys Val Val Val Asp Val Ser His Glu Asp Pro Glu Val
 290 295 300
 Gln Phe Asn Trp Tyr Val Asp Gly Val Glu Val His Asn Ala Lys Thr
 305 310 315 320
 Lys Pro Arg Glu Glu Gln Phe Asn Ser Thr Phe Arg Val Val Ser Val
 325 330 335
 Leu Thr Val Val His Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys
 340 345 350
 Lys Val Ser Asn Lys Gly Leu Pro Ala Pro Ile Glu Lys Thr Ile Ser
 355 360 365
 Lys Thr Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro
 370 375 380
 Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser Leu Thr Cys Leu Val
 385 390 395 400
 Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn Gly
 405 410 415
 Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Met Leu Asp Ser Asp
 420 425 430
 Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val Asp Lys Ser Arg Trp
 435 440 445
 Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu His
 450 455 460
 Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly Lys
 465 470 475

<210> SEQ ID NO 26

<211> LENGTH: 241

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 26

Met Asp Met Arg Val Pro Ala Gln Leu Leu Gly Leu Leu Leu Trp
 1 5 10 15
 Leu Arg Gly Ala Arg Cys Asp Ile Val Met Thr Gln Ser Pro Leu Ser
 20 25 30
 Leu Pro Val Thr Pro Gly Glu Pro Ala Ser Ile Ser Cys Arg Ser Ser

-continued

35					40					45					
Gln	Ser	Leu	Leu	His	Ser	Asn	Gly	Tyr	Asn	Tyr	Leu	Asp	Trp	Tyr	Leu
50					55					60					
Gln	Lys	Pro	Gly	Gln	Ser	Pro	Gln	Leu	Leu	Ile	Tyr	Leu	Gly	Ser	Asn
65					70					75					80
Arg	Ala	Ser	Gly	Val	Pro	Asp	Arg	Phe	Ser	Gly	Ser	Gly	Ser	Gly	Thr
				85					90					95	
Asp	Phe	Thr	Leu	Lys	Ile	Ser	Arg	Val	Glu	Ala	Glu	Asp	Val	Gly	Val
			100					105					110		
Tyr	Tyr	Cys	Met	Gln	Ala	Leu	Gln	Thr	Pro	Leu	Thr	Phe	Gly	Gly	Gly
		115					120					125			
Thr	Lys	Val	Glu	Ile	Lys	Arg	Thr	Val	Ala	Ala	Pro	Ser	Val	Phe	Ile
		130					135					140			
Phe	Pro	Pro	Ser	Asp	Glu	Gln	Leu	Lys	Ser	Gly	Thr	Ala	Ser	Val	Val
145					150					155					160
Cys	Leu	Leu	Asn	Asn	Phe	Tyr	Pro	Arg	Glu	Ala	Lys	Val	Gln	Trp	Lys
			165						170					175	
Val	Asp	Asn	Ala	Leu	Gln	Ser	Gly	Asn	Ser	Gln	Glu	Ser	Val	Thr	Glu
		180						185					190		
Gln	Asp	Ser	Lys	Asp	Ser	Thr	Tyr	Ser	Leu	Ser	Ser	Thr	Leu	Thr	Leu
		195					200					205			
Ser	Lys	Ala	Asp	Tyr	Glu	Lys	His	Lys	Val	Tyr	Ala	Cys	Glu	Val	Thr
		210					215					220			
His	Gln	Gly	Leu	Ser	Ser	Pro	Val	Thr	Lys	Ser	Phe	Asn	Arg	Gly	Glu
225					230					235					240

Cys

<210> SEQ ID NO 27

<211> LENGTH: 475

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 27

Met	Asp	Met	Arg	Val	Pro	Ala	Gln	Leu	Leu	Gly	Leu	Leu	Leu	Trp
1				5						10				15
Leu	Arg	Gly	Ala	Arg	Cys	Glu	Val	Gln	Leu	Val	Glu	Ser	Gly	Gly
			20					25					30	
Leu	Val	Gln	Pro	Gly	Gly	Ser	Leu	Arg	Leu	Ser	Cys	Ala	Ala	Ser
		35					40					45		
Phe	Thr	Phe	Ser	Ser	Tyr	Trp	Met	Ser	Trp	Val	Arg	Gln	Ala	Pro
		50				55					60			
Lys	Gly	Leu	Glu	Trp	Val	Ala	Ser	Ile	Lys	Gln	Asp	Gly	Ser	Glu
65					70					75				80
Tyr	Tyr	Val	Asp	Ser	Val	Lys	Gly	Arg	Phe	Thr	Ile	Ser	Arg	Asp
			85					90					95	
Ala	Arg	Asn	Ser	Leu	Tyr	Leu	Gln	Met	Asn	Ser	Leu	Arg	Ala	Glu
			100					105					110	
Thr	Ala	Val	Tyr	Tyr	Cys	Ala	Arg	Asp	Leu	Val	Leu	Met	Val	Tyr
		115					120					125		
Ile	Asp	Tyr	Tyr	Tyr	Tyr	Gly	Met	Asp	Val	Trp	Gly	Gln	Gly	Thr
		130				135					140			
Val	Thr	Val	Ser	Ser	Ala	Ser	Thr	Lys	Gly	Pro	Ser	Val	Phe	Pro
145					150					155				160
Ala	Pro	Cys	Ser	Arg	Ser	Thr	Ser	Glu	Ser	Thr	Ala	Ala	Leu	Gly
														Cys

-continued

Gln	Lys	Pro	Gly	Gln	Ser	Pro	Gln	Leu	Leu	Ile	Tyr	Leu	Gly	Ser	Asn
65					70					75					80
Arg	Ala	Ser	Gly	Val	Pro	Asp	Arg	Phe	Ser	Gly	Ser	Gly	Ser	Gly	Thr
			85					90						95	
His	Leu	Thr	Leu	Lys	Ile	Ser	Arg	Val	Glu	Ala	Glu	Asp	Val	Gly	Val
			100					105					110		
Tyr	Tyr	Cys	Met	Gln	Thr	Leu	Gln	Thr	Pro	Leu	Thr	Phe	Gly	Gly	Gly
		115					120					125			
Thr	Lys	Val	Glu	Ile	Lys	Arg	Thr	Val	Ala	Ala	Pro	Ser	Val	Phe	Ile
	130					135					140				
Phe	Pro	Pro	Ser	Asp	Glu	Gln	Leu	Lys	Ser	Gly	Thr	Ala	Ser	Val	Val
145					150					155					160
Cys	Leu	Leu	Asn	Asn	Phe	Tyr	Pro	Arg	Glu	Ala	Lys	Val	Gln	Trp	Lys
			165						170					175	
Val	Asp	Asn	Ala	Leu	Gln	Ser	Gly	Asn	Ser	Gln	Glu	Ser	Val	Thr	Glu
			180					185					190		
Gln	Asp	Ser	Lys	Asp	Ser	Thr	Tyr	Ser	Leu	Ser	Ser	Thr	Leu	Thr	Leu
		195					200					205			
Ser	Lys	Ala	Asp	Tyr	Glu	Lys	His	Lys	Val	Tyr	Ala	Cys	Glu	Val	Thr
	210					215					220				
His	Gln	Gly	Leu	Ser	Ser	Pro	Val	Thr	Lys	Ser	Phe	Asn	Arg	Gly	Glu
225					230					235					240

Cys

<210> SEQ ID NO 29

<211> LENGTH: 475

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 29

Met	Asp	Met	Arg	Val	Pro	Ala	Gln	Leu	Leu	Gly	Leu	Leu	Leu	Leu	Trp
1				5					10					15	
Leu	Arg	Gly	Ala	Arg	Cys	Glu	Val	Gln	Leu	Val	Glu	Ser	Gly	Gly	Gly
		20						25					30		
Leu	Val	Gln	Pro	Gly	Gly	Ser	Leu	Arg	Leu	Ser	Cys	Ala	Ala	Ser	Gly
		35					40					45			
Phe	Thr	Phe	Ser	Asn	Tyr	Trp	Met	Ser	Trp	Val	Arg	Gln	Ala	Pro	Gly
	50					55				60					
Lys	Gly	Leu	Glu	Trp	Val	Ala	Ser	Ile	Lys	Gln	Asp	Gly	Ser	Glu	Lys
65					70					75				80	
Tyr	Tyr	Val	Asp	Ser	Val	Lys	Gly	Arg	Phe	Ala	Ile	Ser	Arg	Asp	Asn
			85					90						95	
Ala	Lys	Asn	Ser	Leu	Phe	Leu	Gln	Met	Asn	Ser	Leu	Arg	Ala	Glu	Asp
			100					105					110		
Thr	Ala	Val	Tyr	Tyr	Cys	Ala	Arg	Asp	Leu	Val	Leu	Met	Val	Tyr	Asp
		115					120					125			
Ile	Asp	Tyr	Tyr	Tyr	Tyr	Gly	Met	Asp	Val	Trp	Gly	Gln	Gly	Thr	Thr
	130					135					140				
Val	Thr	Val	Ser	Ser	Ala	Ser	Thr	Lys	Gly	Pro	Ser	Val	Phe	Pro	Leu
145					150					155					160
Ala	Pro	Cys	Ser	Arg	Ser	Thr	Ser	Glu	Ser	Thr	Ala	Ala	Leu	Gly	Cys
			165					170						175	
Leu	Val	Lys	Asp	Tyr	Phe	Pro	Glu	Pro	Val	Thr	Val	Ser	Trp	Asn	Ser
			180					185						190	

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Gly Leu Gln Thr Gly Asp Glu Ala Asp Tyr Tyr Cys Gly Thr Trp Asp
 100 105 110
 Ser Arg Leu Ser Ala Val Val Phe Gly Gly Gly Thr Lys Leu Thr Val
 115 120 125
 Leu Gly Gln Pro Lys Ala Asn Pro Thr Val Thr Leu Phe Pro Pro Ser
 130 135 140
 Ser Glu Glu Leu Gln Ala Asn Lys Ala Thr Leu Val Cys Leu Ile Ser
 145 150 155 160
 Asp Phe Tyr Pro Gly Ala Val Thr Val Ala Trp Lys Ala Asp Gly Ser
 165 170 175
 Pro Val Lys Ala Gly Val Glu Thr Thr Lys Pro Ser Lys Gln Ser Asn
 180 185 190
 Asn Lys Tyr Ala Ala Ser Ser Tyr Leu Ser Leu Thr Pro Glu Gln Trp
 195 200 205
 Lys Ser His Arg Ser Tyr Ser Cys Gln Val Thr His Glu Gly Ser Thr
 210 215 220
 Val Glu Lys Thr Val Ala Pro Thr Glu Cys Ser
 225 230 235

<210> SEQ ID NO 31

<211> LENGTH: 475

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 31

Met Glu Phe Gly Leu Ser Trp Val Phe Leu Val Ala Leu Leu Arg Gly
 1 5 10 15
 Val Gln Cys Gln Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln
 20 25 30
 Pro Gly Arg Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe
 35 40 45
 Ser Ser Phe Gly Met His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu
 50 55 60
 Glu Trp Val Ala Val Ile Ser Phe Asp Gly Ser Ile Lys Tyr Ser Val
 65 70 75 80
 Asp Ser Val Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn
 85 90 95
 Thr Leu Phe Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val
 100 105 110
 Tyr Tyr Cys Ala Arg Asp Arg Leu Asn Tyr Tyr Asp Ser Ser Gly Tyr
 115 120 125
 Tyr His Tyr Lys Tyr Tyr Gly Met Ala Val Trp Gly Gln Gly Thr Thr
 130 135 140
 Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro Ser Val Phe Pro Leu
 145 150 155 160
 Ala Pro Cys Ser Arg Ser Thr Ser Glu Ser Thr Ala Ala Leu Gly Cys
 165 170 175
 Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val Ser Trp Asn Ser
 180 185 190
 Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala Val Leu Gln Ser
 195 200 205
 Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val Pro Ser Ser Asn
 210 215 220
 Phe Gly Thr Gln Thr Tyr Thr Cys Asn Val Asp His Lys Pro Ser Asn

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225	230	235	240
Thr Lys Val Asp Lys Thr Val Glu Arg Lys Cys Cys Val Glu Cys Pro	245	250	255
Pro Cys Pro Ala Pro Pro Val Ala Gly Pro Ser Val Phe Leu Phe Pro	260	265	270
Pro Lys Pro Lys Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr	275	280	285
Cys Val Val Val Asp Val Ser His Glu Asp Pro Glu Val Gln Phe Asn	290	295	300
Trp Tyr Val Asp Gly Val Glu Val His Asn Ala Lys Thr Lys Pro Arg	305	310	315
Glu Glu Gln Phe Asn Ser Thr Phe Arg Val Val Ser Val Leu Thr Val	325	330	335
Val His Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser	340	345	350
Asn Lys Gly Leu Pro Ala Pro Ile Glu Lys Thr Ile Ser Lys Thr Lys	355	360	365
Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg Glu	370	375	380
Glu Met Thr Lys Asn Gln Val Ser Leu Thr Cys Leu Val Lys Gly Phe	385	390	395
Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu	405	410	415
Asn Asn Tyr Lys Thr Thr Pro Pro Met Leu Asp Ser Asp Gly Ser Phe	420	425	430
Phe Leu Tyr Ser Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly	435	440	445
Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu His Asn His Tyr	450	455	460
Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly Lys	465	470	475

<210> SEQ ID NO 32
 <211> LENGTH: 234
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 32

Met Asp Met Arg Val Pro Ala Gln Leu Leu Gly Leu Leu Leu Trp	1	5	10	15
Leu Arg Gly Ala Arg Cys Tyr Glu Leu Thr Gln Pro Pro Ser Val Ser	20	25	30	
Val Ser Pro Gly Gln Thr Ala Thr Ile Thr Cys Ser Gly Asp Lys Leu	35	40	45	
Gly Glu Arg Tyr Ala Ser Trp Tyr Gln Gln Arg Pro Gly Gln Ser Pro	50	55	60	
Val Leu Val Ile Tyr Gln Asp Ile Lys Arg Pro Ser Gly Ile Pro Glu	65	70	75	80
Arg Phe Ser Gly Ser Asn Ser Gly Asn Thr Ala Thr Leu Thr Ile Ser	85	90	95	
Gly Thr Gln Ala Met Asp Glu Ala Asp Tyr Phe Cys Gln Ala Trp Tyr	100	105	110	
Ser Ser Thr Asn Val Leu Phe Gly Gly Gly Thr Lys Leu Thr Val Leu	115	120	125	

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Gly	Gln	Pro	Lys	Ala	Ala	Pro	Ser	Val	Thr	Leu	Phe	Pro	Pro	Ser	Ser
130						135					140				
Glu	Glu	Leu	Gln	Ala	Asn	Lys	Ala	Thr	Leu	Val	Cys	Leu	Ile	Ser	Asp
145				150					155						160
Phe	Tyr	Pro	Gly	Ala	Val	Thr	Val	Ala	Trp	Lys	Ala	Asp	Ser	Ser	Pro
			165						170					175	
Val	Lys	Ala	Gly	Val	Glu	Thr	Thr	Thr	Pro	Ser	Lys	Gln	Ser	Asn	Asn
			180					185					190		
Lys	Tyr	Ala	Ala	Ser	Ser	Tyr	Leu	Ser	Leu	Thr	Pro	Glu	Gln	Trp	Lys
		195					200					205			
Ser	His	Arg	Ser	Tyr	Ser	Cys	Gln	Val	Thr	His	Glu	Gly	Ser	Thr	Val
	210					215					220				
Glu	Lys	Thr	Val	Ala	Pro	Thr	Glu	Cys	Ser						
225					230										

<210> SEQ ID NO 33

<211> LENGTH: 466

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 33

Met	Glu	Phe	Gly	Leu	Ser	Trp	Val	Phe	Leu	Val	Ala	Leu	Leu	Arg	Gly
1			5					10						15	
Val	Gln	Cys	Gln	Val	Gln	Leu	Val	Glu	Ser	Gly	Gly	Gly	Val	Val	Gln
		20						25					30		
Pro	Gly	Arg	Ser	Leu	Arg	Leu	Ser	Cys	Ala	Ala	Ser	Gly	Phe	Thr	Phe
		35				40						45			
Ser	Ser	Tyr	Gly	Met	His	Trp	Val	Arg	Gln	Ala	Pro	Gly	Lys	Gly	Leu
	50					55					60				
Glu	Trp	Val	Ala	Val	Ile	Trp	Tyr	Ala	Glu	Ser	Asn	Lys	Tyr	Tyr	Ala
65				70					75						80
Asp	Ser	Val	Lys	Gly	Arg	Phe	Thr	Ile	Ser	Arg	Asp	Asn	Ser	Lys	Asn
			85						90					95	
Thr	Leu	Tyr	Leu	Gln	Met	Asn	Ser	Leu	Arg	Ala	Glu	Asp	Thr	Ala	Val
		100						105					110		
Tyr	Tyr	Cys	Ala	Arg	Ala	Gln	Glu	Gly	Ile	Ala	Pro	Asp	Ala	Phe	Asp
		115					120					125			
Ile	Trp	Gly	Gln	Gly	Thr	Met	Val	Thr	Val	Ser	Ser	Ala	Ser	Thr	Lys
	130					135						140			
Gly	Pro	Ser	Val	Phe	Pro	Leu	Ala	Pro	Cys	Ser	Arg	Ser	Thr	Ser	Glu
145				150						155					160
Ser	Thr	Ala	Ala	Leu	Gly	Cys	Leu	Val	Lys	Asp	Tyr	Phe	Pro	Glu	Pro
			165						170					175	
Val	Thr	Val	Ser	Trp	Asn	Ser	Gly	Ala	Leu	Thr	Ser	Gly	Val	His	Thr
			180					185					190		
Phe	Pro	Ala	Val	Leu	Gln	Ser	Ser	Gly	Leu	Tyr	Ser	Leu	Ser	Ser	Val
		195					200					205			
Val	Thr	Val	Pro	Ser	Ser	Asn	Phe	Gly	Thr	Gln	Thr	Tyr	Thr	Cys	Asn
	210					215					220				
Val	Asp	His	Lys	Pro	Ser	Asn	Thr	Lys	Val	Asp	Lys	Thr	Val	Glu	Arg
225					230					235				240	
Lys	Cys	Cys	Val	Glu	Cys	Pro	Pro	Cys	Pro	Ala	Pro	Pro	Val	Ala	Gly
			245						250					255	
Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile
		260						265					270		

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Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His Glu
 275 280 285
 Asp Pro Glu Val Gln Phe Asn Trp Tyr Val Asp Gly Val Glu Val His
 290 295 300
 Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Phe Asn Ser Thr Phe Arg
 305 310 315 320
 Val Val Ser Val Leu Thr Val Val His Gln Asp Trp Leu Asn Gly Lys
 325 330 335
 Glu Tyr Lys Cys Lys Val Ser Asn Lys Gly Leu Pro Ala Pro Ile Glu
 340 345 350
 Lys Thr Ile Ser Lys Thr Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr
 355 360 365
 Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser Leu
 370 375 380
 Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp
 385 390 395 400
 Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Met
 405 410 415
 Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val Asp
 420 425 430
 Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His
 435 440 445
 Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro
 450 455 460
 Gly Lys
 465

<210> SEQ ID NO 34
 <211> LENGTH: 234
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 34

Met Gly Ser Thr Ala Ile Leu Gly Leu Leu Leu Ala Val Leu Gln Gly
 1 5 10 15
 Gly Arg Ala Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Leu
 20 25 30
 Ser Pro Gly Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser Val
 35 40 45
 Ser Ser Ser Tyr Leu Ala Trp His Gln Gln Lys Pro Gly Gln Ala Pro
 50 55 60
 Arg Leu Leu Ile Tyr Gly Ala Ser Ser Arg Ala Thr Gly Ile Pro Asp
 65 70 75 80
 Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser
 85 90 95
 Arg Leu Glu Pro Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Gly
 100 105 110
 Ser Ser Pro Trp Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Arg
 115 120 125
 Thr Val Ala Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln
 130 135 140
 Leu Lys Ser Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr
 145 150 155 160
 Pro Arg Glu Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser

-continued

Thr Lys Pro Arg Glu Gln Phe Asn Ser Thr Phe Arg Val Val Ser
 305 310 315 320
 Val Leu Thr Val Val His Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys
 325 330 335
 Cys Lys Val Ser Asn Lys Gly Leu Pro Ala Pro Ile Glu Lys Thr Ile
 340 345 350
 Ser Lys Thr Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro
 355 360 365
 Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser Leu Thr Cys Leu
 370 375 380
 Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn
 385 390 395 400
 Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Met Leu Asp Ser
 405 410 415
 Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val Asp Lys Ser Arg
 420 425 430
 Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu
 435 440 445
 His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly Lys
 450 455 460

<210> SEQ ID NO 36
 <211> LENGTH: 239
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 36

Met Arg Leu Leu Ala Gln Leu Leu Gly Leu Leu Met Leu Trp Val Pro
 1 5 10 15
 Gly Ser Ser Gly Asp Ile Val Met Thr Gln Thr Pro Leu Ser Ser Pro
 20 25 30
 Val Thr Leu Gly Gln Pro Ala Ser Ile Ser Cys Arg Ser Ser Gln Ser
 35 40 45
 Leu Val His Ser Asp Gly Asn Thr Tyr Leu Ser Trp Leu Gln Gln Arg
 50 55 60
 Pro Gly Gln Pro Pro Arg Leu Leu Ile Tyr Lys Lys Phe Asn Arg Phe
 65 70 75 80
 Ser Gly Val Pro Asp Arg Phe Ser Gly Ser Gly Ala Gly Thr Asp Phe
 85 90 95
 Thr Leu Lys Ile Ser Arg Val Glu Ala Glu Asp Val Gly Val Tyr Tyr
 100 105 110
 Cys Met Gln Ala Thr Gln Ile Pro Leu Thr Phe Gly Pro Gly Thr Lys
 115 120 125
 Val Asp Ile Lys Arg Thr Val Ala Ala Pro Ser Val Phe Ile Phe Pro
 130 135 140
 Pro Ser Asp Glu Gln Leu Lys Ser Gly Thr Ala Ser Val Val Cys Leu
 145 150 155 160
 Leu Asn Asn Phe Tyr Pro Arg Glu Ala Lys Val Gln Trp Lys Val Asp
 165 170 175
 Asn Ala Leu Gln Ser Gly Asn Ser Gln Glu Ser Val Thr Glu Gln Asp
 180 185 190
 Ser Lys Asp Ser Thr Tyr Ser Leu Ser Ser Thr Leu Thr Leu Ser Lys
 195 200 205
 Ala Asp Tyr Glu Lys His Lys Val Tyr Ala Cys Glu Val Thr His Gln
 210 215 220

-continued

Gly Leu Ser Ser Pro Val Thr Lys Ser Phe Asn Arg Gly Glu Cys
 225 230 235

<210> SEQ ID NO 37
 <211> LENGTH: 472
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 37

Met Glu Phe Gly Leu Ser Trp Val Phe Leu Val Ala Leu Leu Arg Gly
 1 5 10 15
 Val Gln Cys Gln Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln
 20 25 30
 Pro Gly Arg Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe
 35 40 45
 Ser Phe Tyr Gly Met His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu
 50 55 60
 Glu Trp Val Ala Val Ile Trp Tyr Asp Gly Ser Asn Lys Tyr Tyr Ala
 65 70 75 80
 Asp Ser Val Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn
 85 90 95
 Thr Leu Tyr Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val
 100 105 110
 Tyr Tyr Cys Ala Arg Gly Gly Tyr Asp Tyr Val Trp Gly Ser Tyr Arg
 115 120 125
 Arg Asn Ser Asp Phe Asp Tyr Trp Gly Gln Gly Thr Leu Val Thr Val
 130 135 140
 Ser Ser Ala Ser Thr Lys Gly Pro Ser Val Phe Pro Leu Ala Pro Cys
 145 150 155 160
 Ser Arg Ser Thr Ser Glu Ser Thr Ala Ala Leu Gly Cys Leu Val Lys
 165 170 175
 Asp Tyr Phe Pro Glu Pro Val Thr Val Ser Trp Asn Ser Gly Ala Leu
 180 185 190
 Thr Ser Gly Val His Thr Phe Pro Ala Val Leu Gln Ser Ser Gly Leu
 195 200 205
 Tyr Ser Leu Ser Ser Val Val Thr Val Pro Ser Ser Asn Phe Gly Thr
 210 215 220
 Gln Thr Tyr Thr Cys Asn Val Asp His Lys Pro Ser Asn Thr Lys Val
 225 230 235 240
 Asp Lys Thr Val Glu Arg Lys Cys Cys Val Glu Cys Pro Pro Cys Pro
 245 250 255
 Ala Pro Pro Val Ala Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro
 260 265 270
 Lys Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val
 275 280 285
 Val Asp Val Ser His Glu Asp Pro Glu Val Gln Phe Asn Trp Tyr Val
 290 295 300
 Asp Gly Val Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln
 305 310 315 320
 Phe Asn Ser Thr Phe Arg Val Val Ser Val Leu Thr Val Val His Gln
 325 330 335
 Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Gly
 340 345 350
 Leu Pro Ala Pro Ile Glu Lys Thr Ile Ser Lys Thr Lys Gly Gln Pro

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355	360	365
Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg Glu Glu Met Thr		
370	375	380
Lys Asn Gln Val Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser		
385	390	395 400
Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr		
	405	410 415
Lys Thr Thr Pro Pro Met Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr		
	420	425 430
Ser Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe		
	435	440 445
Ser Cys Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys		
	450	455 460
Ser Leu Ser Leu Ser Pro Gly Lys		
465	470	

<210> SEQ ID NO 38

<211> LENGTH: 235

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 38

Met Glu Ala Pro Ala Gln Leu Leu Phe Leu Leu Leu Trp Leu Pro		
1	5	10 15
Asp Thr Thr Gly Glu Ile Val Met Thr Gln Ser Pro Ala Thr Leu Ser		
	20	25 30
Val Ser Pro Gly Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser		
	35	40 45
Val Asp Ser Asn Leu Ala Trp Tyr Arg Gln Lys Pro Gly Gln Ala Pro		
	50	55 60
Arg Leu Leu Ile Tyr Gly Ala Ser Thr Arg Ala Thr Gly Ile Pro Ala		
	65	70 75 80
Arg Phe Ser Gly Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Ser		
	85	90 95
Ser Leu Gln Ser Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Ile		
	100	105 110
Asn Trp Pro Pro Ile Thr Phe Gly Gln Gly Thr Arg Leu Glu Ile Lys		
	115	120 125
Arg Thr Val Ala Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu		
	130	135 140
Gln Leu Lys Ser Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe		
	145	150 155 160
Tyr Pro Arg Glu Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln		
	165	170 175
Ser Gly Asn Ser Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser		
	180	185 190
Thr Tyr Ser Leu Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu		
	195	200 205
Lys His Lys Val Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser		
	210	215 220
Pro Val Thr Lys Ser Phe Asn Arg Gly Glu Cys		
225	230	235

<210> SEQ ID NO 39

<211> LENGTH: 465

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<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 39

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Met Lys His Leu Trp Phe Phe Leu Leu Leu Val Ala Ala Pro Arg Trp
1      5      10      15
Val Leu Ser Gln Val Gln Leu Gln Glu Ser Gly Pro Gly Leu Val Lys
20      25      30
Pro Ser Glu Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Gly Ser Ile
35      40      45
Ser Ile Tyr Tyr Trp Ser Trp Ile Arg Gln Pro Pro Gly Lys Gly Leu
50      55      60
Glu Trp Ile Gly Tyr Val Tyr Tyr Ser Gly Ser Thr Asn Tyr Asn Pro
65      70      75      80
Ser Leu Lys Ser Arg Val Thr Ile Ser Val Asp Thr Ser Lys Asn Gln
85      90      95
Phe Ser Leu Lys Leu Asn Ser Val Thr Ala Ala Asp Thr Ala Val Tyr
100     105     110
Tyr Cys Ala Arg Gly Gly Tyr Asp Phe Trp Ser Gly Tyr Phe Asp Tyr
115     120     125
Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly
130     135     140
Pro Ser Val Phe Pro Leu Ala Pro Cys Ser Arg Ser Thr Ser Glu Ser
145     150     155     160
Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val
165     170     175
Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe
180     185     190
Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val
195     200     205
Thr Val Pro Ser Ser Asn Phe Gly Thr Gln Thr Tyr Thr Cys Asn Val
210     215     220
Asp His Lys Pro Ser Asn Thr Lys Val Asp Lys Thr Val Glu Arg Lys
225     230     235     240
Cys Cys Val Glu Cys Pro Pro Cys Pro Ala Pro Pro Val Ala Gly Pro
245     250     255
Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser
260     265     270
Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His Glu Asp
275     280     285
Pro Glu Val Gln Phe Asn Trp Tyr Val Asp Gly Val Glu Val His Asn
290     295     300
Ala Lys Thr Lys Pro Arg Glu Glu Gln Phe Asn Ser Thr Phe Arg Val
305     310     315     320
Val Ser Val Leu Thr Val Val His Gln Asp Trp Leu Asn Gly Lys Glu
325     330     335
Tyr Lys Cys Lys Val Ser Asn Lys Gly Leu Pro Ala Pro Ile Glu Lys
340     345     350
Thr Ile Ser Lys Thr Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr
355     360     365
Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser Leu Thr
370     375     380
Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu
385     390     395     400

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Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Met	Leu
405									410			415			
Asp	Ser	Asp	Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Lys	Leu	Thr	Val	Asp	Lys
420								425			430				
Ser	Arg	Trp	Gln	Gln	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val	Met	His	Glu
435							440			445					
Ala	Leu	His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser	Leu	Ser	Pro	Gly
450						455			460						
Lys															
465															

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<210> SEQ ID NO 40
<211> LENGTH: 240
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens
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<400> SEQUENCE: 40

Met	Val	Leu	Gln	Thr						Ile	Ser	Leu	Leu	Leu	Trp	Ile	Ser
1				5		Gln	Val	Phe	Ile	Ser	10					15	
Gly	Ala	Tyr	Gly	Asp	Ile	Val	Leu	Thr	Gln	Ser	25	Pro	Asp	Ser	Leu	Ala	
			20						25					30			
Val	Ser	Leu	Gly	Glu	Arg	Ala	Thr	Ile	Asn	Cys	Lys	Ser	Ser	Gln	Ser		
		35					40					45					
Ile	Leu	Tyr	Ser	Ser	Ser	Asn	Glu	Asn	Phe	Leu	Thr	Trp	Tyr	Gln	Gln		
	50					55					60						
Lys	Pro	Gly	Gln	Pro	Pro	Lys	Leu	Leu	Ile	Tyr	Trp	Ala	Ser	Thr	Arg		
65					70					75					80		
Glu	Ser	Gly	Val	Pro	Asp	Arg	Phe	Ser	Gly	Ser	Gly	Ser	Gly	Thr	Asp		
			85						90					95			
Phe	Thr	Leu	Thr	Ile	Ser	Ser	Leu	Gln	Pro	Glu	Asp	Val	Ala	Val	Tyr		
			100					105					110				
Tyr	Cys	Gln	Gln	Tyr	Phe	Ser	Val	Phe	Arg	Thr	Phe	Gly	Gln	Gly	Thr		
		115					120					125					
Arg	Val	Glu	Ile	Lys	Arg	Thr	Val	Ala	Ala	Pro	Ser	Val	Phe	Ile	Phe		
	130					135					140						
Pro	Pro	Ser	Asp	Glu	Gln	Leu	Lys	Ser	Gly	Thr	Ala	Ser	Val	Val	Cys		
145					150					155					160		
Leu	Leu	Asn	Asn	Phe	Tyr	Pro	Arg	Glu	Ala	Lys	Val	Gln	Trp	Lys	Val		
			165						170					175			
Asp	Asn	Ala	Leu	Gln	Ser	Gly	Asn	Ser	Gln	Glu	Ser	Val	Thr	Glu	Gln		
			180				185						190				
Asp	Ser	Lys	Asp	Ser	Thr	Tyr	Ser	Leu	Ser	Ser	Thr	Leu	Thr	Leu	Ser		
		195				200						205					
Lys	Ala	Asp	Tyr	Glu	Lys	His	Lys	Val	Tyr	Ala	Cys	Glu	Val	Thr	His		
	210					215					220						
Gln	Gly	Leu	Ser	Ser	Pro	Val	Thr	Lys	Ser	Phe	Asn	Arg	Gly	Glu	Cys		
225					230					235					240		

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<210> SEQ ID NO 41
<211> LENGTH: 465
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens
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<400> SEQUENCE: 41

Met Asp Trp Thr Trp Arg Ile Leu Phe Leu Val Ala Ala Ala Thr Gly
1 5 10 15

Ala	His	Ser	Gln	Val	Gln	Leu	Val	Gln	Ser	Gly	Ala	Glu	Val	Lys	Lys	
			20					25					30			
Pro	Gly	Ala	Ser	Val	Lys	Val	Ser	Cys	Lys	Ala	Ser	Gly	Tyr	Thr	Phe	
		35					40					45				
Thr	Gly	Tyr	Tyr	Ile	His	Trp	Val	Arg	Gln	Ala	Pro	Gly	Gln	Gly	Leu	
	50					55					60					
Glu	Trp	Met	Gly	Trp	Ile	Asn	Pro	Asn	Ser	Gly	Gly	Thr	Asn	Ser	Ala	
65					70					75					80	
Gln	Lys	Phe	Arg	Gly	Arg	Val	Thr	Met	Thr	Arg	Asp	Thr	Ser	Ile	Ser	
				85					90					95		
Thr	Ala	Tyr	Met	Glu	Leu	Ser	Arg	Leu	Arg	Ser	Asp	Asp	Thr	Ala	Val	
			100					105					110			
Tyr	Tyr	Cys	Ala	Arg	Glu	Gly	Gly	Tyr	Ser	Tyr	Gly	Tyr	Phe	Asp	Tyr	
		115					120					125				
Trp	Gly	Gln	Gly	Thr	Leu	Val	Thr	Val	Ser	Ser	Ala	Ser	Thr	Lys	Gly	
	130					135					140					
Pro	Ser	Val	Phe	Pro	Leu	Ala	Pro	Cys	Ser	Arg	Ser	Thr	Ser	Glu	Ser	
145					150					155				160		
Thr	Ala	Ala	Leu	Gly	Cys	Leu	Val	Lys	Asp	Tyr	Phe	Pro	Glu	Pro	Val	
				165					170					175		
Thr	Val	Ser	Trp	Asn	Ser	Gly	Ala	Leu	Thr	Ser	Gly	Val	His	Thr	Phe	
			180					185					190			
Pro	Ala	Val	Leu	Gln	Ser	Ser	Gly	Leu	Tyr	Ser	Leu	Ser	Ser	Val	Val	
		195					200					205				
Thr	Val	Pro	Ser	Ser	Asn	Phe	Gly	Thr	Gln	Thr	Tyr	Thr	Cys	Asn	Val	
	210					215					220					
Asp	His	Lys	Pro	Ser	Asn	Thr	Lys	Val	Asp	Lys	Thr	Val	Glu	Arg	Lys	
225					230					235					240	
Cys	Cys	Val	Glu	Cys	Pro	Pro	Cys	Pro	Ala	Pro	Pro	Val	Ala	Gly	Pro	
				245					250					255		
Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile	Ser	
			260					265					270			
Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	His	Glu	Asp	
		275					280					285				
Pro	Glu	Val	Gln	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val	His	Asn	
	290					295					300					
Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu	Gln	Phe	Asn	Ser	Thr	Phe	Arg	Val	
305					310					315					320	
Val	Ser	Val	Leu	Thr	Val	Val	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	
				325					330					335		
Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	Gly	Leu	Pro	Ala	Pro	Ile	Glu	Lys	
			340					345					350			
Thr	Ile	Ser	Lys	Thr	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln				

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Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His Glu
435 440 445

Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly
450 455 460

Lys
465

<210> SEQ ID NO 42
<211> LENGTH: 236
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 42

Met Asp Met Arg Val Pro Ala Gln Leu Leu Gly Leu Leu Leu Leu Trp
1 5 10 15

Phe Pro Gly Ala Arg Cys Asp Ile Gln Met Thr Gln Ser Pro Ser Ser
20 25 30

Leu Ser Ala Ser Val Gly Asp Arg Val Thr Ile Thr Cys Arg Ala Ser
35 40 45

Gln Gly Ile Arg Asn Asp Leu Gly Trp Tyr Gln Gln Lys Pro Gly Lys
50 55 60

Ala Pro Lys Arg Leu Ile Tyr Ala Ala Ser Ser Leu Gln Ser Gly Val
65 70 75 80

Pro Ser Arg Phe Ser Gly Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr
85 90 95

Ile Ser Ser Val Gln Pro Glu Asp Phe Val Thr Tyr Tyr Cys Leu Gln
100 105 110

His Asn Ser Asn Pro Leu Thr Phe Gly Gly Gly Thr Lys Val Glu Ile
115 120 125

Lys Arg Thr Val Ala Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp
130 135 140

Glu Gln Leu Lys Ser Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn
145 150 155 160

Phe Tyr Pro Arg Glu Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu
165 170 175

Gln Ser Gly Asn Ser Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp
180 185 190

Ser Thr Tyr Ser Leu Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr
195 200 205

Glu Lys His Lys Val Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser
210 215 220

Ser Pro Val Thr Lys Ser Phe Asn Arg Gly Glu Cys
225 230 235

<210> SEQ ID NO 43
<211> LENGTH: 473
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 43

Met Glu Phe Gly Leu Ser Trp Val Phe Leu Val Ala Leu Leu Arg Gly
1 5 10 15

Val Gln Cys Gln Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln
20 25 30

Pro Gly Arg Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe
35 40 45

Ser 50	Tyr	Gly	Met	His	Trp 55	Val	Arg	Gln	Ala	Pro 60	Gly	Lys	Gly	Leu
Glu 65	Trp	Val	Ala	Val	Met 70	Trp	Tyr	Asp	Gly	Ser 75	Asn	Lys	Asp	Val 80
Asp	Ser	Val	Lys	Gly 85	Arg	Phe	Thr	Ile	Ser 90	Arg	Asp	Asn	Ser	Asn 95
Thr	Leu	Tyr	Leu	Gln 100	Met	Asn	Arg	Leu	Arg 105	Ala	Glu	Asp	Thr	Val 110
Tyr	Tyr	Cys 115	Ala	Arg	Glu	Lys	Asp 120	His	Tyr	Asp	Ile	Leu 125	Thr	Tyr
Asn	Tyr 130	Tyr	Tyr	Gly	Leu	Asp 135	Val	Trp	Gly	Gln	Gly 140	Thr	Thr	Thr
Val 145	Ser	Ser	Ala	Ser	Thr 150	Lys	Gly	Pro	Ser	Val 155	Phe	Pro	Leu	Pro 160
Cys	Ser	Arg	Ser	Thr 165	Ser	Glu	Ser	Thr	Ala 170	Ala	Leu	Gly	Cys	Val 175
Lys	Asp	Tyr	Phe 180	Pro	Glu	Pro	Val	Thr	Val 185	Ser	Trp	Asn	Ser	Ala
Leu	Thr 195	Ser	Gly	Val	His	Thr	Phe 200	Pro	Ala	Val	Leu	Gln 205	Ser	Gly
Leu 210	Tyr	Ser	Leu	Ser	Ser	Val 215	Val	Thr	Val	Pro	Ser 220	Ser	Asn	Gly
Thr 225	Gln	Thr	Tyr	Thr	Cys 230	Asn	Val	Asp	His 235	Lys	Pro	Ser	Asn	Lys
Val	Asp	Lys	Thr 245	Val	Glu	Arg	Lys	Cys	Cys 250	Val	Glu	Cys	Pro	Cys
Pro	Ala	Pro	Pro 260	Val	Ala	Gly	Pro	Ser 265	Val	Phe	Leu	Phe	Pro	Lys
Pro	Lys 275	Asp	Thr	Leu	Met	Ile	Ser 280	Arg	Thr	Pro	Glu	Val 285	Thr	Val
Val 290	Val	Asp	Val	Ser	His	Glu 295	Asp	Pro	Glu	Val	Gln 300	Phe	Asn	Tyr
Val 305	Asp	Gly	Val	Glu 310	Val	His	Asn	Ala	Lys 315	Thr	Lys	Pro	Arg	Glu
Gln	Phe	Asn	Ser 325	Thr	Phe	Arg	Val	Val	Ser 330	Val	Leu	Thr	Val	His
Gln	Asp	Trp 340	Leu	Asn	Gly	Lys	Glu 345	Tyr	Lys	Cys	Lys	Val 350	Ser	Lys
Gly 355	Leu	Pro	Ala	Pro	Ile	Glu	Lys 360	Thr	Ile	Ser	Lys	Thr 365	Lys	Gln
Pro 370	Arg	Glu	Pro	Gln	Val	Tyr 375	Thr	Leu	Pro	Pro	Ser 380	Arg	Glu	Met
Thr 385	Lys	Asn	Gln	Val 390	Ser	Leu	Thr	Cys	Leu 395	Val	Lys	Gly	Phe	Pro
Ser	Asp	Ile	Ala 405	Val	Glu	Trp	Glu	Ser	Asn 410	Gly	Gln	Pro	Glu	Asn
Tyr	Lys	Thr	Thr 420	Pro	Pro	Met	Leu	Asp	Ser 425	Asp	Gly	Ser	Phe	Leu
Tyr	Ser	Lys 435	Leu	Thr	Val	Asp	Lys 440	Ser	Arg	Trp	Gln	Gln 445	Gly	Val
Phe 450	Ser	Cys	Ser	Val	Met	His 455	Glu	Ala	Leu	His	Asn 460	His	Tyr	Gln
Lys	Ser	Leu	Ser	Leu	Ser	Pro	Gly	Lys						

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465

470

<210> SEQ ID NO 44
 <211> LENGTH: 235
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 44

```

Met Glu Thr Pro Ala Gln Leu Leu Phe Leu Leu Leu Trp Leu Pro
 1          5          10          15
Asp Thr Thr Gly Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser
          20          25          30
Leu Ser Pro Gly Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Tyr
          35          40          45
Ile Ser Asn Thr Tyr Leu Ala Trp Phe Gln Gln Lys Pro Gly Gln Ala
          50          55          60
Pro Arg Leu Leu Ile Tyr Gly Ala Ala Thr Arg Ala Thr Gly Ile Pro
          65          70          75          80
Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Phe Thr Ile
          85          90          95
Ser Arg Leu Glu Pro Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr
          100          105          110
Gly Ser Ser Pro Trp Thr Phe Gly Gln Gly Thr Thr Val Glu Ile Lys
          115          120          125
Arg Thr Val Ala Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu
          130          135          140
Gln Leu Lys Ser Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe
          145          150          155          160
Tyr Pro Arg Glu Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln
          165          170          175
Ser Gly Asn Ser Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser
          180          185          190
Thr Tyr Ser Leu Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu
          195          200          205
Lys His Lys Val Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser
          210          215          220
Pro Val Thr Lys Ser Phe Asn Arg Gly Glu Cys
          225          230          235

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<210> SEQ ID NO 45
 <211> LENGTH: 463
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 45

```

Met Asp Trp Thr Trp Arg Ile Leu Phe Leu Val Ala Ala Ala Thr Gly
 1          5          10          15
Ala His Ser Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys
          20          25          30
Pro Gly Ala Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Thr Phe
          35          40          45
Thr Gly Tyr Tyr Met His Trp Val Arg Gln Ala Pro Gly Gln Gly Leu
          50          55          60
Glu Trp Met Gly Trp Ile Asn Pro Asn Ser Gly Gly Thr Asn Tyr Ala
          65          70          75          80
Gln Arg Phe Arg Gly Arg Val Thr Met Thr Arg Asp Thr Ser Ile Ser

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85					90					95					
Thr	Ala	Tyr	Met	Glu	Leu	Ser	Arg	Leu	Arg	Ser	Asp	Asp	Thr	Ala	Val
			100					105					110		
Tyr	Tyr	Cys	Ala	Arg	Ala	Pro	Tyr	Asp	Trp	Thr	Phe	Asp	Tyr	Trp	Gly
		115					120					125			
Gln	Gly	Thr	Leu	Val	Thr	Val	Ser	Ser	Ala	Ser	Thr	Lys	Gly	Pro	Ser
		130					135					140			
Val	Phe	Pro	Leu	Ala	Pro	Cys	Ser	Arg	Ser	Thr	Ser	Glu	Ser	Thr	Ala
145						150					155				160
Ala	Leu	Gly	Cys	Leu	Val	Lys	Asp	Tyr	Phe	Pro	Glu	Pro	Val	Thr	Val
			165						170					175	
Ser	Trp	Asn	Ser	Gly	Ala	Leu	Thr	Ser	Gly	Val	His	Thr	Phe	Pro	Ala
		180						185					190		
Val	Leu	Gln	Ser	Ser	Gly	Leu	Tyr	Ser	Leu	Ser	Ser	Val	Val	Thr	Val
		195					200					205			
Pro	Ser	Ser	Asn	Phe	Gly	Thr	Gln	Thr	Tyr	Thr	Cys	Asn	Val	Asp	His
		210				215					220				
Lys	Pro	Ser	Asn	Thr	Lys	Val	Asp	Lys	Thr	Val	Glu	Arg	Lys	Cys	Cys
225						230					235				240
Val	Glu	Cys	Pro	Pro	Cys	Pro	Ala	Pro	Pro	Val	Ala	Gly	Pro	Ser	Val
			245						250					255	
Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg	Thr
		260					265						270		
Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	His	Glu	Asp	Pro	Glu
		275					280					285			
Val	Gln	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys
		290				295					300				
Thr	Lys	Pro	Arg	Glu	Gln	Phe	Asn	Ser	Thr	Phe	Arg	Val	Val	Ser	
305						310					315			320	
Val	Leu	Thr	Val	Val	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys
			325						330					335	
Cys	Lys	Val	Ser	Asn	Lys	Gly	Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile
		340					345						350		
Ser	Lys	Thr	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro
		355					360					365			
Pro	Ser	Arg	Glu	Glu	Met	Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Leu
		370				375						380			
Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Asn
385						390					395				400
Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Met	Leu	Asp	Ser
			405						410					415	
Asp	Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Lys	Leu	Thr	Val	Asp	Lys	Ser	Arg
			420					425					430		
Trp	Gln	Gln	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val	Met	His	Glu	Ala	Leu
		435					440					445			
His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser	Leu	Ser	Pro	Gly	Lys	
		450				455						460			

<210> SEQ ID NO 46

<211> LENGTH: 236

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 46

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Met Asp Met Arg Val Pro Ala Gln Leu Leu Gly Leu Leu Leu Leu Trp
1      5      10      15
Leu Arg Gly Ala Arg Cys Asp Ile Gln Met Thr Gln Ser Pro Ser Ser
20      25      30
Val Ser Ala Ser Val Gly Asp Arg Val Thr Ile Thr Cys Arg Ala Ser
35      40      45
Gln Gly Ile Ser Asn Trp Leu Ala Trp Tyr Gln Gln Lys Pro Gly Thr
50      55      60
Ala Pro Lys Leu Leu Ile Tyr Ala Ala Ser Ser Leu Gln Ser Gly Val
65      70      75      80
Pro Ser Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr
85      90      95
Ile Ser Ser Leu Gln Pro Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln
100     105     110
Ala Asn Ser Phe Pro Phe Thr Phe Gly Pro Gly Thr Lys Val Asp Ile
115     120     125
Lys Arg Thr Val Ala Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp
130     135     140
Glu Gln Leu Lys Ser Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn
145     150     155     160
Phe Tyr Pro Arg Glu Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu
165     170     175
Gln Ser Gly Asn Ser Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp
180     185     190
Ser Thr Tyr Ser Leu Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr
195     200     205
Glu Lys His Lys Val Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser
210     215     220
Ser Pro Val Thr Lys Ser Phe Asn Arg Gly Glu Cys
225     230     235

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<210> SEQ ID NO 47

<211> LENGTH: 469

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 47

```

Met Asp Met Arg Val Pro Ala Gln Leu Leu Gly Leu Leu Leu Leu Trp
1      5      10      15
Leu Arg Gly Ala Arg Cys Gln Val Gln Leu Val Gln Ser Gly Thr Glu
20      25      30
Val Lys Lys Pro Gly Ala Ser Met Lys Val Ser Cys Lys Ala Ser Gly
35      40      45
Tyr Thr Phe Thr Ser Tyr Tyr Met His Trp Val Arg Gln Ala Pro Gly
50      55      60
Gln Gly Leu Glu Trp Met Gly Ile Ile Asn Pro Ser Gly Asp Ser Thr
65      70      75      80
Ser Tyr Ala Gln Lys Phe Gln Gly Arg Val Thr Met Thr Arg Asp Thr
85      90      95
Ser Thr Asn Thr Val Tyr Met Glu Leu Ser Ser Leu Arg Ser Glu Asp
100     105     110
Thr Ala Met Tyr Tyr Cys Ala Arg Asp Val Glu Val Arg Gly Ile Ser
115     120     125
His Phe Asp Tyr Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser Ala
130     135     140

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Ser Thr Lys Gly Pro Ser Val Phe Pro Leu Ala Pro Cys Ser Arg Ser
 145 150 155 160
 Thr Ser Glu Ser Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe
 165 170 175
 Pro Glu Pro Val Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly
 180 185 190
 Val His Thr Phe Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu
 195 200 205
 Ser Ser Val Val Thr Val Pro Ser Ser Asn Phe Gly Thr Gln Thr Tyr
 210 215 220
 Thr Cys Asn Val Asp His Lys Pro Ser Asn Thr Lys Val Asp Lys Thr
 225 230 235 240
 Val Glu Arg Lys Cys Cys Val Glu Cys Pro Pro Cys Pro Ala Pro Pro
 245 250 255
 Val Ala Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr
 260 265 270
 Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val
 275 280 285
 Ser His Glu Asp Pro Glu Val Gln Phe Asn Trp Tyr Val Asp Gly Val
 290 295 300
 Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Phe Asn Ser
 305 310 315 320
 Thr Phe Arg Val Val Ser Val Leu Thr Val Val His Gln Asp Trp Leu
 325 330 335
 Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Gly Leu Pro Ala
 340 345 350
 Pro Ile Glu Lys Thr Ile Ser Lys Thr Lys Gly Gln Pro Arg Glu Pro
 355 360 365
 Gln Val Tyr Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln
 370 375 380
 Val Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala
 385 390 395 400
 Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr
 405 410 415
 Pro Pro Met Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu
 420 425 430
 Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser
 435 440 445
 Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser
 450 455 460
 Leu Ser Pro Gly Lys
 465

<210> SEQ ID NO 48

<211> LENGTH: 238

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 48

Met Gly Trp Ser Cys Ile Ile Leu Phe Leu Val Ala Thr Ala Thr Gly
 1 5 10 15
 Val His Ser Asp Ile Gln Leu Thr Gln Ser Pro Ser Ser Leu Ser Ala
 20 25 30
 Ser Val Gly Asp Arg Val Thr Met Ser Cys Lys Ser Ser Gln Ser Val

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35	40	45
Leu Tyr Ser Ala Asn His	Lys Asn Tyr Leu Ala	Trp Tyr Gln Gln Lys
50	55	60
Pro Gly Lys Ala Pro Lys	Leu Leu Ile Tyr Trp	Ala Ser Thr Arg Glu
65	70	75 80
Ser Gly Val Pro Ser Arg Phe	Ser Gly Ser Gly Ser Gly	Thr Asp Phe
	85	90 95
Thr Phe Thr Ile Ser Ser	Leu Gln Pro Glu Asp	Ile Ala Thr Tyr Tyr
	100	105 110
Cys His Gln Tyr Leu Ser	Ser Trp Thr Phe Gly Gly	Gly Thr Lys Val
	115	120 125
Gln Ile Lys Arg Thr Val	Ala Ala Pro Ser Val	Phe Ile Phe Pro Pro
	130	135 140
Ser Asp Glu Gln Leu Lys	Ser Gly Thr Ala Ser	Val Val Cys Leu Leu
145	150	155 160
Asn Asn Phe Tyr Pro Arg	Glu Ala Lys Val Gln	Trp Lys Val Asp Asn
	165	170 175
Ala Leu Gln Ser Gly Asn	Ser Gln Glu Ser Val	Thr Glu Gln Asp Ser
	180	185 190
Lys Asp Ser Thr Tyr Ser	Leu Ser Ser Thr Leu	Thr Leu Ser Lys Ala
	195	200 205
Asp Tyr Glu Lys His Lys	Val Tyr Ala Cys Glu	Val Thr His Gln Gly
	210	215 220
Leu Ser Ser Pro Val Thr	Lys Ser Phe Asn Arg	Gly Glu Cys
225	230	235

<210> SEQ ID NO 49

<211> LENGTH: 465

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 49

Met Gly Trp Ser Cys Ile	Ile Leu Phe Leu Val	Ala Thr Ala Thr Gly
1	5	10 15
Val His Ser Gln Val Gln	Leu Val Gln Ser Gly	Ala Glu Val Lys Lys
	20	25 30
Pro Gly Ser Ser Val Lys	Val Ser Cys Lys Ala	Ser Gly Tyr Thr Phe
	35	40 45
Thr Ser Tyr Trp Leu His	Trp Val Arg Gln Ala	Pro Gly Gln Gly Leu
	50	55 60
Glu Trp Ile Gly Tyr Ile	Asn Pro Arg Asn Asp	Tyr Thr Glu Tyr Asn
65	70	75 80
Gln Asn Phe Lys Asp Lys	Ala Thr Ile Thr Ala	Asp Glu Ser Thr Asn
	85	90 95
Thr Ala Tyr Met Glu Leu	Ser Ser Leu Arg Ser	Glu Asp Thr Ala Phe
	100	105 110
Tyr Phe Cys Ala Arg Arg	Asp Ile Thr Thr Phe	Tyr Trp Gly Gln Gly
	115	120 125
Thr Thr Val Thr Val Ser	Ser Ala Ser Thr Lys	Gly Pro Ser Val Phe
	130	135 140
Pro Leu Ala Pro Ser Ser	Lys Ser Thr Ser Gly	Gly Thr Ala Ala Leu
145	150	155 160
Gly Cys Leu Val Lys Asp	Tyr Phe Pro Glu Pro	Val Thr Val Ser Trp
	165	170 175

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Asn	Ser	Gly	Ala	Leu	Thr	Ser	Gly	Val	His	Thr	Phe	Pro	Ala	Val	Leu
			180					185					190		
Gln	Ser	Ser	Gly	Leu	Tyr	Ser	Leu	Ser	Ser	Val	Val	Thr	Val	Pro	Ser
	195						200					205			
Ser	Ser	Leu	Gly	Thr	Gln	Thr	Tyr	Ile	Cys	Asn	Val	Asn	His	Lys	Pro
	210					215					220				
Ser	Asn	Thr	Lys	Val	Asp	Lys	Arg	Val	Glu	Pro	Lys	Ser	Cys	Asp	Lys
225					230					235					240
Thr	His	Thr	Cys	Pro	Pro	Cys	Pro	Ala	Pro	Glu	Leu	Leu	Gly	Gly	Pro
				245					250					255	
Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile	Ser
			260					265					270		
Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	His	Glu	Asp
	275						280					285			
Pro	Glu	Val	Lys	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val	His	Asn
	290					295					300				
Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu	Gln	Tyr	Asn	Ser	Thr	Tyr	Arg	Val
305					310					315					320
Val	Ser	Val	Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu
			325					330						335	
Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	Ala	Leu	Pro	Ala	Pro	Ile	Glu	Lys
			340					345					350		
Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr
	355						360					365			
Leu	Pro	Pro	Ser	Arg	Glu	Glu	Met	Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr
	370					375					380				
Cys	Leu	Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu
385					390					395					400
Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Val	Leu
			405					410						415	
Asp	Ser	Asp	Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Lys	Leu	Thr	Val	Asp	Lys
			420					425					430		
Ser	Arg	Trp	Gln	Gln	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val	Met	His	Glu
	435					440						445			
Ala	Leu	His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser	Leu	Ser	Pro	Gly
	450					455					460				

Lys
465

<210> SEQ ID NO 50
 <211> LENGTH: 235
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 50

Met	Glu	Thr	Pro	Ala	Gln	Leu	Leu	Phe	Leu	Leu	Leu	Leu	Trp	Leu	Pro
1				5				10					15		
Asp	Thr	Thr	Gly	Glu	Ile	Val	Leu	Thr	Gln	Ser	Pro	Gly	Thr	Leu	Ser
	20							25					30		
Leu	Ser	Pro	Gly	Glu	Arg	Ala	Thr	Leu	Ser	Cys	Arg	Ala	Ser	Gln	Ser
		35					40					45			
Val	Ser	Ser	Ser	Tyr	Leu	Ala	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	Gln	Ala
	50					55					60				
Pro	Arg	Leu	Leu	Ile	Tyr	Gly	Ala	Ser	Ser	Arg	Ala	Thr	Gly	Ile	Pro
65				70						75				80	

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Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile
      85                      90                      95
Ser Arg Leu Glu Pro Glu Asp Phe Ala Val Tyr Tyr Cys Gln Arg Ser
      100                    105                    110
Gly Gly Ser Ser Phe Thr Phe Gly Pro Gly Thr Lys Val Asp Ile Lys
      115                    120                    125
Arg Thr Val Ala Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu
      130                    135                    140
Gln Leu Lys Ser Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe
      145                    150                    155                    160
Tyr Pro Arg Glu Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln
      165                    170                    175
Ser Gly Asn Ser Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser
      180                    185                    190
Thr Tyr Ser Leu Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu
      195                    200                    205
Lys His Lys Val Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser
      210                    215                    220
Pro Val Thr Lys Ser Phe Asn Arg Gly Glu Cys
      225                    230                    235

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<210> SEQ ID NO 51
<211> LENGTH: 466
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

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<400> SEQUENCE: 51

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Met Gly Ser Thr Ala Ile Leu Ala Leu Leu Leu Ala Val Leu Gln Gly
1      5                      10                      15
Val Cys Ala Glu Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys
      20                    25                    30
Pro Gly Glu Ser Leu Lys Ile Ser Cys Lys Gly Ser Gly Tyr Asn Phe
      35                    40                    45
Thr Ser Tyr Trp Ile Gly Trp Val Arg Gln Met Pro Gly Lys Gly Leu
      50                    55                    60
Glu Leu Met Gly Ile Ile Tyr Pro Gly Asp Ser Asp Thr Arg Tyr Ser
      65                    70                    75                    80
Pro Ser Phe Gln Gly Gln Val Thr Ile Ser Ala Asp Lys Ser Ile Ser
      85                    90                    95
Thr Ala Tyr Leu Gln Trp Ser Ser Leu Lys Ala Ser Asp Thr Ala Met
      100                   105                   110
Tyr Tyr Cys Gly Ser Gly Ser Tyr Phe Tyr Phe Asp Leu Trp Gly Arg
      115                   120                   125
Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro Ser Val
      130                   135                   140
Phe Pro Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly Thr Ala Ala
      145                   150                   155                   160
Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val Ser
      165                   170                   175
Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala Val
      180                   185                   190
Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val Pro
      195                   200                   205
Ser Ser Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val Asn His Lys

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210	215	220
Pro Ser Asn Thr Lys Val Asp Lys Lys Val Glu Pro Lys Ser Cys Asp		
225	230	235 240
Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly Gly		
	245	250 255
Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile		
	260	265 270
Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His Glu		
	275	280 285
Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val His		
	290	295 300
Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr Arg		
305	310	315 320
Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys		
	325	330 335
Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile Glu		
	340	345 350
Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr		
	355	360 365
Thr Leu Pro Pro Ser Arg Asp Glu Leu Thr Lys Asn Gln Val Ser Leu		
	370	375 380
Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp		
385	390	395 400
Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val		
	405	410 415
Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val Asp		
	420	425 430
Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His		
	435	440 445
Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro		
	450	455 460
Gly Lys		
465		

<210> SEQ ID NO 52

<211> LENGTH: 241

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 52

Met Asp Met Arg Val Pro Ala Gln Leu Leu Gly Leu Leu Leu Leu Trp		
1	5	10 15
Leu Arg Gly Ala Arg Cys Asp Ile Val Met Thr Gln Ser Pro Leu Ser		
	20	25 30
Leu Pro Val Thr Pro Gly Glu Pro Ala Ser Ile Ser Cys Arg Ser Ser		
	35	40 45
Gln Ser Leu Leu His Ser His Gly Tyr Asn Tyr Leu Asp Trp Tyr Leu		
	50	55 60
Gln Lys Pro Gly Gln Ser Pro Gln Leu Leu Ile Tyr Leu Gly Ser Asn		
	65	70 75 80
Arg Ala Ser Gly Val Pro Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr		
	85	90 95
Asp Phe Thr Leu Lys Ile Ser Arg Val Glu Ala Glu Asp Val Gly Val		
	100	105 110

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Tyr Tyr Cys Met Gln Gly Thr His Trp Pro Pro Thr Phe Gly Gln Gly
 115                      120                      125

Thr Lys Leu Glu Ile Lys Arg Thr Val Ala Ala Pro Ser Val Phe Ile
 130                      135                      140

Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser Gly Thr Ala Ser Val Val
 145                      150                      155                      160

Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu Ala Lys Val Gln Trp Lys
                      165                      170                      175

Val Asp Asn Ala Leu Gln Ser Gly Asn Ser Gln Glu Ser Val Thr Glu
                      180                      185                      190

Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu Ser Ser Thr Leu Thr Leu
 195                      200                      205

Ser Lys Ala Asp Tyr Glu Lys His Lys Val Tyr Ala Cys Glu Val Thr
 210                      215                      220

His Gln Gly Leu Ser Ser Pro Val Thr Lys Ser Phe Asn Arg Gly Glu
 225                      230                      235                      240

Cys

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<210> SEQ ID NO 53
<211> LENGTH: 470
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

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<400> SEQUENCE: 53

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Met Asp Met Arg Val Pro Ala Gln Leu Leu Gly Leu Leu Leu Trp
 1      5      10      15

Leu Arg Gly Ala Arg Cys Glu Val Gln Leu Val Gln Ser Gly Gly Gly
      20      25      30

Val Val Gln Pro Gly Arg Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly
 35      40      45

Phe Thr Phe Ser Ser Tyr Gly Met His Trp Val Arg Gln Ala Pro Gly
 50      55      60

Lys Gly Leu Glu Trp Val Ser Tyr Ile Ser Ser Ser Gly Ser Thr Ile
 65      70      75      80

Tyr Tyr Ala Asp Ser Val Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn
      85      90      95

Ala Lys Asn Ser Leu Tyr Leu Gln Met Asn Ser Leu Arg Ala Glu Asp
      100      105      110

Thr Ala Val Tyr Tyr Cys Ala Arg Asp Leu Leu Asp Tyr Asp Leu Leu
 115      120      125

Thr Gly Tyr Gly Tyr Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
 130      135      140

Ala Ser Thr Lys Gly Pro Ser Val Phe Pro Leu Ala Pro Cys Ser Arg
 145      150      155      160

Ser Thr Ser Glu Ser Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr
      165      170      175

Phe Pro Glu Pro Val Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser
      180      185      190

Gly Val His Thr Phe Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser
      195      200      205

Leu Ser Ser Val Val Thr Val Pro Ser Ser Asn Phe Gly Thr Gln Thr
 210      215      220

Tyr Thr Cys Asn Val Asp His Lys Pro Ser Asn Thr Lys Val Asp Lys
 225      230      235      240

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Thr	Val	Glu	Arg	Lys	Cys	Cys	Val	Glu	Cys	Pro	Pro	Cys	Pro	Ala	Pro
				245					250					255	
Pro	Val	Ala	Gly	Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp
			260					265					270		
Thr	Leu	Met	Ile	Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp
		275					280					285			
Val	Ser	His	Glu	Asp	Pro	Glu	Val	Gln	Phe	Asn	Trp	Tyr	Val	Asp	Gly
	290					295					300				
Val	Glu	Val	His	Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu	Gln	Phe	Asn
305					310					315					320
Ser	Thr	Phe	Arg	Val	Val	Ser	Val	Leu	Thr	Val	Val	His	Gln	Asp	Trp
				325					330					335	
Leu	Asn	Gly	Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	Gly	Leu	Pro
			340					345					350		
Ala	Pro	Ile	Glu	Lys	Thr	Ile	Ser	Lys	Thr	Lys	Gly	Gln	Pro	Arg	Glu
		355					360					365			
Pro	Gln	Val	Tyr	Thr	Leu	Pro	Pro	Ser	Arg	Glu	Glu	Met	Thr	Lys	Asn
	370					375						380			
Gln	Val	Ser	Leu	Thr	Cys	Leu	Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile
385					390					395					400
Ala	Val	Glu	Trp	Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr
			405						410					415	
Thr	Pro	Pro	Met	Leu	Asp	Ser	Asp	Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Lys
			420					425					430		
Leu	Thr	Val	Asp	Lys	Ser	Arg	Trp	Gln	Gln	Gly	Asn	Val	Phe	Ser	Cys
	435						440					445			
Ser	Val	Met	His	Glu	Ala	Leu	His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu
	450					455					460				
Ser	Leu	Ser	Pro	Gly	Lys										
465					470										

<210> SEQ ID NO 54

<211> LENGTH: 235

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 54

Met	Glu	Thr	Pro	Ala	Gln	Leu	Leu	Phe	Leu	Leu	Leu	Trp	Leu	Pro
1				5					10				15	
Asp	Thr	Thr	Gly	Glu	Ile	Val	Leu	Thr	Gln	Ser	Pro	Gly	Thr	Leu
		20					25					30		
Leu	Ser	Pro	Gly	Glu	Arg	Ala	Thr	Leu	Ser	Cys	Arg	Ala	Ser	Gln
	35					40						45		
Ile	Ser	Arg	Ser	Tyr	Leu	Ala	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	Gln
	50				55						60			
Pro	Ser	Leu	Leu	Ile	Tyr	Gly	Ala	Ser	Ser	Arg	Ala	Thr	Gly	Ile
65				70					75				80	
Asp	Arg	Phe	Ser	Gly	Ser	Gly	Ser	Gly	Thr	Asp	Phe	Thr	Leu	Thr
			85					90					95	
Ser	Arg	Leu	Glu	Pro	Glu	Asp	Phe	Ala	Val	Tyr	Tyr	Cys	Gln	Gln
		100						105					110	
Gly	Ser	Ser	Pro	Trp	Thr	Phe	Gly	Gln	Gly	Thr	Lys	Val	Glu	Ile
	115						120					125		
Arg	Thr	Val	Ala	Ala	Pro	Ser	Val	Phe	Ile	Phe	Pro	Pro	Ser	Asp
	130					135					140			

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Gln Leu Lys Ser Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe
 145 150 155 160

Tyr Pro Arg Glu Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln
 165 170 175

Ser Gly Asn Ser Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser
 180 185 190

Thr Tyr Ser Leu Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu
 195 200 205

Lys His Lys Val Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser
 210 215 220

Pro Val Thr Lys Ser Phe Asn Arg Gly Glu Cys
 225 230 235

<210> SEQ ID NO 55

<211> LENGTH: 471

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 55

Met Lys His Leu Trp Phe Phe Leu Leu Leu Val Ala Ala Pro Arg Trp
 1 5 10 15

Val Leu Ser Gln Val Gln Leu Gln Glu Ser Gly Pro Gly Leu Val Lys
 20 25 30

Pro Ser Gln Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Gly Ser Ile
 35 40 45

Ser Ser Gly Asp Tyr Phe Trp Ser Trp Ile Arg Gln Leu Pro Gly Lys
 50 55 60

Gly Leu Glu Trp Ile Gly His Ile His Asn Ser Gly Thr Thr Tyr Tyr
 65 70 75 80

Asn Pro Ser Leu Lys Ser Arg Val Thr Ile Ser Val Asp Thr Ser Lys
 85 90 95

Lys Gln Phe Ser Leu Arg Leu Ser Ser Val Thr Ala Ala Asp Thr Ala
 100 105 110

Val Tyr Tyr Cys Ala Arg Asp Arg Gly Gly Asp Tyr Tyr Tyr Gly Met
 115 120 125

Asp Val Trp Gly Gln Gly Thr Thr Val Thr Val Ser Ser Ala Ser Thr
 130 135 140

Lys Gly Pro Ser Val Phe Pro Leu Ala Pro Ser Ser Lys Ser Thr Ser
 145 150 155 160

Gly Gly Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu
 165 170 175

Pro Val Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His
 180 185 190

Thr Phe Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser
 195 200 205

Val Val Thr Val Pro Ser Ser Ser Leu Gly Thr Gln Thr Tyr Ile Cys
 210 215 220

Asn Val Asn His Lys Pro Ser Asn Thr Lys Val Asp Lys Arg Val Glu
 225 230 235 240

Pro Lys Ser Cys Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro
 245 250 255

Glu Leu Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys
 260 265 270

Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val

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275	280	285
Asp Val Ser His Glu Asp	Pro Glu Val Lys Phe	Asn Trp Tyr Val Asp
290	295	300
Gly Val Glu Val His Asn Ala Lys Thr Lys	Pro Arg Glu Glu Gln Tyr	
305	310	315
Asn Ser Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu His Gln Asp		335
325	330	
Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu		350
340	345	
Pro Ala Pro Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg		365
355	360	
Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys		380
370	375	
Asn Gln Val Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp		400
385	390	395
Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys		415
405	410	
Thr Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser		430
420	425	
Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser		445
435	440	
Cys Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser		460
450	455	
Leu Ser Leu Ser Pro Gly Lys		
465	470	
<210> SEQ ID NO 56		
<211> LENGTH: 236		
<212> TYPE: PRT		
<213> ORGANISM: Homo sapiens		
<400> SEQUENCE: 56		
Met Asp Met Arg Val Pro Ala Gln Leu Leu Gly Leu Leu Leu Trp		
1	5	10
Leu Ser Gly Ala Arg Cys Asp Ile Gln Met Thr Gln Ser Pro Ser Ser		
20	25	30
Leu Ser Ala Ser Val Gly Asp Arg Val Thr Ile Thr Cys Gln Ala Ser		
35	40	45
Gln Asp Ile Ser Asn Tyr Leu Asn Trp Tyr Gln Gln Lys Pro Gly Lys		
50	55	60
Ala Pro Lys Leu Leu Ile Tyr Asp Ala Ser Asn Leu Glu Thr Gly Val		
65	70	75
Pro Ser Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Phe Thr		
85	90	95
Ile Ser Ser Leu Gln Pro Glu Asp Ile Ala Thr Tyr Phe Cys Gln His		
100	105	110
Phe Asp His Leu Pro Leu Ala Phe Gly Gly Gly Thr Lys Val Glu Ile		
115	120	125
Lys Arg Thr Val Ala Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp		
130	135	140
Glu Gln Leu Lys Ser Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn		
145	150	155
Phe Tyr Pro Arg Glu Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu		
165	170	175

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Gln Ser Gly Asn Ser Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp
 180 185 190

Ser Thr Tyr Ser Leu Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr
 195 200 205

Glu Lys His Lys Val Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser
 210 215 220

Ser Pro Val Thr Lys Ser Phe Asn Arg Gly Glu Cys
 225 230 235

<210> SEQ ID NO 57
 <211> LENGTH: 464
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 57

Met Lys His Leu Trp Phe Phe Leu Leu Leu Val Ala Ala Pro Arg Trp
 1 5 10 15

Val Leu Ser Gln Val Gln Leu Gln Glu Ser Gly Pro Gly Leu Val Lys
 20 25 30

Pro Ser Glu Thr Leu Ser Leu Thr Cys Thr Val Ser Gly Gly Ser Val
 35 40 45

Ser Ser Gly Asp Tyr Tyr Trp Thr Trp Ile Arg Gln Ser Pro Gly Lys
 50 55 60

Gly Leu Glu Trp Ile Gly His Ile Tyr Tyr Ser Gly Asn Thr Asn Tyr
 65 70 75 80

Asn Pro Ser Leu Lys Ser Arg Leu Thr Ile Ser Ile Asp Thr Ser Lys
 85 90 95

Thr Gln Phe Ser Leu Lys Leu Ser Ser Val Thr Ala Ala Asp Thr Ala
 100 105 110

Ile Tyr Tyr Cys Val Arg Asp Arg Val Thr Gly Ala Phe Asp Ile Trp
 115 120 125

Gly Gln Gly Thr Met Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro
 130 135 140

Ser Val Phe Pro Leu Ala Pro Cys Ser Arg Ser Thr Ser Glu Ser Thr
 145 150 155 160

Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr
 165 170 175

Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro
 180 185 190

Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr
 195 200 205

Val Pro Ser Ser Asn Phe Gly Thr Gln Thr Tyr Thr Cys Asn Val Asp
 210 215 220

His Lys Pro Ser Asn Thr Lys Val Asp Lys Thr Val Glu Arg Lys Cys
 225 230 235 240

Cys Val Glu Cys Pro Pro Cys Pro Ala Pro Pro Val Ala Gly Pro Ser
 245 250 255

Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser Arg
 260 265 270

Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His Glu Asp Pro
 275 280 285

Glu Val Gln Phe Asn Trp Tyr Val Asp Gly Val Glu Val His Asn Ala
 290 295 300

Lys Thr Lys Pro Arg Glu Glu Gln Phe Asn Ser Thr Phe Arg Val Val
 305 310 315 320

Ser	Val	Leu	Thr	Val	Val	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	
				325						330		335				
Lys	Cys	Lys	Val	Ser	Asn	Lys	Gly	Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr	
				340						345		350				
Ile	Ser	Lys	Thr	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	
				355						360		365				
Pro	Pro	Ser	Arg	Glu	Glu	Met	Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	
				370						375		380				
Leu	Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	
385						390						395		400		
Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Met	Leu	Asp	
				405						410		415				
Ser	Asp	Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Lys	Leu	Thr	Val	Asp	Lys	Ser	
				420						425		430				
Arg	Trp	Gln	Gln	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val	Met	His	Glu	Ala	
				435						440		445				
Leu	His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser	Leu	Ser	Pro	Gly	Lys	
				450						455		460				

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<210> SEQ ID NO 58
<211> LENGTH: 236
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens
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<400> SEQUENCE: 58

Met 1	Glu	Thr	Pro 5			Ala	Gln	Leu	Leu	Phe 10	Leu	Leu	Leu	Trp	Leu 15	Pro
Asp	Thr	Ala	Ser 20	Glu	Ile	Val	Leu	Thr 25	Gln	Ser	Pro	Gly	Thr 30	Leu	Ser	
Leu	Ser	Pro	Gly 35	Glu	Arg	Ala	Thr 40	Leu	Ser	Cys	Arg	Ala 45	Ser	Gln	Ser	
Val 50	Ser	Asn	Ser	Tyr	Leu	Ala 55	Trp	Tyr	Gln	Gln	Lys 60	Pro	Gly	Gln	Ala	
Pro 65	Arg	Leu	Leu	Ile	Tyr 70	Gly	Ala	Ser	Ser	Arg 75	Ala	Pro	Gly	Ile	Pro 80	
Asp	Arg	Phe	Ser 85	Gly	Ser	Gly	Ser	Gly	Thr 90	Asp	Phe	Thr	Leu 95	Thr	Ile	
Ser	Arg	Leu	Glu 100	Pro	Glu	Asp	Phe	Ala 105	Val	Tyr	Tyr	Cys 110	Gln	Gln	Tyr	
Asp	His 115	Ser	Ala	Gly	Trp	Thr	Phe 120	Gly	Gln	Gly	Thr	Lys 125	Val	Glu	Ile	
Lys 130	Arg	Thr	Val	Ala	Ala	Pro 135	Ser	Val	Phe	Ile	Phe 140	Pro	Pro	Ser	Asp	
Glu 145	Gln	Leu	Lys	Ser	Gly 150	Thr	Ala	Ser	Val	Val 155	Cys	Leu	Leu	Asn 160	Asn	
Phe	Tyr	Pro	Arg	Glu 165	Ala	Lys	Val	Gln	Trp 170	Lys	Val	Asp	Asn 175	Ala	Leu	
Gln	Ser	Gly	Asn 180	Ser	Gln	Glu	Ser	Val 185	Thr	Glu	Gln	Asp	Ser 190	Lys	Asp	
Ser	Thr 195	Tyr	Ser	Leu	Ser	Ser	Thr 200	Leu	Thr	Leu	Ser	Lys 205	Ala	Asp	Tyr	
Glu 210	Lys	His	Lys	Val	Tyr	Ala 215	Cys	Glu	Val	Thr	His 220	Gln	Gly	Leu	Ser	
Ser	Pro	Val	Thr	Lys	Ser	Phe	Asn	Arg	Gly	Glu	Cys					

[illegible]

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Glu Glu Met Thr Lys Asn Gln Val Ser Leu Thr Cys Leu Val Lys Gly
 370 375 380

Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro
 385 390 395 400

Glu Asn Asn Tyr Lys Thr Thr Pro Pro Met Leu Asp Ser Asp Gly Ser
 405 410 415

Phe Phe Leu Tyr Ser Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln
 420 425 430

Gly Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu His Asn His
 435 440 445

Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly Lys
 450 455 460

<210> SEQ ID NO 60
 <211> LENGTH: 234
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 60

Met Glu Thr Pro Ala Gln Leu Leu Phe Leu Leu Leu Trp Leu Pro
 1 5 10 15

Asp Thr Thr Gly Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser
 20 25 30

Leu Ser Pro Gly Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser
 35 40 45

Val Ser Ser Ser Tyr Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala
 50 55 60

Pro Arg Leu Leu Ile Tyr Gly Ala Ser Arg Arg Ala Thr Gly Ile Pro
 65 70 75 80

Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile
 85 90 95

Ser Arg Leu Glu Pro Glu Asp Phe Ala Val Tyr Tyr Cys Gln Arg Tyr
 100 105 110

Gly Ser Ser His Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile Ser Arg
 115 120 125

Thr Val Ala Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln
 130 135 140

Leu Lys Ser Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr
 145 150 155 160

Pro Arg Glu Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser
 165 170 175

Gly Asn Ser Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr
 180 185 190

Tyr Ser Leu Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys
 195 200 205

His Lys Val Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro
 210 215 220

Val Thr Lys Ser Phe Asn Arg Gly Glu Cys
 225 230

<210> SEQ ID NO 61
 <211> LENGTH: 467
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 61

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Met	Gly	Ser	Thr	Ala	Ile	Leu	Ala	Leu	Leu	Leu	Ala	Val	Leu	Gln	Gly	1	5	10	15
Val	Cys	Ala	Glu	Val	Gln	Leu	Val	Gln	Ser	Gly	Ala	Glu	Val	Lys	Lys	20	25	30	
Pro	Gly	Glu	Ser	Leu	Lys	Ile	Ser	Cys	Lys	Val	Ser	Gly	Tyr	Phe	Phe	35	40	45	
Thr	Thr	Tyr	Trp	Ile	Gly	Trp	Val	Arg	Gln	Met	Pro	Gly	Lys	Gly	Leu	50	55	60	
Glu	Tyr	Met	Gly	Ile	Ile	Tyr	Pro	Gly	Asp	Ser	Asp	Thr	Arg	Tyr	Ser	65	70	75	80
Pro	Ser	Phe	Gln	Gly	Gln	Val	Thr	Ile	Ser	Ala	Asp	Lys	Ser	Ile	Ser	85	90	95	
Thr	Ala	Tyr	Leu	Gln	Trp	Ser	Ser	Leu	Lys	Ala	Ser	Asp	Thr	Ala	Met	100	105	110	
Tyr	Tyr	Cys	Ala	Arg	Gly	Gly	Asn	Trp	Asn	Cys	Phe	Asp	Tyr	Trp	Gly	115	120	125	
Gln	Gly	Thr	Leu	Val	Thr	Val	Ser	Ser	Ala	Ser	Thr	Lys	Gly	Pro	Ser	130	135	140	
Val	Phe	Pro	Leu	Ala	Pro	Ser	Ser	Lys	Ser	Thr	Ser	Gly	Gly	Thr	Ala	145	150	155	160
Ala	Leu	Gly	Cys	Leu	Val	Lys	Asp	Tyr	Phe	Pro	Glu	Pro	Val	Thr	Val	165	170	175	
Ser	Trp	Asn	Ser	Gly	Ala	Leu	Thr	Ser	Gly	Val	His	Thr	Phe	Pro	Ala	180	185	190	
Val	Leu	Gln	Ser	Ser	Gly	Leu	Tyr	Ser	Leu	Ser	Ser	Val	Val	Thr	Val	195	200	205	
Pro	Ser	Ser	Ser	Leu	Gly	Thr	Gln	Thr	Tyr	Ile	Cys	Asn	Val	Asn	His	210	215	220	
Lys	Pro	Ser	Asn	Thr	Lys	Val	Asp	Lys	Arg	Val	Glu	Pro	Lys	Ser	Cys	225	230	235	240
Asp	Lys	Thr	His	Thr	Cys	Pro	Pro	Cys	Pro	Ala	Pro	Glu	Leu	Leu	Gly	245	250	255	
Gly	Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met	260	265	270	
Ile	Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	His	275	280	285	
Glu	Asp	Pro	Glu	Val	Lys	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val	290	295	300	
His	Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu	Gln	Tyr	Asn	Ser	Thr	Tyr	305	310	315	320
Arg	Val	Val	Ser	Val	Leu	Thr	Val	Leu	His	Gln	Asp	Trp	Leu	Asn	Gly	325	330	335	
Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	Ala	Leu	Pro	Ala	Pro	Ile	340	345	350	
Glu	Lys	Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val	355	360	365	
Tyr	Thr	Leu	Pro	Pro	Ser	Arg	Asp	Glu	Leu	Thr	Lys	Asn	Gln	Val	Ser	370	375	380	
Leu	Thr	Cys	Leu	Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu	385	390	395	400
Trp	Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro	405	410	415	
Val	Leu	Asp	Ser	Asp	Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Lys	Leu	Thr	Val				

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420	425	430
Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met		
435	440	445
His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser		
450	455	460
Pro Gly Lys		
465		

<210> SEQ ID NO 62
 <211> LENGTH: 241
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 62

Met Asp Met Arg Val Pro Ala Gln Leu Leu Gly Leu Leu Leu Leu Trp		
1	5	10 15
Leu Arg Gly Ala Arg Cys Asp Val Val Met Thr Gln Ser Pro Leu Ser		
	20	25 30
Leu Pro Val Thr Pro Gly Glu Pro Ala Ser Ile Ser Cys Arg Ser Ser		
	35	40 45
Gln Ser Leu Leu His Ser Asn Gly Tyr Asn Tyr Leu Asp Trp Tyr Leu		
	50	55 60
Gln Lys Pro Gly Gln Ser Pro Gln Leu Leu Ile Tyr Leu Gly Ser Asn		
	65	70 75 80
Arg Ala Ser Gly Val Pro Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr		
	85	90 95
Asp Phe Thr Leu Lys Ile Ser Arg Val Glu Ala Glu Asp Val Gly Val		
	100	105 110
Tyr Tyr Cys Met Gln Gly Thr His Trp Pro Leu Thr Phe Gly Gln Gly		
	115	120 125
Thr Lys Val Glu Ile Lys Arg Thr Val Ala Ala Pro Ser Val Phe Ile		
	130	135 140
Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser Gly Thr Ala Ser Val Val		
	145	150 155 160
Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu Ala Lys Val Gln Trp Lys		
	165	170 175
Val Asp Asn Ala Leu Gln Ser Gly Asn Ser Gln Glu Ser Val Thr Glu		
	180	185 190
Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu Ser Ser Thr Leu Thr Leu		
	195	200 205
Ser Lys Ala Asp Tyr Glu Lys His Lys Val Tyr Ala Cys Glu Val Thr		
	210	215 220
His Gln Gly Leu Ser Ser Pro Val Thr Lys Ser Phe Asn Arg Gly Glu		
	225	230 235 240
Cys		

<210> SEQ ID NO 63
 <211> LENGTH: 471
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 63

Met Asp Met Arg Val Pro Ala Gln Leu Leu Gly Leu Leu Leu Leu Trp		
1	5	10 15
Leu Arg Gly Ala Arg Cys Gln Val Gln Leu Gln Glu Ser Gly Pro Gly		
	20	25 30

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Leu Val Lys Pro Ser Gly Thr Leu Ser Leu Thr Cys Ala Val Ser Gly
 35 40 45
 Gly Ser Ile Ser Ser Ser Asn Trp Trp Ser Trp Val Arg Gln Pro Pro
 50 55 60
 Gly Lys Gly Leu Glu Trp Ile Gly Glu Ile Tyr His Ser Gly Ser Thr
 65 70 75 80
 Asn Tyr Asn Pro Ser Leu Lys Ser Arg Val Thr Ile Ser Val Asp Lys
 85 90 95
 Ser Lys Asn Gln Phe Ser Leu Lys Leu Ser Ser Val Thr Ala Ala Asp
 100 105 110
 Thr Ala Val Tyr Tyr Cys Ala Arg Trp Thr Gly Arg Thr Asp Ala Phe
 115 120 125
 Asp Ile Trp Gly Gln Gly Thr Met Val Thr Val Ser Ser Ala Ser Thr
 130 135 140
 Lys Gly Pro Ser Val Phe Pro Leu Ala Pro Ser Ser Lys Ser Thr Ser
 145 150 155 160
 Gly Gly Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu
 165 170 175
 Pro Val Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His
 180 185 190
 Thr Phe Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser
 195 200 205
 Val Val Thr Val Pro Ser Ser Ser Leu Gly Thr Gln Thr Tyr Ile Cys
 210 215 220
 Asn Val Asn His Lys Pro Ser Asn Thr Lys Val Asp Lys Lys Val Glu
 225 230 235 240
 Pro Lys Ser Cys Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro
 245 250 255
 Glu Leu Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys
 260 265 270
 Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val
 275 280 285
 Asp Val Ser His Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp
 290 295 300
 Gly Val Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr
 305 310 315 320
 Asn Ser Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu His Gln Asp
 325 330 335
 Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu
 340 345 350
 Pro Ala Pro Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg
 355 360 365
 Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg Asp Glu Leu Thr Lys
 370 375 380
 Asn Gln Val Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp
 385 390 395 400
 Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys
 405 410 415
 Thr Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser
 420 425 430
 Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser
 435 440 445

-continued

Cys Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser
450 455 460

Leu Ser Leu Ser Pro Gly Lys
465 470

<210> SEQ ID NO 64
<211> LENGTH: 234
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 64

Met Glu Ala Pro Ala Gln Leu Leu Phe Leu Leu Leu Trp Leu Pro
1 5 10 15

Asp Thr Thr Gly Glu Ile Val Met Thr Gln Ser Pro Ala Thr Leu Ser
20 25 30

Val Ser Pro Gly Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser
35 40 45

Val Ser Ser Asn Leu Ala Trp Phe Gln Gln Lys Pro Gly Gln Ala Pro
50 55 60

Arg Pro Leu Ile Tyr Asp Ala Ser Thr Arg Ala Thr Gly Val Pro Ala
65 70 75 80

Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser
85 90 95

Ser Leu Gln Ser Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Asp
100 105 110

Asn Trp Pro Leu Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys Arg
115 120 125

Thr Val Ala Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln
130 135 140

Leu Lys Ser Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr
145 150 155 160

Pro Arg Glu Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser
165 170 175

Gly Asn Ser Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr
180 185 190

Tyr Ser Leu Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys
195 200 205

His Lys Val Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro
210 215 220

Val Thr Lys Ser Phe Asn Arg Gly Glu Cys
225 230

<210> SEQ ID NO 65
<211> LENGTH: 461
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 65

Met Glu Trp Thr Trp Arg Val Leu Phe Leu Val Ala Ala Ala Thr Gly
1 5 10 15

Ala His Ser Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys
20 25 30

Pro Gly Ala Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Thr Phe
35 40 45

Thr Arg Tyr Gly Ile Ser Trp Val Arg Gln Ala Pro Gly Gln Gly Leu
50 55 60

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Glu	Trp	Met	Gly	Trp	Ile	Ser	Thr	Tyr	Ser	Gly	Asn	Thr	Asn	Tyr	Ala	
65					70					75					80	
Gln	Lys	Leu	Gln	Gly	Arg	Val	Thr	Met	Thr	Thr	Asp	Thr	Ser	Thr	Ser	
			85						90					95		
Thr	Ala	Tyr	Met	Glu	Leu	Arg	Ser	Leu	Arg	Ser	Asp	Asp	Thr	Ala	Val	
			100					105					110			
Tyr	Tyr	Cys	Ala	Arg	Arg	Gln	Leu	Tyr	Phe	Asp	Tyr	Trp	Gly	Gln	Gly	
		115					120					125				
Thr	Leu	Val	Thr	Val	Ser	Ser	Ala	Ser	Thr	Lys	Gly	Pro	Ser	Val	Phe	
	130					135					140					
Pro	Leu	Ala	Pro	Cys	Ser	Arg	Ser	Thr	Ser	Glu	Ser	Thr	Ala	Ala	Leu	
145					150					155					160	
Gly	Cys	Leu	Val	Lys	Asp	Tyr	Phe	Pro	Glu	Pro	Val	Thr	Val	Ser	Trp	
			165						170					175		
Asn	Ser	Gly	Ala	Leu	Thr	Ser	Gly	Val	His	Thr	Phe	Pro	Ala	Val	Leu	
		180						185					190			
Gln	Ser	Ser	Gly	Leu	Tyr	Ser	Leu	Ser	Ser	Val	Val	Thr	Val	Pro	Ser	
		195					200					205				
Ser	Asn	Phe	Gly	Thr	Gln	Thr	Tyr	Thr	Cys	Asn	Val	Asp	His	Lys	Pro	
	210					215					220					
Ser	Asn	Thr	Lys	Val	Asp	Lys	Thr	Val	Glu	Arg	Lys	Cys	Cys	Val	Glu	
225					230					235					240	
Cys	Pro	Pro	Cys	Pro	Ala	Pro	Pro	Val	Ala	Gly	Pro	Ser	Val	Phe	Leu	
			245						250					255		
Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg	Thr	Pro	Glu	
			260					265					270			
Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	His	Glu	Asp	Pro	Glu	Val	Gln	
		275					280					285				
Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys	Thr	Lys	
	290					295					300					
Pro	Arg	Glu	Glu	Gln	Phe	Asn	Ser	Thr	Phe	Arg	Val	Val	Ser	Val	Leu	
305					310					315					320	
Thr	Val	Val	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys	Cys	Lys	
			325						330					335		
Val	Ser	Asn	Lys	Gly	Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile	Ser	Lys	
		340						345					350			
Thr	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro	Pro	Ser	
		355					360					365				
Arg	Glu	Glu	Met	Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Leu	Val	Lys	
	370					375					380					
Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Asn	Gly	Gln	
385					390					395					400	
Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Met	Leu	Asp	Ser	Asp	Gly	
			405						410					415		
Ser	Phe	Phe	Leu	Tyr	Ser	Lys	Leu	Thr	Val	Asp	Lys	Ser	Arg	Trp	Gln	
		420						425					430			
Gln	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val	Met	His	Glu	Ala	Leu	His	Asn	
		435					440					445				
His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser	Leu	Ser	Pro	Gly	Lys				
	450					455					460					

<210> SEQ ID NO 66

<211> LENGTH: 239

<212> TYPE: PRT

-continued

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 66

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Met Arg Leu Pro Ala Gln Leu Leu Gly Leu Leu Met Leu Trp Ile Pro
 1           5           10           15
Gly Ser Ser Ala Asp Ile Val Met Thr Gln Thr Pro Leu Ser Leu Ser
          20           25           30
Val Thr Pro Gly Gln Pro Ala Ser Ile Ser Cys Lys Ser Gly Gln Ser
          35           40           45
Leu Leu His Ser Asp Gly Lys Thr Tyr Leu Tyr Trp Tyr Leu Gln Lys
 50           55           60
Pro Gly Gln Pro Pro Gln Phe Leu Ile Tyr Glu Val Ser Asn Arg Phe
65           70           75           80
Ser Arg Val Pro Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe
          85           90           95
Thr Leu Arg Ile Ser Arg Val Glu Ala Glu Asp Val Gly Ile Tyr Tyr
          100          105          110
Cys Met Gln Ser Ile Gln Leu Pro Trp Thr Phe Gly Gln Gly Thr Gln
          115          120          125
Val Glu Ile Lys Arg Thr Val Ala Ala Pro Ser Val Phe Ile Phe Pro
          130          135          140
Pro Ser Asp Glu Gln Leu Lys Ser Gly Thr Ala Ser Val Val Cys Leu
145          150          155          160
Leu Asn Asn Phe Tyr Pro Arg Glu Ala Lys Val Gln Trp Lys Val Asp
          165          170          175
Asn Ala Leu Gln Ser Gly Asn Ser Gln Glu Ser Val Thr Glu Gln Asp
          180          185          190
Ser Lys Asp Ser Thr Tyr Ser Leu Ser Ser Thr Leu Thr Leu Ser Lys
          195          200          205
Ala Asp Tyr Glu Lys His Lys Val Tyr Ala Cys Glu Val Thr His Gln
          210          215          220
Gly Leu Ser Ser Pro Val Thr Lys Ser Phe Asn Arg Gly Glu Cys
225          230          235

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<210> SEQ ID NO 67

<211> LENGTH: 459

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 67

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Met Glu Phe Gly Leu Ser Trp Val Phe Leu Val Ala Leu Leu Arg Gly
 1           5           10           15
Val Gln Cys Gln Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln
          20           25           30
Pro Gly Arg Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe
          35           40           45
Ser Gly Tyr Gly Met His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu
          50           55           60
Glu Trp Val Ala Val Ile Ser Tyr Asp Gly Asn Asp Lys Tyr Tyr Ala
65           70           75           80
Asp Ser Val Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Asn
          85           90           95
Thr Leu Tyr Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val
          100          105          110
Tyr Tyr Cys Ala Arg Glu Leu Arg Val Leu Trp Gly Gln Gly Thr Leu

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115					120					125					
Val	Thr	Val	Ser	Ser	Ala	Ser	Thr	Lys	Gly	Pro	Ser	Val	Phe	Pro	Leu
130						135					140				
Ala	Pro	Cys	Ser	Arg	Ser	Thr	Ser	Glu	Ser	Thr	Ala	Ala	Leu	Gly	Cys
145					150					155					160
Leu	Val	Lys	Asp	Tyr	Phe	Pro	Glu	Pro	Val	Thr	Val	Ser	Trp	Asn	Ser
				165					170					175	
Gly	Ala	Leu	Thr	Ser	Gly	Val	His	Thr	Phe	Pro	Ala	Val	Leu	Gln	Ser
			180					185						190	
Ser	Gly	Leu	Tyr	Ser	Leu	Ser	Ser	Val	Val	Thr	Val	Pro	Ser	Ser	Asn
		195					200					205			
Phe	Gly	Thr	Gln	Thr	Tyr	Thr	Cys	Asn	Val	Asp	His	Lys	Pro	Ser	Asn
	210					215					220				
Thr	Lys	Val	Asp	Lys	Thr	Val	Glu	Arg	Lys	Cys	Cys	Val	Glu	Cys	Pro
225					230					235					240
Pro	Cys	Pro	Ala	Pro	Pro	Val	Ala	Gly	Pro	Ser	Val	Phe	Leu	Phe	Pro
				245					250					255	
Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile	Ser	Arg	Thr	Pro	Glu	Val	Thr
			260					265						270	
Cys	Val	Val	Val	Asp	Val	Ser	His	Glu	Asp	Pro	Glu	Val	Gln	Phe	Asn
		275					280					285			
Trp	Tyr	Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys	Thr	Lys	Pro	Arg
	290					295					300				
Glu	Glu	Gln	Phe	Asn	Ser	Thr	Phe	Arg	Val	Val	Ser	Val	Leu	Thr	Val
305					310					315					320
Val	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser
				325					330					335	
Asn	Lys	Gly	Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile	Ser	Lys	Thr	Lys
			340					345					350		
Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro	Pro	Ser	Arg	Glu
		355					360					365			
Glu	Met	Thr	Lys	Asn	Gln	Val	Ser	Leu	Thr	Cys	Leu	Val	Lys	Gly	Phe
	370					375					380				
Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Asn	Gly	Gln	Pro	Glu
385					390					395					400
Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Met	Leu	Asp	Ser	Asp	Gly	Ser	Phe
				405					410					415	
Phe	Leu	Tyr	Ser	Lys	Leu	Thr	Val	Asp	Lys	Ser	Arg	Trp	Gln	Gln	Gly
			420					425					430		
Asn	Val	Phe	Ser	Cys	Ser	Val	Met	His	Glu	Ala	Leu	His	Asn	His	Tyr
		435					440					445			
Thr	Gln	Lys	Ser	Leu	Ser	Leu	Ser	Pro	Gly	Lys					
	450					455									

<210> SEQ ID NO 68

<211> LENGTH: 235

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 68

Met	Asp	Met	Arg	Val	Pro	Ala	Gln	Leu	Leu	Gly	Leu	Leu	Leu	Leu	Trp
1				5				10						15	
Leu	Arg	Gly	Ala	Arg	Cys	Asp	Ile	Gln	Met	Thr	Gln	Ser	Pro	Ser	Ser
	20						25					30			

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Leu Ser Ala Ser Val Gly Asp Arg Val Thr Ile Thr Cys Arg Ala Ser
   35                               40               45

Gln Asp Ile Ser Ser Tyr Leu Asn Trp Tyr Gln Gln Lys Pro Gly Lys
   50                               55               60

Ala Pro Lys Leu Leu Ile Tyr Ser Thr Ser Arg Leu Asn Ser Gly Val
   65                               70               75               80

Pro Ser Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr
                        85                               90               95

Ile Ser Ser Leu Gln Pro Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln
   100                               105               110

Asp Ile Lys His Pro Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys
   115                               120               125

Arg Thr Val Ala Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu
   130                               135               140

Gln Leu Lys Ser Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe
   145                               150               155               160

Tyr Pro Arg Glu Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln
                        165                               170               175

Ser Gly Asn Ser Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser
   180                               185               190

Thr Tyr Ser Leu Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu
   195                               200               205

Lys His Lys Val Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser
   210                               215               220

Pro Val Thr Lys Ser Phe Asn Arg Gly Glu Cys
   225                               230               235

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<210> SEQ ID NO 69
<211> LENGTH: 465
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

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<400> SEQUENCE: 69

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Met Asp Trp Thr Trp Arg Ile Leu Phe Leu Val Ala Ala Ala Thr Gly
  1      5      10      15

Ala His Ser Glu Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys
   20      25      30

Pro Gly Ser Ser Val Lys Val Ser Cys Lys Ala Ser Gly Phe Thr Phe
   35      40      45

Thr Asp Tyr Ile Met His Trp Val Arg Gln Ala Pro Gly Gln Gly Leu
   50      55      60

Glu Trp Met Gly Tyr Ile Asn Pro Tyr Asn Asp Thr Glu Tyr Asn
   65      70      75      80

Glu Lys Phe Lys Gly Arg Val Thr Ile Thr Ala Asp Lys Ser Thr Ser
   85      90      95

Thr Ala Tyr Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val
  100      105      110

Tyr Tyr Cys Ala Arg Ser Ile Tyr Tyr Tyr Asp Ala Pro Phe Ala Tyr
  115      120      125

Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly
  130      135      140

Pro Ser Val Phe Pro Leu Ala Pro Cys Ser Arg Ser Thr Ser Glu Ser
  145      150      155      160

Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val
  165      170      175

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Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe
 180 185 190
 Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val
 195 200 205
 Thr Val Pro Ser Ser Asn Phe Gly Thr Gln Thr Tyr Thr Cys Asn Val
 210 215 220
 Asp His Lys Pro Ser Asn Thr Lys Val Asp Lys Thr Val Glu Arg Lys
 225 230 235 240
 Cys Cys Val Glu Cys Pro Pro Cys Pro Ala Pro Pro Val Ala Gly Pro
 245 250 255
 Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser
 260 265 270
 Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His Glu Asp
 275 280 285
 Pro Glu Val Gln Phe Asn Trp Tyr Val Asp Gly Val Glu Val His Asn
 290 295 300
 Ala Lys Thr Lys Pro Arg Glu Glu Gln Phe Asn Ser Thr Phe Arg Val
 305 310 315 320
 Val Ser Val Leu Thr Val Val His Gln Asp Trp Leu Asn Gly Lys Glu
 325 330 335
 Tyr Lys Cys Lys Val Ser Asn Lys Gly Leu Pro Ala Pro Ile Glu Lys
 340 345 350
 Thr Ile Ser Lys Thr Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr
 355 360 365
 Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser Leu Thr
 370 375 380
 Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu
 385 390 395 400
 Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Met Leu
 405 410 415
 Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val Asp Lys
 420 425 430
 Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His Glu
 435 440 445
 Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly
 450 455 460
 Lys
 465

<210> SEQ ID NO 70
 <211> LENGTH: 233
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 70

Met Gly Ser Thr Ala Ile Leu Gly Leu Leu Leu Ala Val Leu Gln Gly
 1 5 10 15
 Gly Arg Ala Ser Tyr Val Leu Thr Gln Pro Pro Ser Val Ser Val Ala
 20 25 30
 Pro Gly Gln Thr Ala Arg Ile Thr Cys Gly Gly Asn Asn Leu Gly Ser
 35 40 45
 Lys Ser Val His Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Val Leu
 50 55 60
 Val Val Tyr Asp Asp Ser Asp Arg Pro Ser Trp Ile Pro Glu Arg Phe

-continued

65	70	75	80
Ser Gly Ser Asn Ser Gly Asn Thr Ala Thr Leu Thr Ile Ser Arg Gly			
	85	90	95
Glu Ala Gly Asp Glu Ala Asp Tyr Tyr Cys Gln Val Trp Asp Ser Ser			
	100	105	110
Ser Asp His Val Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu Gly			
	115	120	125
Gln Pro Lys Ala Ala Pro Ser Val Thr Leu Phe Pro Pro Ser Ser Glu			
	130	135	140
Glu Leu Gln Ala Asn Lys Ala Thr Leu Val Cys Leu Ile Ser Asp Phe			
	145	150	155
Tyr Pro Gly Ala Val Thr Val Ala Trp Lys Ala Asp Ser Ser Pro Val			
	165	170	175
Lys Ala Gly Val Glu Thr Thr Thr Pro Ser Lys Gln Ser Asn Asn Lys			
	180	185	190
Tyr Ala Ala Ser Ser Tyr Leu Ser Leu Thr Pro Glu Gln Trp Lys Ser			
	195	200	205
His Arg Ser Tyr Ser Cys Gln Val Thr His Glu Gly Ser Thr Val Glu			
	210	215	220
Lys Thr Val Ala Pro Thr Glu Cys Ser			
	225	230	

<210> SEQ ID NO 71

<211> LENGTH: 467

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 71

Met Gly Ser Thr Ala Ile Leu Gly Leu Leu Leu Ala Val Leu Gln Gly			
1	5	10	15
Gly Arg Ala Gln Met Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln			
	20	25	30
Pro Gly Arg Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe			
	35	40	45
Arg Thr Tyr Gly Met His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu			
	50	55	60
Glu Trp Val Ala Val Ile Trp Tyr Asp Gly Ser Asn Lys His Tyr Ala			
	65	70	75
Asp Ser Val Lys Gly Arg Phe Thr Ile Thr Arg Asp Asn Ser Lys Asn			
	85	90	95
Thr Leu Asn Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val			
	100	105	110
Tyr Tyr Cys Ala Arg Ala Pro Gln Trp Glu Leu Val His Glu Ala Phe			
	115	120	125
Asp Ile Trp Gly Gln Gly Thr Met Val Thr Val Ser Ser Ala Ser Thr			
	130	135	140
Lys Gly Pro Ser Val Phe Pro Leu Ala Pro Cys Ser Arg Ser Thr Ser			
	145	150	155
Glu Ser Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu			
	165	170	175
Pro Val Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His			
	180	185	190
Thr Phe Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser			
	195	200	205

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Val	Val	Thr	Val	Pro	Ser	Ser	Asn	Phe	Gly	Thr	Gln	Thr	Tyr	Thr	Cys
210						215					220				
Asn	Val	Asp	His	Lys	Pro	Ser	Asn	Thr	Lys	Val	Asp	Lys	Thr	Val	Glu
225					230					235					240
Arg	Lys	Cys	Cys	Val	Glu	Cys	Pro	Pro	Cys	Pro	Ala	Pro	Pro	Val	Ala
				245					250					255	
Gly	Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met
			260					265					270		
Ile	Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	His
		275					280					285			
Glu	Asp	Pro	Glu	Val	Gln	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val
	290					295					300				
His	Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu	Gln	Phe	Asn	Ser	Thr	Phe
305					310					315					320
Arg	Val	Val	Ser	Val	Leu	Thr	Val	Val	His	Gln	Asp	Trp	Leu	Asn	Gly
			325						330					335	
Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	Gly	Leu	Pro	Ala	Pro	Ile
			340					345					350		
Glu	Lys	Thr	Ile	Ser	Lys	Thr	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val
		355					360					365			
Tyr	Thr	Leu	Pro	Pro	Ser	Arg	Glu	Glu	Met	Thr	Lys	Asn	Gln	Val	Ser
	370					375					380				
Leu	Thr	Cys	Leu	Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu
385					390					395					400
Trp	Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro
			405						410					415	
Met	Leu	Asp	Ser	Asp	Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Lys	Leu	Thr	Val
			420					425					430		
Asp	Lys	Ser	Arg	Trp	Gln	Gln	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val	Met
		435					440					445			
His	Glu	Ala	Leu	His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser	Leu	Ser
	450				455						460				
Pro	Gly	Lys													
465															

<210> SEQ ID NO 72

<211> LENGTH: 234

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 72

Met	Gly	Val	Pro	Thr	His	Leu	Leu	Gly	Leu	Leu	Leu	Trp	Ile	Thr
1			5					10					15	
His	Ala	Ile	Cys	Asp	Ile	Arg	Met	Thr	Gln	Ser	Pro	Ala	Ser	Leu
	20					25						30		
Ala	Ser	Leu	Gly	Glu	Thr	Val	Asn	Ile	Glu	Cys	Leu	Ala	Ser	Glu
	35					40					45			
Ile	Tyr	Ser	Asp	Leu	Ala	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	Lys	Ser
	50				55					60				
Gln	Leu	Leu	Ile	Tyr	Asn	Ala	Asn	Ser	Leu	Gln	Asn	Gly	Val	Pro
65				70						75				80
Arg	Phe	Ser	Gly	Ser	Gly	Ser	Gly	Thr	Gln	Tyr	Ser	Leu	Lys	Ile
			85					90					95	
Ser	Leu	Gln	Ser	Glu	Asp	Val	Ala	Thr	Tyr	Phe	Cys	Gln	Gln	Tyr
		100						105						110

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Asn Tyr Pro Pro Thr Phe Gly Gly Gly Thr Lys Leu Glu Leu Lys Arg
    115                120                125

Ala Asp Ala Ala Pro Thr Val Ser Ile Phe Pro Pro Ser Thr Glu Gln
    130                135                140

Leu Ala Thr Gly Gly Ala Ser Val Val Cys Leu Met Asn Asn Phe Tyr
    145                150                155                160

Pro Arg Asp Ile Ser Val Lys Trp Lys Ile Asp Gly Thr Glu Arg Arg
    165                170                175

Asp Gly Val Leu Asp Ser Val Thr Asp Gln Asp Ser Lys Asp Ser Thr
    180                185                190

Tyr Ser Met Ser Ser Thr Leu Ser Leu Thr Lys Ala Asp Tyr Glu Ser
    195                200                205

His Asn Leu Tyr Thr Cys Glu Val Val His Lys Thr Ser Ser Ser Pro
    210                215                220

Val Val Lys Ser Phe Asn Arg Asn Glu Cys
    225                230

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<210> SEQ ID NO 73
<211> LENGTH: 465
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

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<400> SEQUENCE: 73

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Met Asp Ile Arg Leu Ser Leu Ala Phe Leu Val Leu Phe Ile Lys Gly
1          5          10          15

Val Gln Cys Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln
20         25         30

Pro Ala Asn Ser Leu Lys Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe
35         40         45

Ser Asp Tyr Ala Met Ala Trp Val Arg Gln Ser Pro Lys Lys Gly Leu
50         55         60

Glu Trp Val Ala Thr Ile Ile Tyr Asp Gly Ser Ser Thr Tyr Tyr Arg
65         70         75         80

Asp Ser Val Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Ser
85         90         95

Thr Leu Tyr Leu Gln Met Asp Ser Leu Arg Ser Glu Asp Thr Ala Thr
100        105        110

Tyr Tyr Cys Ala Thr Gly Leu Gly Ile Ala Thr Asp Tyr Phe Asp Tyr
115        120        125

Trp Gly Gln Gly Val Leu Val Thr Val Ser Ser Ala Glu Thr Thr Ala
130        135        140

Pro Ser Val Tyr Pro Leu Ala Pro Gly Thr Ala Leu Lys Ser Asn Ser
145        150        155        160

Met Val Thr Leu Gly Cys Leu Val Lys Gly Tyr Phe Pro Glu Pro Val
165        170        175

Thr Val Thr Trp Asn Ser Gly Ala Leu Ser Ser Gly Val His Thr Phe
180        185        190

Pro Ala Val Leu Gln Ser Gly Leu Tyr Thr Leu Thr Ser Ser Val Thr
195        200        205

Val Pro Ser Ser Thr Trp Pro Ser Gln Thr Val Thr Cys Asn Val Ala
210        215        220

His Pro Ala Ser Ser Thr Lys Val Asp Lys Lys Ile Val Pro Arg Asn
225        230        235        240

Cys Gly Gly Asp Cys Lys Pro Cys Ile Cys Thr Gly Ser Glu Val Ser

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										245				250				255			
Ser	Val	Phe	Ile	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Val	Leu	Thr	Ile	Thr						
			260				265						270								
Leu	Thr	Pro	Lys	Val	Thr	Cys	Val	Val	Val	Asp	Ile	Ser	Gln	Asp	Asp						
			275				280						285								
Pro	Glu	Val	His	Phe	Ser	Trp	Phe	Val	Asp	Asp	Val	Glu	Val	His	Thr						
			290				295						300								
Ala	Gln	Thr	Arg	Pro	Pro	Glu	Glu	Gln	Phe	Asn	Ser	Thr	Phe	Arg	Ser						
			305				310						315			320					
Val	Ser	Glu	Leu	Pro	Ile	Leu	His	Gln	Asp	Trp	Leu	Asn	Gly	Arg	Thr						
			325				330						335								
Phe	Arg	Cys	Lys	Val	Thr	Ser	Ala	Ala	Phe	Pro	Ser	Pro	Ile	Glu	Lys						
			340				345						350								
Thr	Ile	Ser	Lys	Pro	Glu	Gly	Arg	Thr	Gln	Val	Pro	His	Val	Tyr	Thr						
			355				360						365								
Met	Ser	Pro	Thr	Lys	Glu	Glu	Met	Thr	Gln	Asn	Glu	Val	Ser	Ile	Thr						
			370				375						380								
Cys	Met	Val	Lys	Gly	Phe	Tyr	Pro	Pro	Asp	Ile	Tyr	Val	Glu	Trp	Gln						
			385				390						395			400					
Met	Asn	Gly	Gln	Pro	Gln	Glu	Asn	Tyr	Lys	Asn	Thr	Pro	Pro	Thr	Met						
			405				410						415								
Asp	Thr	Asp	Gly	Ser	Tyr	Phe	Leu	Tyr	Ser	Lys	Leu	Asn	Val	Lys	Lys						
			420				425						430								
Glu	Lys	Trp	Gln	Gln	Gly	Asn	Thr	Phe	Thr	Cys	Ser	Val	Leu	His	Glu						
			435				440						445								
Gly	Leu	His	Asn	His	His	Thr	Glu	Lys	Ser	Leu	Ser	His	Ser	Pro	Gly						
			450				455						460								
Lys																					
465																					

<400> SEQUENCE: 74

Met	Asp	Met	Arg	Val	Pro	Ala	Gln	Leu	Leu	Gly	Leu	Leu	Leu	Leu	Trp
1				5					10					15	
Leu	Arg	Gly	Ala	Arg	Cys	Glu	Ser	Ala	Leu	Thr	Gln	Pro	Ala	Ser	Val
			20					25					30		
Ser	Gly	Ser	Pro	Gly	Gln	Ser	Ile	Thr	Ile	Ser	Cys	Thr	Gly	Thr	Ser
		35					40					45			
Ser	Asp	Val	Gly	Gly	Tyr	Asn	Ser	Val	Ser	Trp	Tyr	Gln	Gln	His	Pro
	50					55					60				
Gly	Lys	Ala	Pro	Lys	Leu	Met	Ile	Tyr	Glu	Val	Ser	Asn	Arg	Pro	Ser
65					70					75				80	
Gly	Val	Ser	Asn	Arg	Phe	Ser	Gly	Ser	Lys	Ser	Gly	Asn	Thr	Ala	Ser
				85					90					95	
Leu	Thr	Ile	Ser	Gly	Leu	Gln	Ala	Glu	Asp	Glu	Ala	Asp	Tyr	Tyr	Cys
			100					105					110		
Asn	Ser	Tyr	Thr	Ser	Thr	Ser	Met	Val	Phe	Gly	Gly	Gly	Thr	Lys	Leu
		115					120					125			
Thr	Val	Leu	Gly	Gln	Pro	Lys	Ala	Ala	Pro	Ser	Val	Thr	Leu	Phe	Pro
	130					135					140				

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Pro Ser Ser Glu Glu Leu Gln Ala Asn Lys Ala Thr Leu Val Cys Leu
145          150          155          160

Ile Ser Asp Phe Tyr Pro Gly Ala Val Thr Val Ala Trp Lys Ala Asp
          165          170          175

Ser Ser Pro Val Lys Ala Gly Val Glu Thr Thr Thr Pro Ser Lys Gln
          180          185          190

Ser Asn Asn Lys Tyr Ala Ala Ser Ser Tyr Leu Ser Leu Thr Pro Glu
          195          200          205

Gln Trp Lys Ser His Arg Ser Tyr Ser Cys Gln Val Thr His Glu Gly
210          215          220

Ser Thr Val Glu Lys Thr Val Ala Pro Thr Glu Cys Ser
225          230          235

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<210> SEQ ID NO 75
<211> LENGTH: 460
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

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<400> SEQUENCE: 75

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Met Asp Trp Thr Trp Arg Ile Leu Phe Leu Val Ala Ala Ala Thr Gly
1          5          10          15

Val His Ser Glu Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys
20          25          30

Pro Gly Ala Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Thr Leu
35          40          45

Thr Ser Tyr Gly Ile Ser Trp Val Arg Gln Ala Pro Gly Gln Gly Leu
50          55          60

Glu Trp Met Gly Trp Val Ser Phe Tyr Asn Gly Asn Thr Asn Tyr Ala
65          70          75          80

Gln Lys Leu Gln Gly Arg Gly Thr Met Thr Thr Asp Pro Ser Thr Ser
85          90          95

Thr Ala Tyr Met Glu Leu Arg Ser Leu Arg Ser Asp Asp Thr Ala Val
100         105         110

Tyr Tyr Cys Ala Arg Gly Tyr Gly Met Asp Val Trp Gly Gln Gly Thr
115         120         125

Thr Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro Ser Val Phe Pro
130         135         140

Leu Ala Pro Cys Ser Arg Ser Thr Ser Glu Ser Thr Ala Ala Leu Gly
145         150         155         160

Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val Ser Trp Asn
165         170         175

Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala Val Leu Gln
180         185         190

Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val Pro Ser Ser
195         200         205

Asn Phe Gly Thr Gln Thr Tyr Thr Cys Asn Val Asp His Lys Pro Ser
210         215         220

Asn Thr Lys Val Asp Lys Thr Val Glu Arg Lys Cys Cys Val Glu Cys
225         230         235         240

Pro Pro Cys Pro Ala Pro Pro Val Ala Gly Pro Ser Val Phe Leu Phe
245         250         255

Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val
260         265         270

Thr Cys Val Val Val Asp Val Ser His Glu Asp Pro Glu Val Gln Phe
275         280         285

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Asn Trp Tyr Val Asp Gly Val Glu Val His Asn Ala Lys Thr Lys Pro
 290                               295                 300

Arg Glu Glu Gln Phe Asn Ser Thr Phe Arg Val Val Ser Val Leu Thr
 305                               310                 315                 320

Val Val His Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val
                               325                 330                 335

Ser Asn Lys Gly Leu Pro Ala Pro Ile Glu Lys Thr Ile Ser Lys Thr
 340                               345                 350

Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg
 355                               360                 365

Glu Glu Met Thr Lys Asn Gln Val Ser Leu Thr Cys Leu Val Lys Gly
 370                               375                 380

Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro
 385                               390                 395                 400

Glu Asn Asn Tyr Lys Thr Thr Pro Pro Met Leu Asp Ser Asp Gly Ser
                               405                 410                 415

Phe Phe Leu Tyr Ser Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln
 420                               425                 430

Gly Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu His Asn His
 435                               440                 445

Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly Lys
 450                               455                 460

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<210> SEQ ID NO 76

<211> LENGTH: 234

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 76

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Met Glu Thr Pro Ala Gln Leu Leu Phe Leu Leu Leu Trp Leu Pro
 1                               5                               10                 15

Asp Thr Thr Gly Glu Ile Val Met Thr Gln Ser Pro Ala Thr Leu Ser
 20                               25                               30

Val Ser Pro Gly Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser
 35                               40                               45

Val Ser Ser Asn Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro
 50                               55                               60

Arg Leu Leu Ile Tyr Gly Ala Ala Thr Arg Ala Thr Gly Ile Pro Ala
 65                               70                               75                 80

Arg Val Ser Gly Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Ser
 85                               90                               95

Ser Leu Gln Ser Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Asn
 100                              105                              110

Asn Trp Pro Leu Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys Arg
 115                              120                              125

Thr Val Ala Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln
 130                              135                              140

Leu Lys Ser Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr
 145                              150                              155                 160

Pro Arg Glu Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser
 165                              170                              175

Gly Asn Ser Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr
 180                              185                              190

Tyr Ser Leu Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys

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195	200	205
His Lys Val Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro		
210	215	220
Val Thr Lys Ser Phe Asn Arg Gly Glu Cys		
225	230	
<210> SEQ ID NO 77		
<211> LENGTH: 472		
<212> TYPE: PRT		
<213> ORGANISM: Homo sapiens		
<400> SEQUENCE: 77		
Met Asp Met Arg Val Pro Ala Gln Leu Leu Gly Leu Leu Leu Leu Trp		
1	5	10
Leu Arg Gly Ala Arg Cys Gln Val Gln Leu Val Glu Ser Gly Gly Gly		
	20	25
Val Val Gln Pro Gly Arg Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly		
	35	40
Phe Thr Phe Ser Asn Tyr Gly Met His Trp Val Arg Gln Ala Pro Gly		
50	55	60
Glu Gly Leu Glu Trp Val Ala Ala Ile Trp Phe Asp Ala Ser Asp Lys		
65	70	75
Tyr Tyr Ala Asp Ala Val Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn		
	85	90
Ser Lys Asn Thr Leu Tyr Leu Gln Met Asn Ser Leu Arg Ala Glu Asp		
	100	105
Thr Ala Val Tyr Tyr Cys Ala Arg Asp Gln Ala Ile Phe Gly Val Val		
	115	120
Pro Asp Tyr Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser Ala Ser		
	130	135
Thr Lys Gly Pro Ser Val Phe Pro Leu Ala Pro Ser Ser Lys Ser Thr		
145	150	155
Ser Gly Gly Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro		
	165	170
Glu Pro Val Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val		
	180	185
His Thr Phe Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser		
	195	200
Ser Val Val Thr Val Pro Ser Ser Ser Leu Gly Thr Gln Thr Tyr Ile		
	210	215
Cys Asn Val Asn His Lys Pro Ser Asn Thr Lys Val Asp Lys Lys Val		
225	230	235
Glu Pro Lys Ser Cys Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala		
	245	250
Pro Glu Leu Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro		
	260	265
Lys Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val		
	275	280
Val Asp Val Ser His Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr Val		
	290	295
Asp Gly Val Glu Val His Asn Ala Lys Thr Lys Pro Cys Glu Glu Gln		
305	310	315
Tyr Gly Ser Thr Tyr Arg Cys Val Ser Val Leu Thr Val Leu His Gln		
	325	330

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Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala
 340 345 350

Leu Pro Ala Pro Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro
 355 360 365

Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg Glu Glu Met Thr
 370 375 380

Lys Asn Gln Val Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser
 385 390 395 400

Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr
 405 410 415

Lys Thr Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr
 420 425 430

Ser Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe
 435 440 445

Ser Cys Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys
 450 455 460

Ser Leu Ser Leu Ser Pro Gly Lys
 465 470

<210> SEQ ID NO 78
 <211> LENGTH: 231
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 78

Met Ala Trp Ile Pro Leu Phe Leu Gly Val Leu Ala Tyr Cys Thr Gly
 1 5 10 15

Ser Val Ala Ser Tyr Glu Val Thr Gln Ala Pro Ser Val Ser Val Ser
 20 25 30

Pro Gly Gln Thr Ala Ser Ile Thr Cys Ser Gly Asp Lys Leu Gly Asp
 35 40 45

Lys Tyr Ala Cys Trp Tyr Gln Gln Lys Pro Gly Gln Ser Pro Val Leu
 50 55 60

Val Ile Tyr Gln Asp Ser Lys Arg Pro Ser Gly Ile Pro Glu Arg Phe
 65 70 75 80

Ser Gly Ser Asn Ser Gly Asn Thr Ala Thr Leu Thr Ile Ser Gly Thr
 85 90 95

Gln Ala Met Asp Glu Ala Asp Tyr Tyr Cys Gln Ala Trp Asp Ser Ser
 100 105 110

Thr Ala Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu Gly Gln Pro
 115 120 125

Lys Ala Ala Pro Ser Val Thr Leu Phe Pro Pro Ser Ser Glu Glu Leu
 130 135 140

Gln Ala Asn Lys Ala Thr Leu Val Cys Leu Ile Ser Asp Phe Tyr Pro
 145 150 155 160

Gly Ala Val Thr Val Ala Trp Lys Ala Asp Ser Ser Pro Val Lys Ala
 165 170 175

Gly Val Glu Thr Thr Thr Pro Ser Lys Gln Ser Asn Asn Lys Tyr Ala
 180 185 190

Ala Ser Ser Tyr Leu Ser Leu Thr Pro Glu Gln Trp Lys Ser His Arg
 195 200 205

Ser Tyr Ser Cys Gln Val Thr His Glu Gly Ser Thr Val Glu Lys Thr
 210 215 220

Val Ala Pro Thr Glu Cys Ser
 225 230

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<210> SEQ ID NO 79
<211> LENGTH: 467
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 79

Met Asp Trp Thr Trp Ser Ile Leu Phe Leu Val Ala Ala Ala Thr Gly
1          5          10          15
Ala His Ser Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys
20          25          30
Pro Gly Ala Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Thr Phe
35          40          45
Thr Ser Tyr Gly Leu Ser Trp Val Arg Gln Ala Pro Gly Gln Gly Leu
50          55          60
Glu Trp Met Gly Trp Ile Ile Pro Tyr Asn Gly Asn Thr Asn Ser Ala
65          70          75          80
Gln Lys Leu Gln Gly Arg Val Thr Met Thr Thr Asp Thr Ser Thr Ser
85          90          95
Thr Ala Tyr Met Glu Leu Arg Ser Leu Arg Ser Asp Asp Thr Ala Val
100         105         110
Tyr Phe Cys Ala Arg Asp Arg Asp Tyr Gly Val Asn Tyr Asp Ala Phe
115        120        125
Asp Ile Trp Gly Gln Gly Thr Met Val Thr Val Ser Ser Ala Ser Thr
130        135        140
Lys Gly Pro Ser Val Phe Pro Leu Ala Pro Cys Ser Arg Ser Thr Ser
145        150        155        160
Glu Ser Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu
165        170        175
Pro Val Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His
180        185        190
Thr Phe Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser
195        200        205
Val Val Thr Val Pro Ser Ser Asn Phe Gly Thr Gln Thr Tyr Thr Cys
210        215        220
Asn Val Asp His Lys Pro Ser Asn Thr Lys Val Asp Lys Thr Val Glu
225        230        235        240
Arg Lys Cys Cys Val Glu Cys Pro Pro Cys Pro Ala Pro Pro Val Ala
245        250        255
Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met
260        265        270
Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His
275        280        285
Glu Asp Pro Glu Val Gln Phe Asn Trp Tyr Val Asp Gly Val Glu Val
290        295        300
His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Phe Asn Ser Thr Phe
305        310        315        320
Arg Val Val Ser Val Leu Thr Val Val His Gln Asp Trp Leu Asn Gly
325        330        335
Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Gly Leu Pro Ala Pro Ile
340        345        350
Glu Lys Thr Ile Ser Lys Thr Lys Gly Gln Pro Arg Glu Pro Gln Val
355        360        365
Tyr Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser

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370	375	380
Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu		
385	390	395 400
Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro		
	405	410 415
Met Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val		
	420	425 430
Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met		
	435	440 445
His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser		
	450	455 460
Pro Gly Lys		
465		

<210> SEQ ID NO 80
 <211> LENGTH: 235
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 80

Met Ala Trp Ala Pro Leu Leu Leu Thr Leu Leu Ala His Cys Thr Gly		
1	5	10 15
Ser Trp Ala Asn Phe Met Leu Thr Gln Pro His Ser Val Ser Glu Ser		
	20	25 30
Pro Gly Lys Thr Val Ala Ile Ser Cys Thr Arg Asn Ser Gly Ser Ile		
	35	40 45
Ala Ser Asn Ser Val Gln Trp Tyr Gln Gln Arg Pro Gly Ser Ser Pro		
	50	55 60
Thr Thr Val Ile Phe Glu Asp Asn Gln Arg Pro Ser Gly Val Pro Asp		
65	70	75 80
Arg Phe Ser Gly Ser Ile Asp Ser Ser Ser Asn Ser Ala Ser Leu Thr		
	85	90 95
Ile Ser Gly Leu Lys Thr Glu Asp Glu Ala Asp Tyr Tyr Cys Gln Ser		
	100	105 110
Tyr Asp Ser Asn Asn Trp Val Phe Gly Gly Gly Thr Lys Leu Thr Val		
	115	120 125
Leu Gly Gln Pro Lys Ala Asn Pro Thr Val Thr Leu Phe Pro Pro Ser		
	130	135 140
Ser Glu Glu Leu Gln Ala Asn Lys Ala Thr Leu Val Cys Leu Ile Ser		
145	150	155 160
Asp Phe Tyr Pro Gly Ala Val Thr Val Ala Trp Lys Ala Asp Gly Ser		
	165	170 175
Pro Val Lys Ala Gly Val Glu Thr Thr Lys Pro Ser Lys Gln Ser Asn		
	180	185 190
Asn Lys Tyr Ala Ala Ser Ser Tyr Leu Ser Leu Thr Pro Glu Gln Trp		
	195	200 205
Lys Ser His Arg Ser Tyr Ser Cys Gln Val Thr His Glu Gly Ser Thr		
	210	215 220
Val Glu Lys Thr Val Ala Pro Thr Glu Cys Ser		
225	230	235

<210> SEQ ID NO 81
 <211> LENGTH: 470
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

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<400> SEQUENCE: 81

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Met Glu Phe Gly Leu Ser Trp Val Phe Leu Val Ala Leu Leu Arg Gly
1      5      10      15
Val Gln Cys Gln Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln
20      25      30
Pro Gly Arg Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe
35      40      45
Ser Ser Tyr Val Met His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu
50      55      60
Glu Trp Val Ala Val Ile Trp Tyr Asp Gly Ser Asn Lys Tyr Tyr Ala
65      70      75      80
Asp Ser Val Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn
85      90      95
Thr Leu Tyr Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val
100     105     110
Tyr Tyr Cys Ala Arg Glu Gly Tyr Asp Tyr Gly Glu Asp Tyr Tyr Tyr
115     120     125
Tyr Gly Met Asp Val Trp Gly Gln Gly Thr Thr Val Thr Val Ser Ser
130     135     140
Ala Ser Thr Lys Gly Pro Ser Val Phe Pro Leu Ala Pro Cys Ser Arg
145     150     155     160
Ser Thr Ser Glu Ser Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr
165     170     175
Phe Pro Glu Pro Val Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser
180     185     190
Gly Val His Thr Phe Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser
195     200     205
Leu Ser Ser Val Val Thr Val Pro Ser Ser Asn Phe Gly Thr Gln Thr
210     215     220
Tyr Thr Cys Asn Val Asp His Lys Pro Ser Asn Thr Lys Val Asp Lys
225     230     235     240
Thr Val Glu Arg Lys Cys Cys Val Glu Cys Pro Pro Cys Pro Ala Pro
245     250     255
Pro Val Ala Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp
260     265     270
Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp
275     280     285
Val Ser His Glu Asp Pro Glu Val Gln Phe Asn Trp Tyr Val Asp Gly
290     295     300
Val Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Phe Asn
305     310     315     320
Ser Thr Phe Arg Val Val Ser Val Leu Thr Val Val His Gln Asp Trp
325     330     335
Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Gly Leu Pro
340     345     350
Ala Pro Ile Glu Lys Thr Ile Ser Lys Thr Lys Gly Gln Pro Arg Glu
355     360     365
Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn
370     375     380
Gln Val Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile
385     390     395     400
Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr
405     410     415

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Thr Pro Pro Met Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys
 420 425 430

Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys
 435 440 445

Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu
 450 455 460

Ser Leu Ser Pro Gly Lys
 465 470

<210> SEQ ID NO 82
 <211> LENGTH: 236
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 82

Met Asp Met Arg Val Pro Ala Gln Leu Leu Gly Leu Leu Leu Trp
 1 5 10 15

Leu Arg Gly Ala Arg Cys Asp Ile Gln Met Thr Gln Ser Pro Ser Ser
 20 25 30

Leu Ser Ala Ser Val Gly Asp Arg Val Thr Ile Thr Cys Arg Ala Ser
 35 40 45

Gln Asp Ile Ser Asn Tyr Leu Asn Trp Tyr Gln Gln Lys Pro Gly Lys
 50 55 60

Ala Pro Lys Leu Leu Ile Tyr Tyr Thr Ser Arg Leu Leu Ser Gly Val
 65 70 75 80

Pro Ser Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr
 85 90 95

Ile Ser Ser Leu Gln Pro Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln
 100 105 110

Gly Asp Thr Leu Pro Tyr Thr Phe Gly Gly Gly Thr Lys Val Glu Ile
 115 120 125

Lys Arg Thr Val Ala Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp
 130 135 140

Glu Gln Leu Lys Ser Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn
 145 150 155 160

Phe Tyr Pro Arg Glu Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu
 165 170 175

Gln Ser Gly Asn Ser Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp
 180 185 190

Ser Thr Tyr Ser Leu Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr
 195 200 205

Glu Lys His Lys Val Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser
 210 215 220

Ser Pro Val Thr Lys Ser Phe Asn Arg Gly Glu Cys
 225 230 235

<210> SEQ ID NO 83
 <211> LENGTH: 468
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 83

Met Asp Trp Thr Trp Arg Ile Leu Phe Leu Val Ala Ala Ala Thr Gly
 1 5 10 15

Ala His Ser Glu Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys
 20 25 30

-continued

-continued

Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu
450 455 460

Ser Pro Gly Lys
465

<210> SEQ ID NO 84
<211> LENGTH: 240
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 84

Met Val Leu Gln Thr Gln Val Phe Ile Ser Leu Leu Leu Trp Ile Ser
1 5 10 15
Gly Ala Tyr Gly Asp Ile Val Met Thr Gln Ser Pro Asp Ser Leu Ala
20 25 30
Val Ser Leu Gly Glu Arg Ala Thr Ile Asn Cys Lys Ser Ser Gln Ser
35 40 45
Val Leu Asp Ser Ser Asp Asn Lys Asn Tyr Leu Ala Trp Tyr Gln Gln
50 55 60
Lys Pro Gly Gln Pro Pro Lys Leu Leu Ile Tyr Trp Ala Ser Asn Arg
65 70 75 80
Glu Ser Gly Val Pro Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp
85 90 95
Phe Thr Leu Thr Ile Ser Ser Leu Gln Ala Glu Asp Val Ala Val Tyr
100 105 110
Tyr Cys Gln Gln Tyr Tyr Ser Asp Pro Phe Thr Phe Gly Pro Gly Thr
115 120 125
Lys Val Asp Ile Lys Arg Thr Val Ala Ala Pro Ser Val Phe Ile Phe
130 135 140
Pro Pro Ser Asp Glu Gln Leu Lys Ser Gly Thr Ala Ser Val Val Cys
145 150 155 160
Leu Leu Asn Asn Phe Tyr Pro Arg Glu Ala Lys Val Gln Trp Lys Val
165 170 175
Asp Asn Ala Leu Gln Ser Gly Asn Ser Gln Glu Ser Val Thr Glu Gln
180 185 190
Asp Ser Lys Asp Ser Thr Tyr Ser Leu Ser Ser Thr Leu Thr Leu Ser
195 200 205
Lys Ala Asp Tyr Glu Lys His Lys Val Tyr Ala Cys Glu Val Thr His
210 215 220
Gln Gly Leu Ser Ser Pro Val Thr Lys Ser Phe Asn Arg Gly Glu Cys
225 230 235 240

<210> SEQ ID NO 85
<211> LENGTH: 465
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 85

Met Asp Trp Thr Trp Ser Ile Leu Phe Leu Val Ala Ala Pro Thr Gly
1 5 10 15
Ala His Ser Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys
20 25 30
Pro Gly Ala Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Thr Phe
35 40 45
Thr Ser Tyr Gly Ile Ser Trp Val Arg Gln Ala Pro Gly Gln Gly Leu
50 55 60

Glu 65	Trp	Met	Gly	Trp	Ile 70	Ser	Ala	Tyr	Asn	Gly 75	Asn	Thr	Asn	Tyr	Ala 80
Gln	Lys	Leu	Gln	Gly 85	Arg	Val	Thr	Met	Thr 90	Thr	Asp	Thr	Ser	Thr 95	Ser
Thr	Ala	Tyr	Met	Glu 100	Leu	Arg	Ser	Leu	Arg 105	Ser	Asp	Asp	Thr	Ala	Val
Tyr	Tyr	Cys	Ala	Arg	Glu	Ser	Trp	Phe	Gly 120	Glu	Val	Phe	Phe	Asp	Tyr
Trp	Gly	Gln	Gly	Thr	Leu	Val	Thr	Val	Ser 135	Ser	Ala	Ser	Thr	Lys	Gly
Pro 145	Ser	Val	Phe	Pro	Leu 150	Ala	Pro	Cys	Ser	Arg 155	Ser	Thr	Ser	Glu	Ser 160
Thr	Ala	Ala	Leu	Gly 165	Cys	Leu	Val	Lys	Asp 170	Tyr	Phe	Pro	Glu	Pro	Val
Thr	Val	Ser	Trp	Asn	Ser	Gly	Ala	Leu	Thr 185	Ser	Gly	Val	His	Thr	Phe
Pro	Ala	Val	Leu	Gln	Ser	Ser	Gly 200	Leu	Tyr	Ser	Leu	Ser	Ser	Val	Val
Thr 210	Val	Pro	Ser	Ser	Asn	Phe	Gly 215	Thr	Gln	Thr	Tyr	Thr	Cys	Asn	Val
Asp 225	His	Lys	Pro	Ser	Asn 230	Thr	Lys	Val	Asp	Lys 235	Thr	Val	Glu	Arg	Lys 240
Cys	Cys	Val	Glu	Cys 245	Pro	Pro	Cys	Pro	Ala 250	Pro	Pro	Val	Ala	Gly	Pro 255
Ser	Val	Phe	Leu	Phe 260	Pro	Pro	Lys	Pro	Lys 265	Asp	Thr	Leu	Met	Ile	Ser 270
Arg	Thr	Pro	Glu	Val	Thr	Cys	Val 280	Val	Val	Asp	Val	Ser	His	Glu	Asp 285
Pro 290	Glu	Val	Gln	Phe	Asn	Trp	Tyr 295	Val	Asp	Gly	Val	Glu	Val	His	Asn 300
Ala 305	Lys	Thr	Lys	Pro	Arg 310	Glu	Glu	Gln	Phe	Asn 315	Ser	Thr	Phe	Arg	Val 320
Val	Ser	Val	Leu	Thr 325	Val	Val	His	Gln	Asp 330	Trp	Leu	Asn	Gly	Lys	Glu 335
Tyr	Lys	Cys	Lys	Val 340	Ser	Asn	Lys	Gly	Leu 345	Pro	Ala	Pro	Ile	Glu	Lys 350
Thr	Ile	Ser	Lys	Thr 355	Lys	Gly	Gln 360	Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr 365
Leu 370	Pro	Pro	Ser	Arg	Glu	Glu 375	Met	Thr	Lys	Asn 380	Gln	Val	Ser	Leu	Thr 385
Cys 385	Leu	Val	Lys	Gly	Phe 390	Tyr	Pro	Ser	Asp	Ile 395	Ala	Val	Glu	Trp	Glu 400
Ser	Asn	Gly	Gln	Pro 405	Glu	Asn	Asn	Tyr	Lys 410	Thr	Thr	Pro	Pro	Met	Leu 415
Asp	Ser	Asp	Gly	Ser 420	Phe	Phe	Leu	Tyr	Ser 425	Lys	Leu	Thr	Val	Asp	Lys 430
Ser	Arg	Trp	Gln	Gln 435	Gly	Asn	Val 440	Phe	Ser	Cys	Ser	Val	Met	His	Glu 445
Ala 450	Leu	His	Asn	His	Tyr	Thr 455	Gln	Lys	Ser	Leu	Ser	Leu	Ser	Pro	Gly 460

Lys
465

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<210> SEQ ID NO 86
 <211> LENGTH: 236
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 86

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Met Asp Met Arg Val Pro Ala Gln Leu Leu Gly Leu Leu Leu Leu Trp
 1             5             10             15

Leu Arg Gly Ala Arg Cys Asp Ile Gln Met Thr Gln Ser Pro Ser Ser
          20             25             30

Val Ser Ala Ser Val Gly Asp Arg Val Thr Ile Thr Cys Arg Ala Ser
          35             40             45

Gln Gly Ile Ser Ser Trp Leu Ala Trp Tyr Gln Gln Lys Pro Gly Lys
 50             55             60

Ala Pro Lys Leu Leu Ile Tyr Gly Ala Ser Asn Leu Glu Ser Gly Val
 65             70             75             80

Pro Ser Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr
          85             90             95

Ile Ser Ser Leu Gln Pro Glu Asp Phe Ala Asn Tyr Tyr Cys Gln Gln
          100            105            110

Ala Asn Ser Phe Pro Trp Thr Phe Gly Gln Gly Thr Lys Val Glu Ile
          115            120            125

Lys Arg Thr Val Ala Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp
          130            135            140

Glu Gln Leu Lys Ser Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn
          145            150            155            160

Phe Tyr Pro Arg Glu Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu
          165            170            175

Gln Ser Gly Asn Ser Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp
          180            185            190

Ser Thr Tyr Ser Leu Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr
          195            200            205

Glu Lys His Lys Val Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser
          210            215            220

Ser Pro Val Thr Lys Ser Phe Asn Arg Gly Glu Cys
          225            230            235

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<210> SEQ ID NO 87
 <211> LENGTH: 466
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 87

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Met Asp Met Arg Val Pro Ala Gln Leu Leu Gly Leu Leu Leu Leu Trp
 1             5             10             15

Leu Arg Gly Ala Arg Cys Gln Val Gln Leu Val Gln Ser Gly Ala Glu
          20             25             30

Val Lys Lys Pro Gly Ala Ser Val Lys Val Ser Cys Lys Val Ser Gly
          35             40             45

Tyr Thr Leu Ser Asp Leu Ser Ile His Trp Val Arg Gln Ala Pro Gly
          50             55             60

Lys Gly Leu Glu Trp Met Gly Gly Phe Asp Pro Gln Asp Gly Glu Thr
          65             70             75             80

Ile Tyr Ala Gln Lys Phe Gln Gly Arg Val Thr Met Thr Glu Asp Thr
          85             90             95

Ser Thr Asp Thr Ala Tyr Met Glu Leu Ser Ser Leu Lys Ser Glu Asp

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100							105					110				
Thr	Ala	Val	Tyr	Tyr	Cys	Ala	Thr	Gly	Ser	Ser	Ser	Ser	Trp	Phe	Asp	
	115						120					125				
Pro	Trp	Gly	Gln	Gly	Thr	Leu	Val	Thr	Val	Ser	Ser	Ala	Ser	Thr	Lys	
	130					135						140				
Gly	Pro	Ser	Val	Phe	Pro	Leu	Ala	Pro	Cys	Ser	Arg	Ser	Thr	Ser	Glu	
145					150					155					160	
Ser	Thr	Ala	Ala	Leu	Gly	Cys	Leu	Val	Lys	Asp	Tyr	Phe	Pro	Glu	Pro	
				165					170					175		
Val	Thr	Val	Ser	Trp	Asn	Ser	Gly	Ala	Leu	Thr	Ser	Gly	Val	His	Thr	
			180					185					190			
Phe	Pro	Ala	Val	Leu	Gln	Ser	Ser	Gly	Leu	Tyr	Ser	Leu	Ser	Ser	Val	
	195					200						205				
Val	Thr	Val	Pro	Ser	Ser	Asn	Phe	Gly	Thr	Gln	Thr	Tyr	Thr	Cys	Asn	
	210					215					220					
Val	Asp	His	Lys	Pro	Ser	Asn	Thr	Lys	Val	Asp	Lys	Thr	Val	Glu	Arg	
225					230					235				240		
Lys	Cys	Cys	Val	Glu	Cys	Pro	Pro	Cys	Pro	Ala	Pro	Pro	Val	Ala	Gly	
			245					250						255		
Pro	Ser	Val	Phe	Leu	Phe	Pro	Pro	Lys	Pro	Lys	Asp	Thr	Leu	Met	Ile	
			260					265					270			
Ser	Arg	Thr	Pro	Glu	Val	Thr	Cys	Val	Val	Val	Asp	Val	Ser	His	Glu	
	275					280						285				
Asp	Pro	Glu	Val	Gln	Phe	Asn	Trp	Tyr	Val	Asp	Gly	Val	Glu	Val	His	
	290					295					300					
Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu	Gln	Phe	Asn	Ser	Thr	Phe	Arg	
305					310					315					320	
Val	Val	Ser	Val	Leu	Thr	Val	Val	His	Gln	Asp	Trp	Leu	Asn	Gly	Lys	
			325					330						335		
Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys	Gly	Leu	Pro	Ala	Pro	Ile	Glu	
		340						345					350			
Lys	Thr	Ile	Ser	Lys	Thr	Lys	Gly	Gln	Pro	Arg	Glu	Pro	Gln	Val	Tyr	
	355						360					365				
Thr	Leu	Pro	Pro	Ser	Arg	Glu	Glu	Met	Thr	Lys	Asn	Gln	Val	Ser	Leu	
	370					375					380					
Thr	Cys	Leu	Val	Lys	Gly	Phe	Tyr	Pro	Ser	Asp	Ile	Ala	Val	Glu	Trp	
385				390						395				400		
Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn	Tyr	Lys	Thr	Thr	Pro	Pro	Met	
			405						410					415		
Leu	Asp	Ser	Asp	Gly	Ser	Phe	Phe	Leu	Tyr	Ser	Lys	Leu	Thr	Val	Asp	
		420						425					430			
Lys	Ser	Arg	Trp	Gln	Gln	Gly	Asn	Val	Phe	Ser	Cys	Ser	Val	Met	His	
	435					440						445				
Glu	Ala	Leu	His	Asn	His	Tyr	Thr	Gln	Lys	Ser	Leu	Ser	Leu	Ser	Pro	
	450					455					460					
Gly	Lys															
465																

What is claimed is:

1. A method of preparing a polypeptide for analysis, comprising: a) cleaving the polypeptide in a sample in a first digestion, wherein the cleaving produces at least two fragments of the polypeptide, b) cleaving at least one of the at least two fragments of the polypeptide of a) with neutrophil

elastase; and c) analyzing the sample after cleaving the at least two fragments of the polypeptide with the neutrophil elastase.

2. A method of preparing a polypeptide for analysis, comprising: a) cleaving the polypeptide in a sample in a first digestion, wherein the cleaving produces at least two frag-

17. The method of claim **11**, wherein the polypeptide is a therapeutic polypeptide.

65 cirumab, ranibizumab, raxibacumab, regavirumab, reslizumab, rilotumumab, rituximab, robatumumab, roledumab, romosozumab, rontalizumab, rovelizumab, ruplizumab, samalizumab, sarilumab, satumomab, pendetide,

209

secukinumab, sevirumab, sibrotuzumab, sifalimumab, sil-
tuximab, simtuzumab, sipilizumab, sirukumab, solan-
ezumab, solitomab, sonepcizumab, sontuzumab, stamu-
lumab, sulesomab, suvizumab, tabalumab, tacatuzumab
tetraxetan, tadocizumab, talizumab, tanezumab, taplitumo-
mab paptox, tefibazumab, telimomab aritox, tenatumomab,
tefibazumab, teneliximab, teplizumab, teprotumumab,
tezepelumab, TGN1412, tremelimumab, ticilimumab, tild-
rakizumab, tigatuzumab, TNX-650, tocilizumab, torali-
zumab, tositumomab, tralokinumab, trastuzumab, TRBS07,
tregalizumab, tucotuzumab celmoleukin, tuvirumab,
ublituximab, urelumab, urtoxazumab, ustekinumab, vapal-
iximab, vatelizumab, vedolizumab, veltuzumab, vepalimo-
mab, vesencumab, visilizumab, volociximab, vorsetuzumab
mafodotin, votumumab, zalutumumab, zanolimumab,
zatuximab, ziralimumab, zolimomab aritox, a glycoprotein,
CD polypeptide, a HER receptor polypeptide, a cell adhe-
sion polypeptide, a growth factor polypeptide, an insulin
polypeptide, an insulin-related polypeptide, a coagulation
polypeptide, a coagulation-related polypeptide, albumin,

210

IgE, a blood group antigen, a colony stimulating factor, a
receptor, a neurotrophic factor, an interferon, an interleukin,
a viral antigen, a lipoprotein, calcitonin, glucagon, atrial
natriuretic factor, lung surfactant, tumor necrosis factor-
alpha and -beta, enkephalinase, mouse gonadotropin-asso-
ciated peptide, DNase, inhibin, activating, an integrin, protein
A, protein D, a rheumatoid factor, an immunotoxin, a bone
morphogenetic protein, a superoxide dismutase, a surface
membrane polypeptide, a decay accelerating factor, an AIDS
envelope, a transport polypeptide, a homing receptor, an
addressin, a regulatory polypeptide, an immunoadhesin, a
myostatin, a TALL polypeptide, an amyloid polypeptide, a
thymic stromal lymphopoietin, a RANK ligand, a c-kit
polypeptide, a TNF receptor, and an angiopoietin, the anti-
bodies shown in Table H and biologically active fragments,
analogs or variants thereof.

20. The method of claim 17, wherein the therapeutic
polypeptide is a bi-specific T-cell engager molecule.

* * * * *